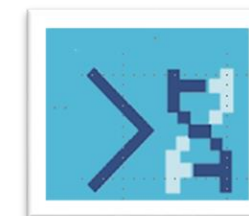


Iniciación al análisis de datos procedentes de técnicas de secuenciación masiva (NGS)

Unidad de Bioinformática (BU-ISCIII)
Unidades Centrales Científico Técnicas – SGSAFI-ISCIII



9 – 13 junio 2025, 12ª Edición
Programa Formación Continua, ISCIII



OBJETIVOS DEL CURSO

- ❖ Aproximación a las técnicas de secuenciación masiva (NGS) y a sus aplicaciones
- ❖ Adquirir conocimientos básicos del entorno linux
- ❖ Familiarizarse con los formatos de ficheros generados en el análisis de datos procedentes de la SM
- ❖ Conocer el flujo del análisis de los datos procedentes de la SM

Sesión 1 - Secuenciación Masiva Plataformas de Secuenciación

Isabel Cuesta

Unidad de Bioinformática
Unidades Centrales Científico Técnicas – SGSAFI-ISCIII

9 – 13 junio 2025, 12ª Edición
Programa Formación Continua, ISCIII

INDICE

❖ Unidad de Bioinformática
Servicios ofertados

❖ Evolución de la secuenciación

❖ Plataformas de secuenciación masiva paralela (NGS o HTS)

Qué es la Bioinformática?

**PROBLEMAS
BIOLÓGICOS**



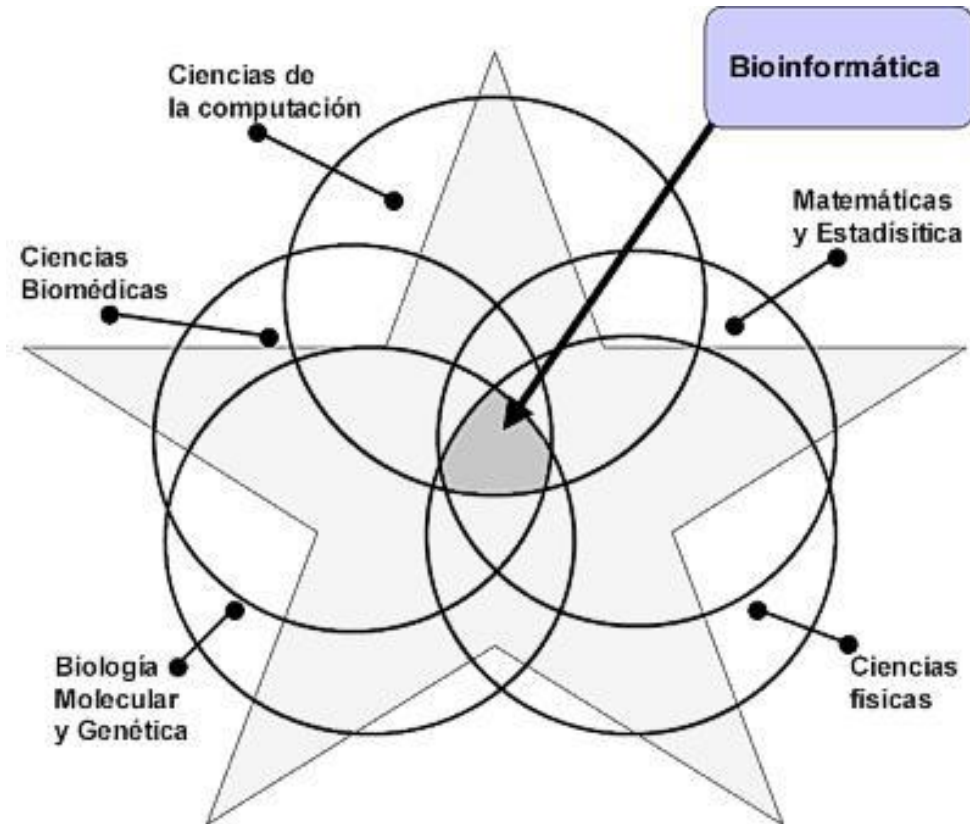
**Procesamiento
de datos**



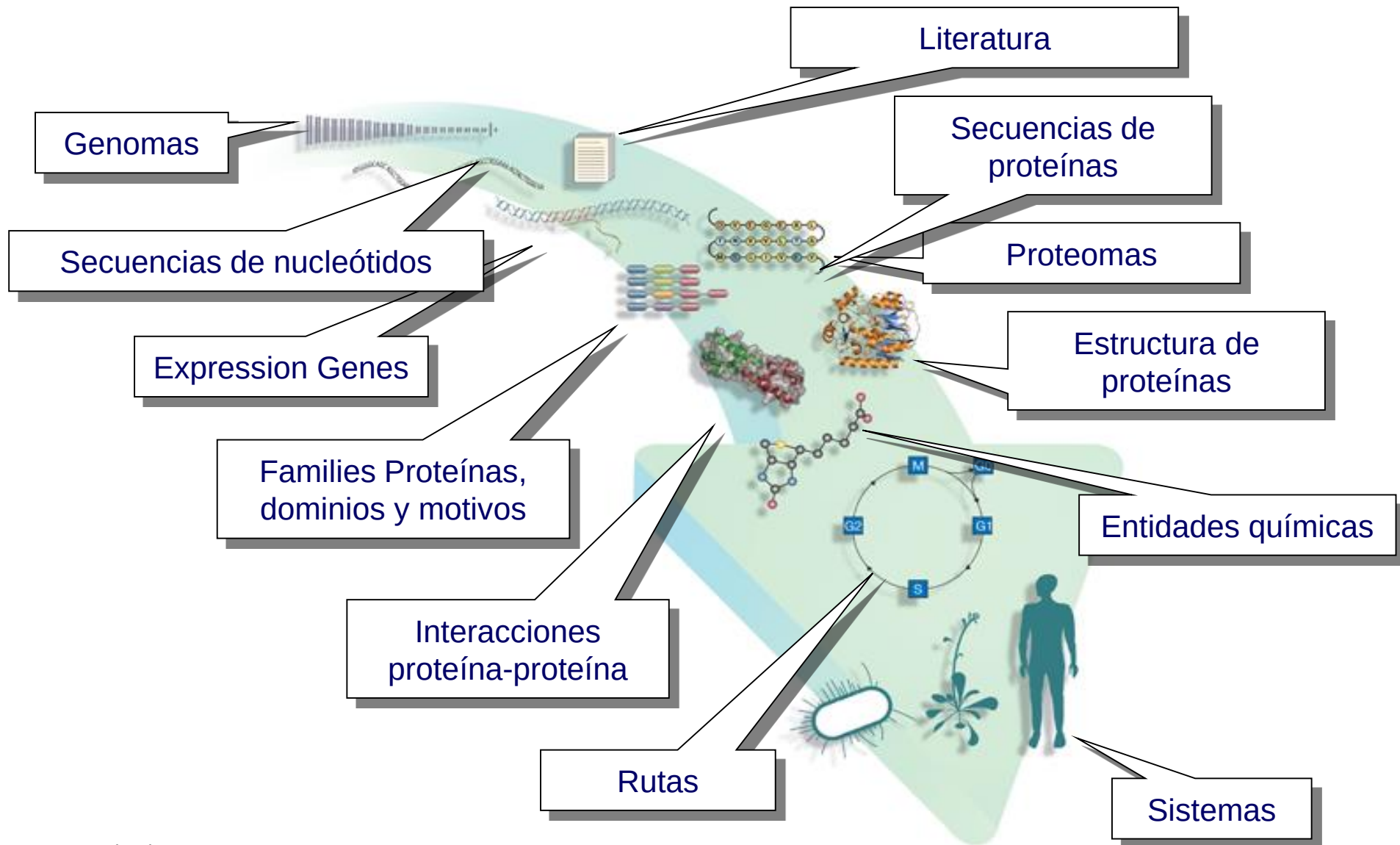
**MÉTODOS
COMPUTACIONALES**



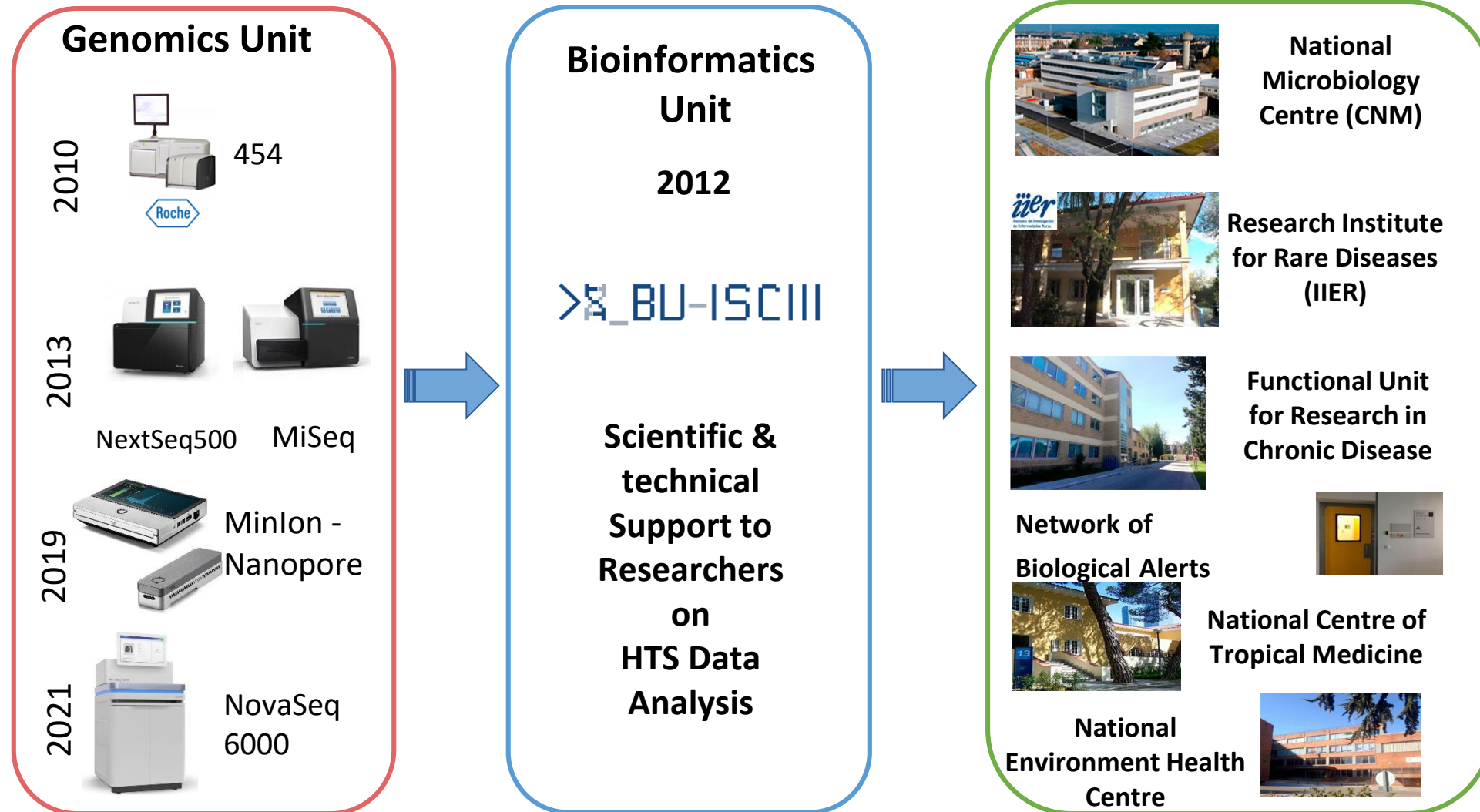
Bioinformática es multidisciplinar



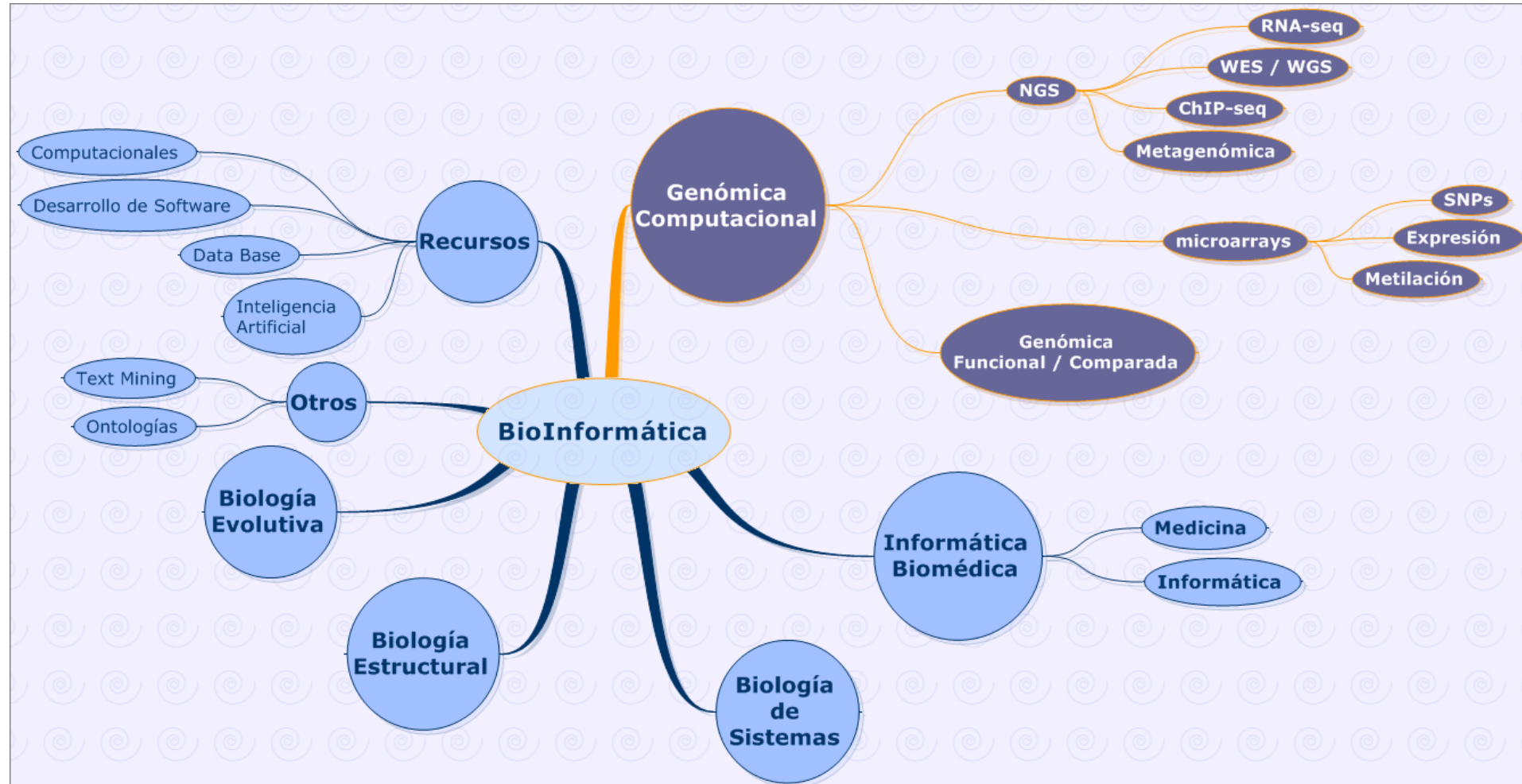
Tipos de datos dan idea de la dimensión de la Bioinformática



Why BU-ISCI III was founded



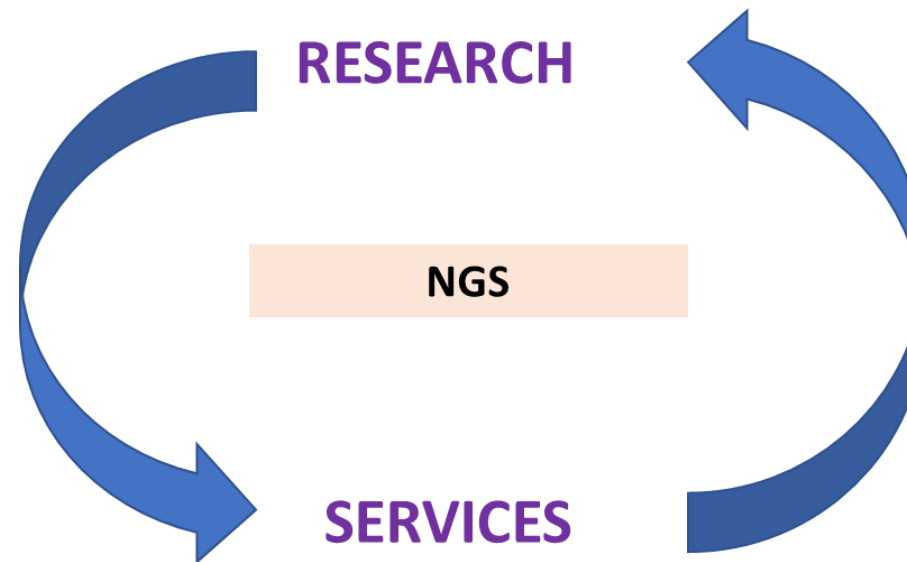
BU-ISCIII Mission - Activities



Bioinformatics Unit Activities

- Identify biological problems (PI / Groups) that could be target of NGS
- Early adopters: establish collaboration with.
- Be strategic providing transversal solutions → reusable tools

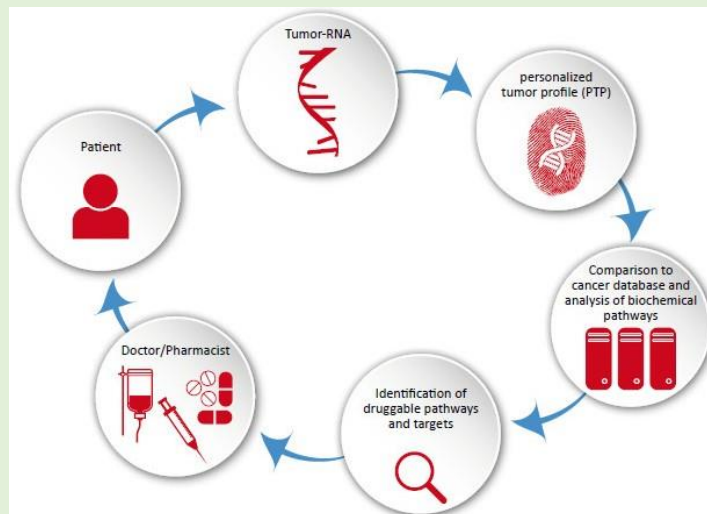
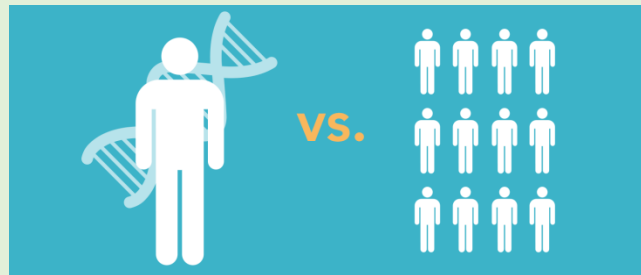
Provide scientific and technical solutions for using NGS in the diagnostic routine or research activity from different ISCI labs



Clinical Bioinformatics - Precision Medicine

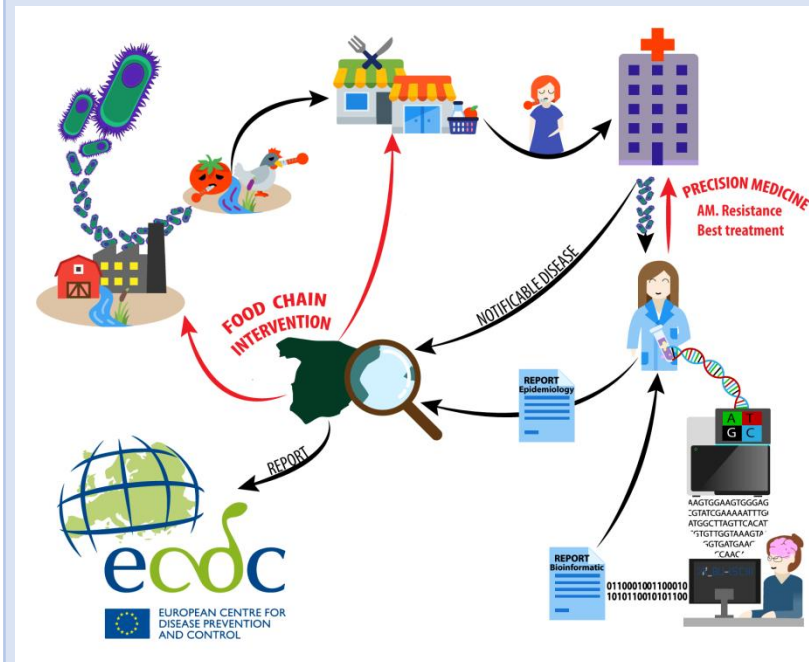
IIER

Tumour or Rare Diseases

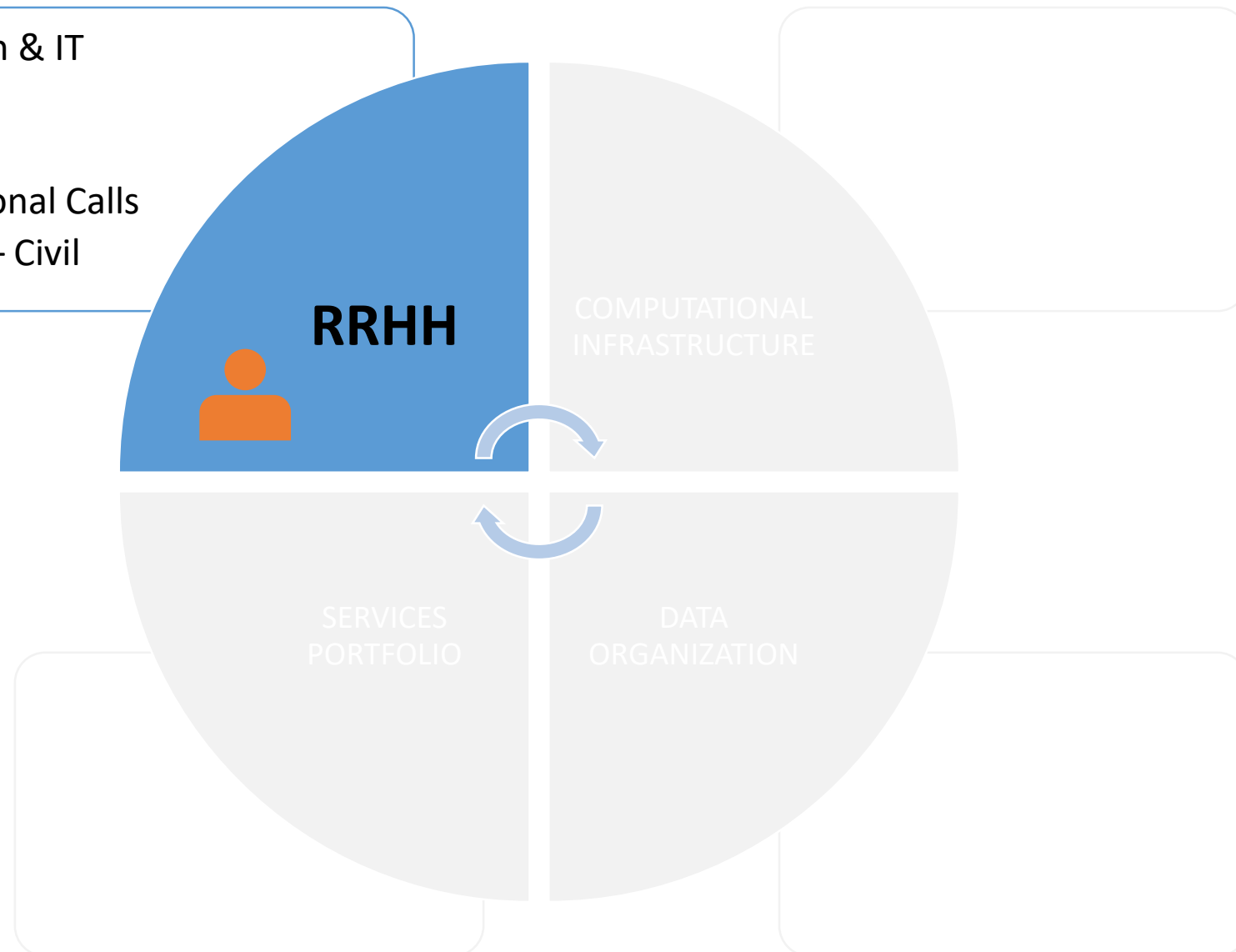


CNM

Pathogen associated



- TEAM: Bioinformatician & IT
- SOURCE OF FUNDING:
 - Research Project
 - National or International Calls
 - Permanent position – Civil Servant



Research Projects



>X_BU-ISCIII



Find us in <https://github.com/BU-ISCIII>

Isabel Cuesta, Coordinador, Dirección Científica

Sara Monzón, Dirección Técnica

Sarai Varona, Responsable Servicio

Juan Ledesma, Científico Titular OPIS

Emi Arjona

Pablo Mata

Jaime Ozaez

Alejandro Bernabeu

Victor López

Daniel Valle

Alba Talavera



Ex- members

Miguel Juliá, Jose Luis García-Pacheco

Pedro Sola, Javier Galiano

Jorge de la Barrera, Alejandro Barrera

Nuria Tubio, Sara López, Luis Chapado,

Guillermo Gorines, Erika Kvaalem, Alberto

Lema, Luis Aranda, Sergio Olmos

Master Students

Emi Arjona, Gregorio Sánchez-Montes, Juan

María Roldan, Esperanza López, Elena

García, Pablo Mata, Ana Hernández, Oscar

Brochado, Andrea Rubio, Enrique Álvarez

Jennifer Villa, Esther Viedma

Jose Miguel Lezana

Genomics Unit

Angel Zaballos

Pilar Jimenez

Staff técnico

utiic

Bruno Lobo

Antonio Arenas

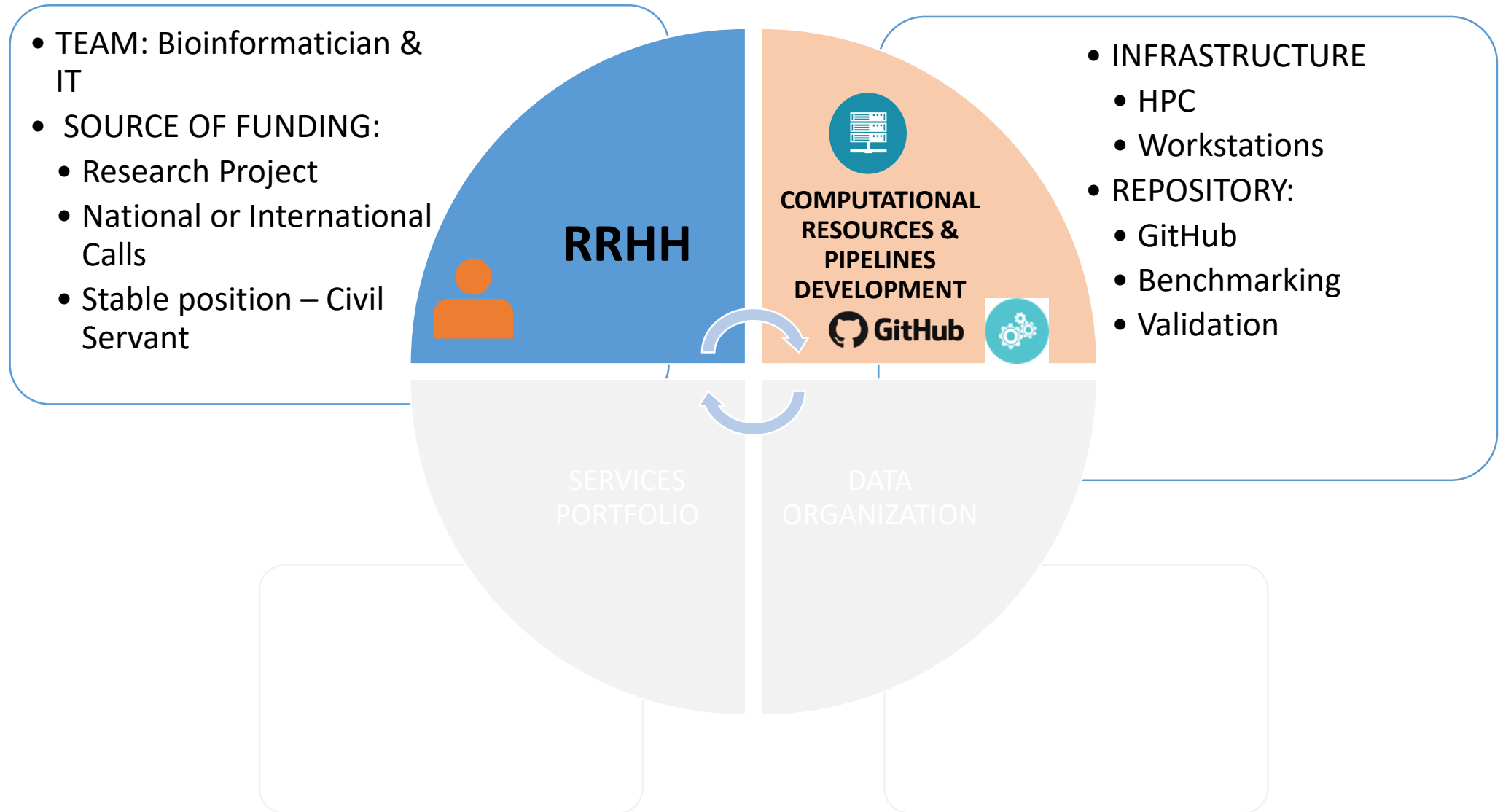
CNM Reference

Laboratories

IIER Human Genetic Lab

RELECOV





Computational Resources

- IT support: establish agreement with IT department including permission for using Linux.



Workstations (5), 4cores, 64Gb, 8TB
Server, 4-quad, 120Gb, 16TB

Data Centre (CPD-ISCI)



HPC 320 cores, 8TB RAM, 10Gbps.
2 flexible and scalable storages,
NetApp, 70 TB and 250TB

- Reproducibility of in-silico pipelines analysis

nextflow



Singularity containers
Admin support & environment independency
Sharing code easier



<https://github.com/BU-ISCI>

Bioinformatic Analysis: Software validation: ECDC EQAs

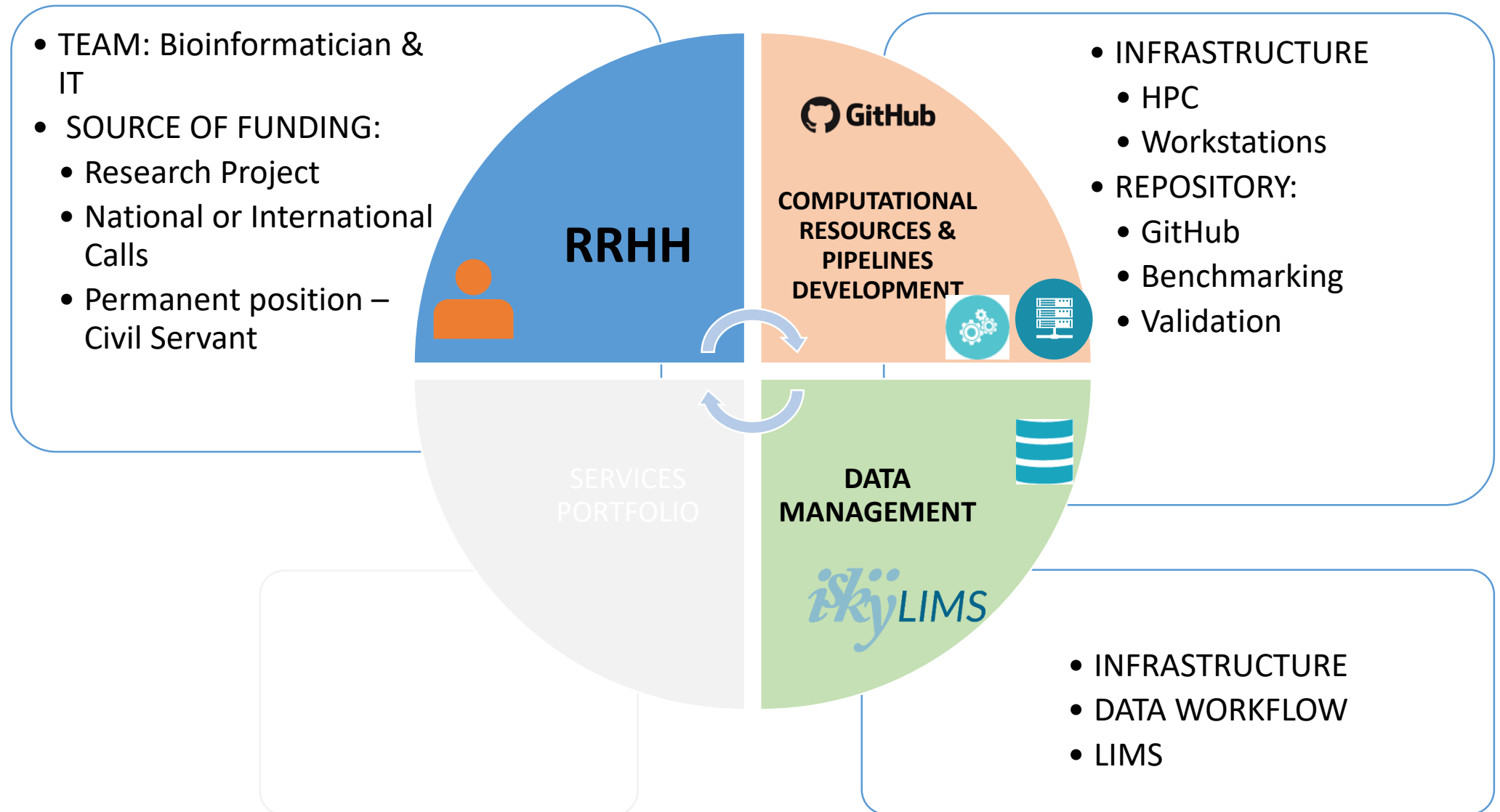
Table 5. Results of allele-based cluster analysis

Lab ID	Allele based analysis					
	Approach	Allelic calling method	Assembler	Scheme	Difference within cluster	Difference outside cluster
EQA provider	BioNumerics	Assembly- and mapping-based	SPAdes	Applied Math (cgMLST/Pasteur)	0-3	24-1112
19	BioNumerics	Assembly- and mapping-based	SPAdes	Applied Math (cgMLST/Pasteur)	0-3	25-1120
35	SeqPhere	Assembly-based only	Velvet	Ruppitsch (cgMLST)	0-2	16-1065
70	SeqPhere	Assembly-based only	Velvet	Ruppitsch (cgMLST)	0-2	16-1062
105*	SeqPhere	Assembly-based only	SPAdes v 3.80	Ruppitsch (cgMLST)	0-1*	23-812
129	SeqPhere	Assembly-based only	Velvet			
135	SeqPhere	Assembly-based only	CLC Genomic Workbench 10			
141	SeqPhere	Assembly-based only	SPAdes 3.9.0			
142	Inhouse	Assembly-based only	SPAdes			
144	SeqPhere	Assembly-based only	Velvet			

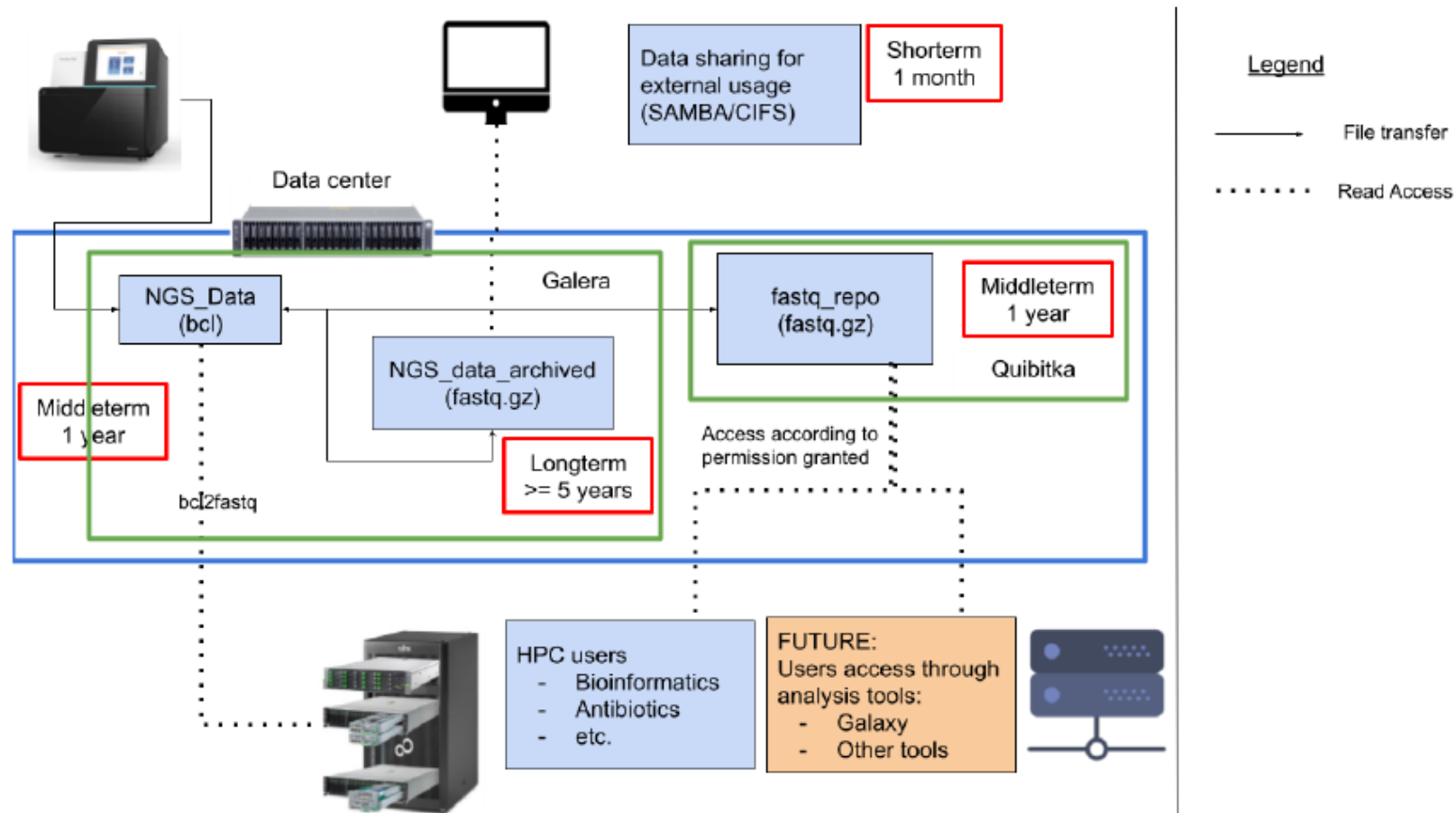
Table 4. Results of SNP-based cluster analysis

Lab ID	SNP-based						
	Approach	Reference	Read mapper	Variant caller	Assembler	Distance within cluster	Distance outside cluster
Provider	Reference-based	ST6 (REF4)	BWA	GATK		0-3	38-71
19*	Reference-based	ST6 ID 2362	BWA	GATK		0-4	43-81
56	Assembly-based			ksnp3	SPAdes	0-57*	561-591 (6109)
105	Reference-based	ST6 J1817	Bowtie2	VARSCAN 2		0-2*	22-42 (1049)
108	Reference-based	In-house strain resp ST	CLC assembly cell v4.4.2	CLC assembly cell v4.4.2		0-2	37-72
142*	Reference-based	Listeria EGDe (cc9)	CLC Bio	CLC Bio		0-1219	1223-2814 (8138)
146	Reference-based	ST6 ref. CP006046 ST1 ref. F2365 ST213/ST382 no ref.	BWA	In-house		0-358	

Fifth external quality assessment scheme for *Listeria monocytogenes* typing



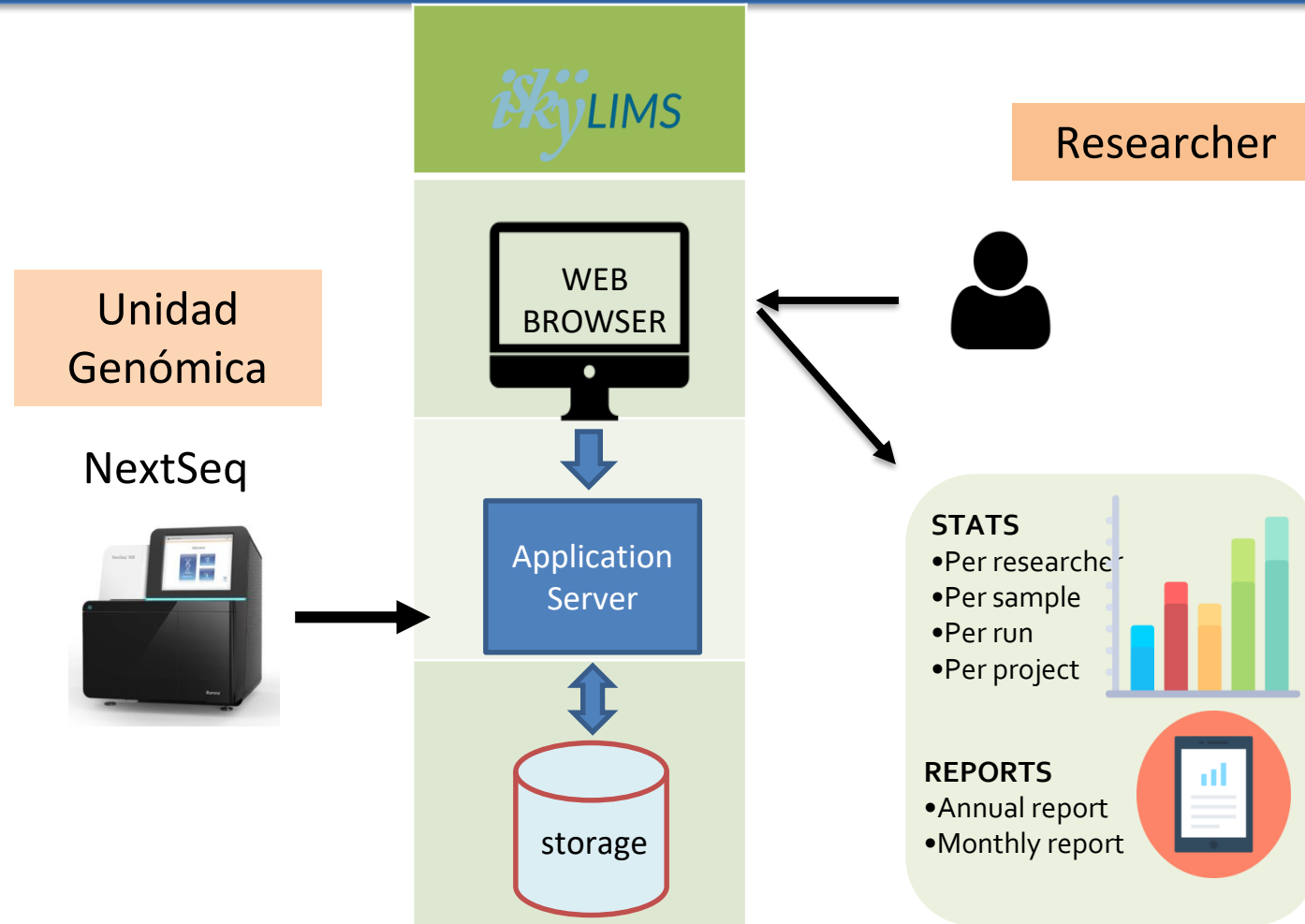
Infrastructure and data management



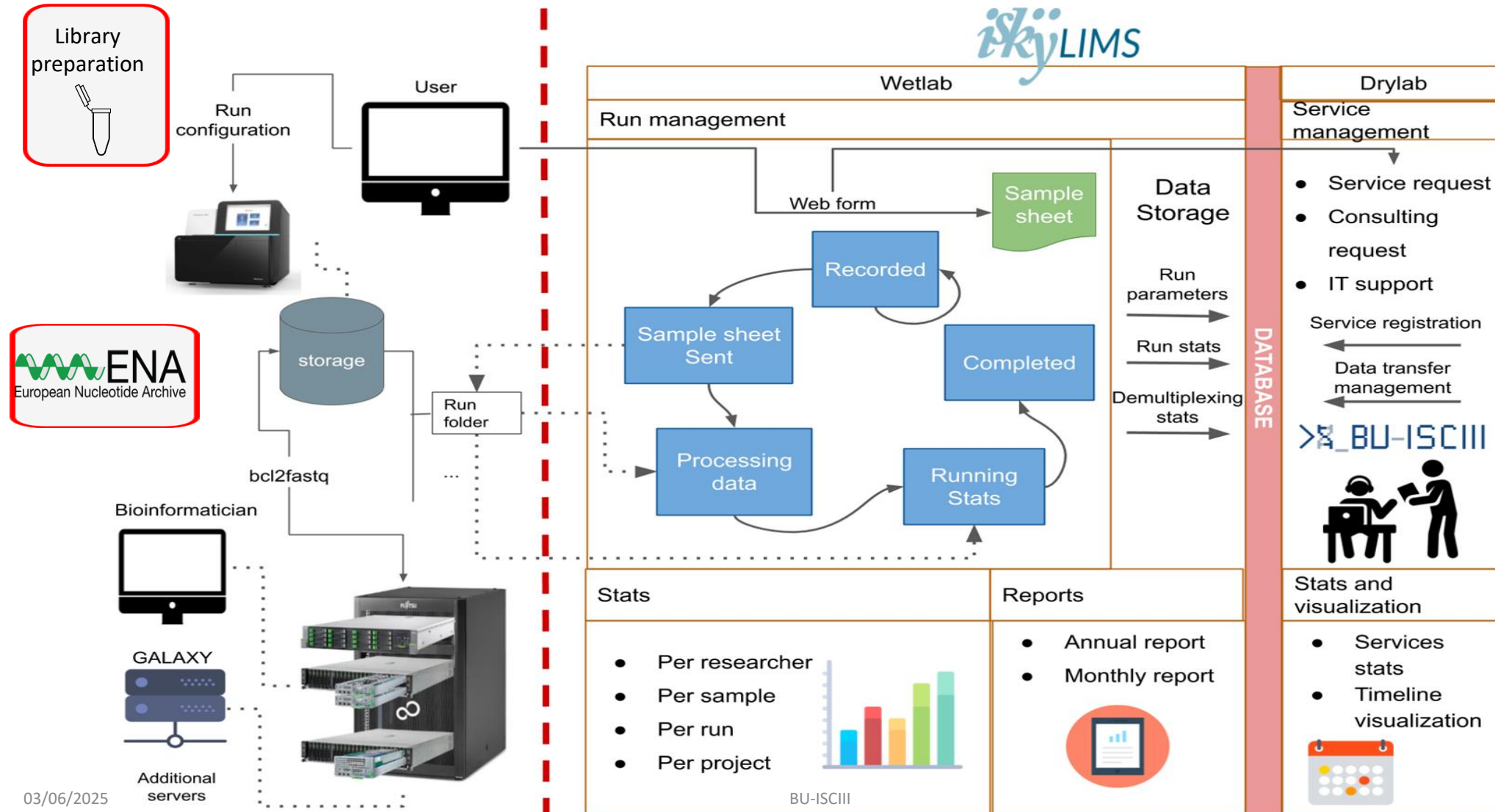
Minimize
I/O issues
Maximize
storage
uses

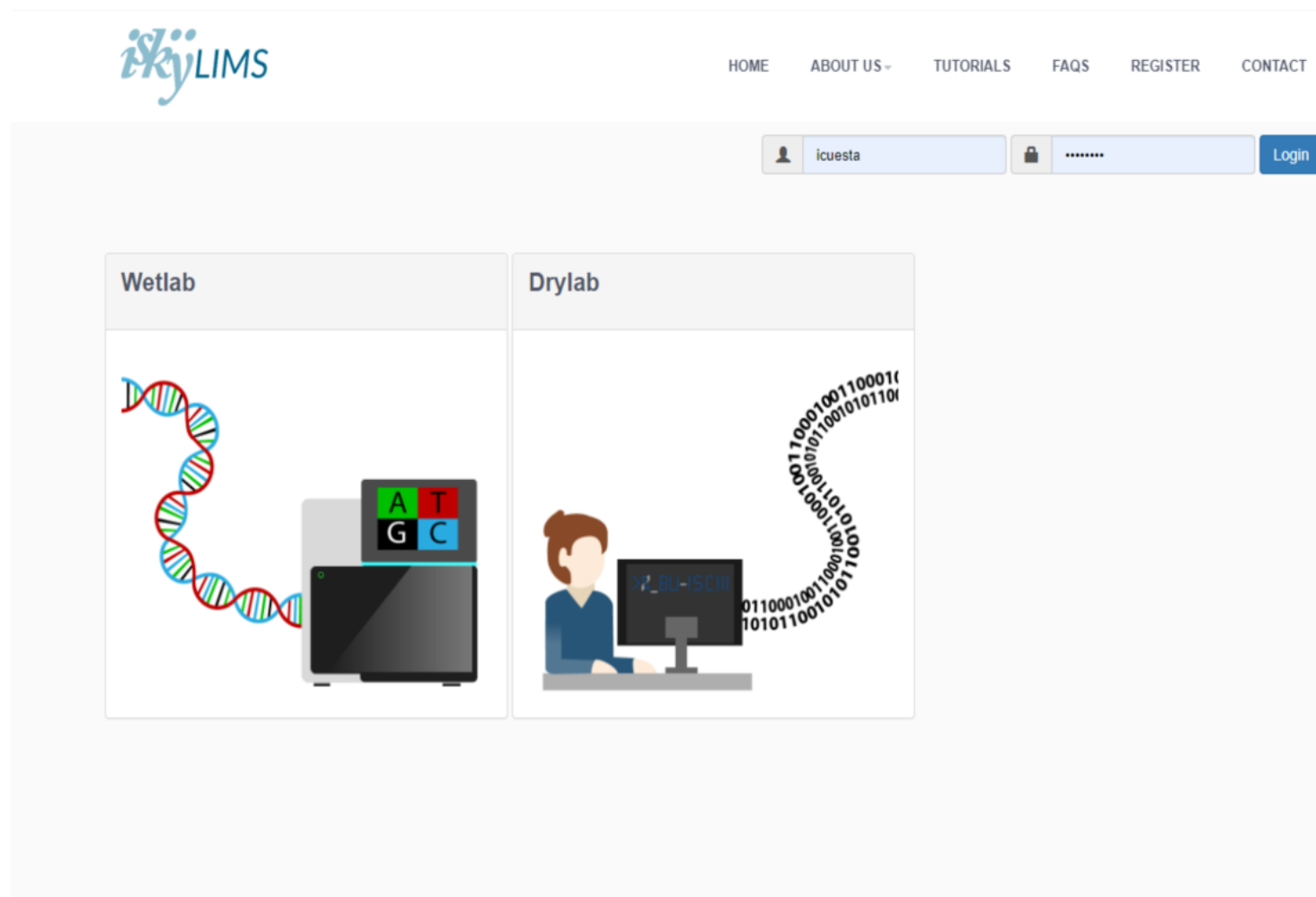
Fig. 3: Esquema de organización datos genómicos

iSkyLIMS <https://iskylims.isciii.es/>



Infrastructure and data management: LIMS





Logos



Connect



Links

- Contact
- Getting started
- FAQs

Sitemap

- iSkyLIMS home
- Drylab page
- Wetlab page

Bioinformatics

iSkyLIMS: DryLab

Welcome

This section will allow you to check BU-ISCIII service activity. Available processes are request new services, colaborations, counseling and infrastructure. You will be able to check the status of your ongoing services.



Services ongoing and queued

Under construction. This will be a table with services ongoing or queued

Timeline of services

Under construction. Kind of diagram with services dates.



Service Request Form

Form for requesting internal service to Bioinformatic Unit

Sequencing Data

User's projects*

BMartinez20161213

EXOMAS_ND_20170303

EXOMAS_ND_20170327_RE

EXOMAS_ND_20170228

MedGen_GNM_050-BM44xxx00470040

Run specifications

File extension

Sequencing platform

- ☐ Genomic Data Analysis
 - ☐ Download and quality analysis
 - ☐ Data download
 - ☐ Sequence quality analysis
 - ☐ Sequence pre-processing (quality filtering)
 - ☐ Next Generation Sequencing data analysis
 - ☐ DNAseq: Exome sequencing (WES) / Genome sequencing (WGS) / Target sequencing
 - ☐ Trio/family variant calling pipeline
 - ☐ Variant calling and annotation pipeline
 - ☐ Microbial: Whole genome outbreak analysis pipeline
 - ☐ Microbial: wgMLST
 - ☐ Microbial: MLST + virulence + AMR + plasmid analysis
 - ☐ Microbial: Assembly + automatic annotation
 - ☐ Microbial: plasmidID pipeline - strain plasmid characterization
 - ☐ RNAseq: Transcriptome sequencing
 - ☐ miRNA-Seq pipeline
 - ☐ mRNA-Seq pipeline
 - ☐ Amplicon sequencing (Deep sequencing)
 - ☐ Low frequency variant detection
 - ☐ Viral: assembly and minor variants detection
 - ☐ Metagenomics
 - ☐ 16S taxonomic profiling
 - ☐ Shotgun metagenomics profiling
 - ☐ Shotgun metagenomics - Virus genome reconstruction
 - ☐ CHIP-SEQ
 - ☐ Peak detection and annotation



Infrastructure and data management



HOME RUN PREPARATION SEARCH STATISTICS REPORTS

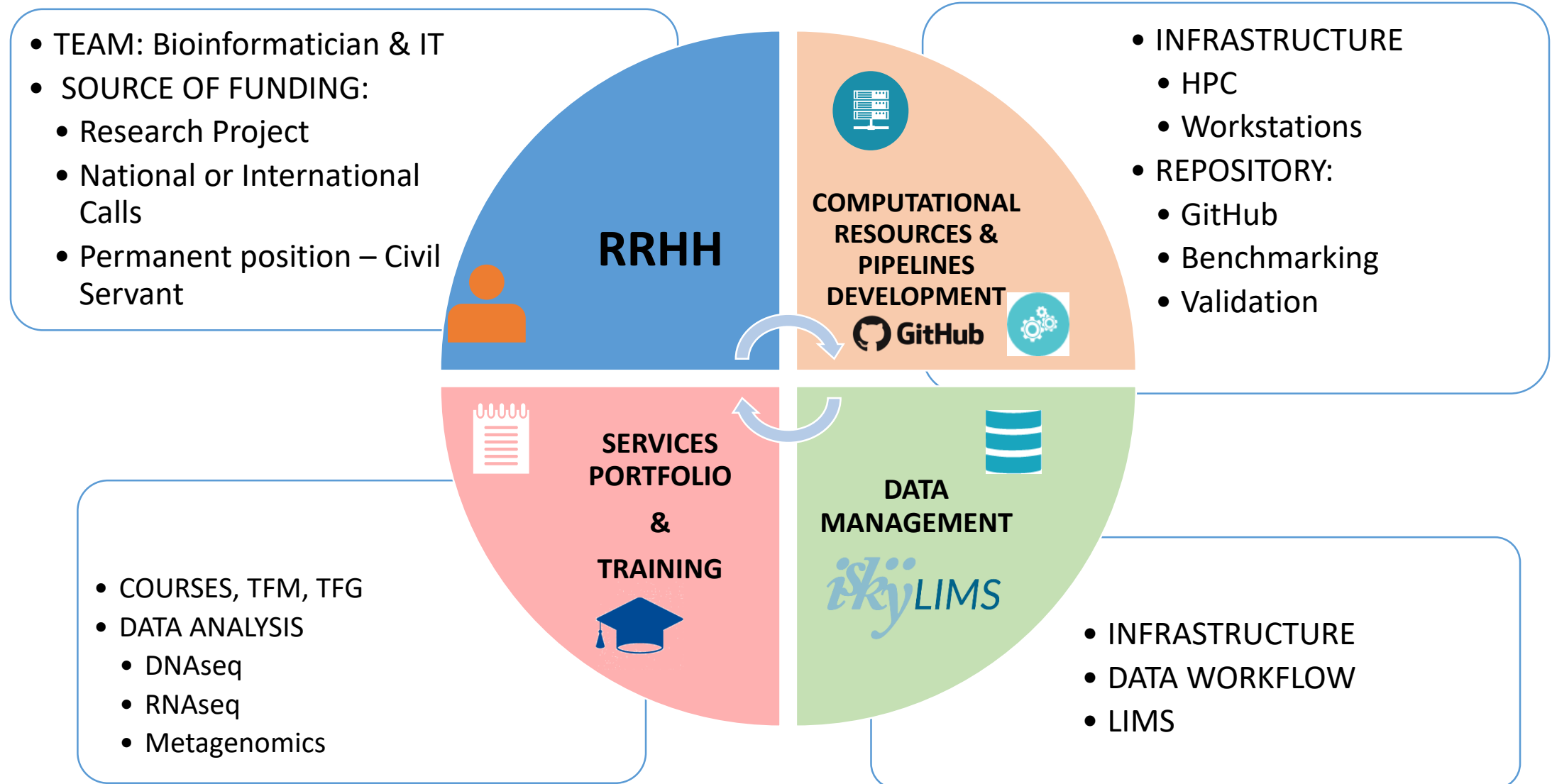
bioinfoadm Logout My account

Statistics results for Investigator rabad

Projects using the sequencer NS500454 :

Export Table To Excel

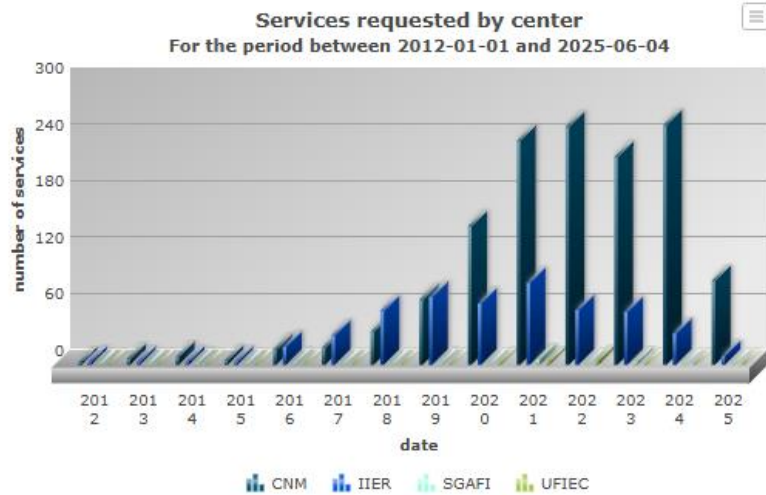
Project name	Date	Libraty Kit	Samples	Cluster PF	Yield Mb	% Q> 30	Mean	Sequencer ID
NextSeq_CNM_191_20191004_RAbad	No Date	Nextera DNA CD Indexes (96 Indexes plated)	48	149,441,968	45,876	89.98	33.70	NS500454
NextSeq_CNM_166_20190528b_Rabad	No Date	Nextera XT v2 Set B	96	139,317,411	43,016	89.58	33.72	NS500454
NextSeq_CNM_166_20190528a_Rabad	No Date	Nextera XT v2 Set A	82	102,267,350	31,623	89.26	33.65	NS500454
NextSeq_CNM_150_20190218B_RAbad	No Date	Nextera XT v2 set B	20	17,335,577	5,352	86.77	33.17	NS500454
NextSeq_CNM_150_20190221A_RAbad	No Date	Nextera XT v2 Set A	96	127,755,164	39,595	85.28	32.86	NS500454
NextSeq_CNM_166_20190528c_Rabad	No Date	Nextera XT v2 Set C	96	152,945,860	47,264	89.38	33.68	NS500454
NextSeq_CNM_170_20190620_RAbad	No Date	IDT-ILMN Nextera UD Index Set A for Nextera DNA FI	47	131,012,486	39,671	90.74	33.94	NS500454
NextSeq_CNM_171_20190624_RAbad	No Date	IDT-ILMN Nextera UD Index Set A for Nextera DNA FI	47	140,488,964	42,597	89.61	33.72	NS500454



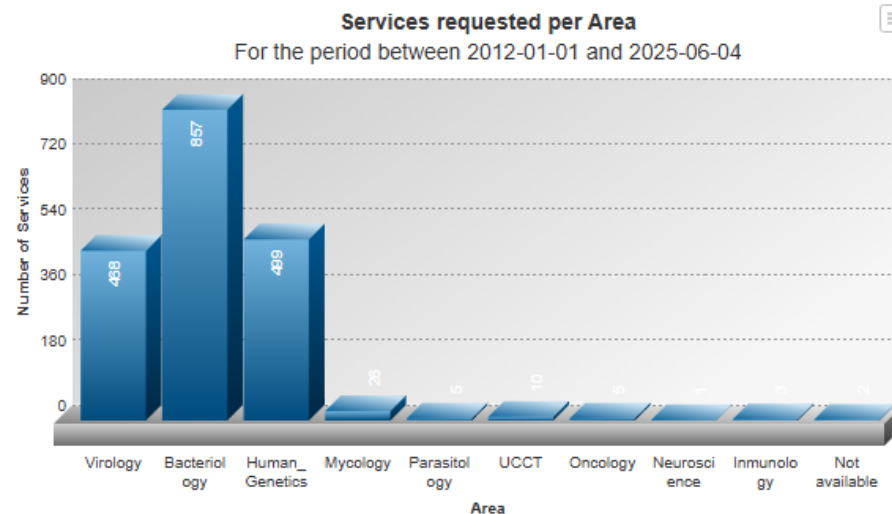
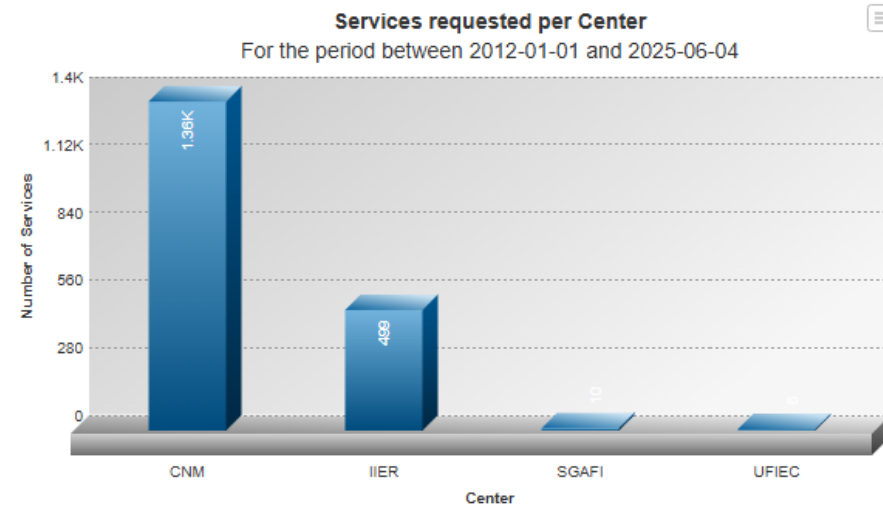
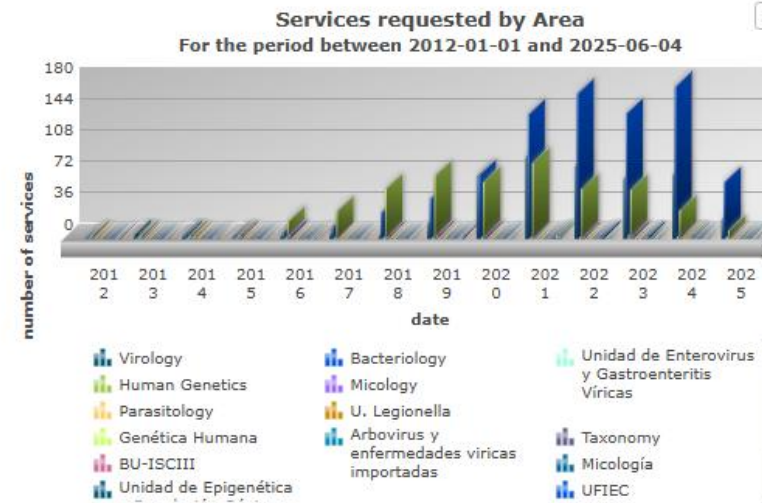
Number of services: 2012 – 2025

13
AÑOS

For the period between 2012-01-01 and 2025-06-04.



For the period between 2012-01-01 and 2025-06-04.



Training

Courses

ISCI

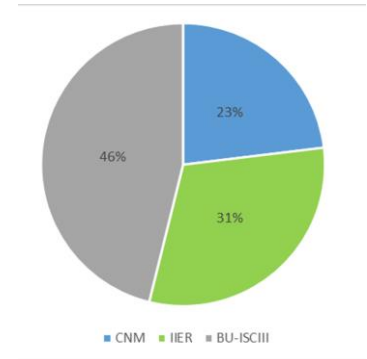
Introduction to massive sequencing data analysis, 2013-2021 (10 editions)

Secuenciación de genomas bacterianos: herramientas y aplicaciones, 2018-2022 (4 editions)

Análisis de genomas virales a través de la plataforma Galaxy, 2021 -2022 (2 editions)

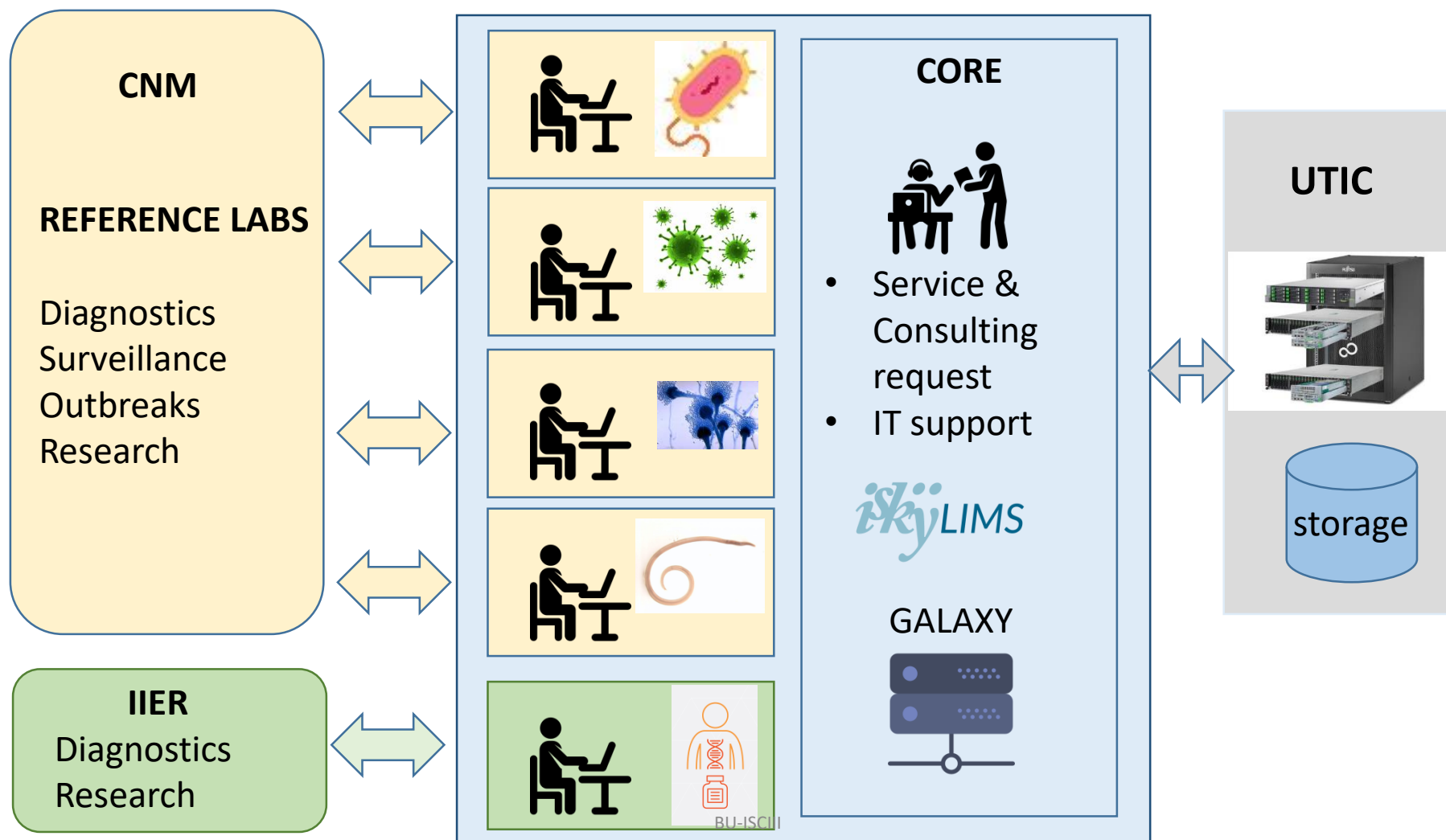
Master & Grade Students

- Bioinformática y Biología Computacional ENS-ISCI
- Bioinformática UAM
- Genética y Biología Molecular UAM
- Microbiología aplicada a la salud pública e investigación en enfermedades infecciosas, U. Alcala de Henares
- Sciences in Omics Data Analysis, Universidad de VIC, U. Central de Cataluña
- Complutense University

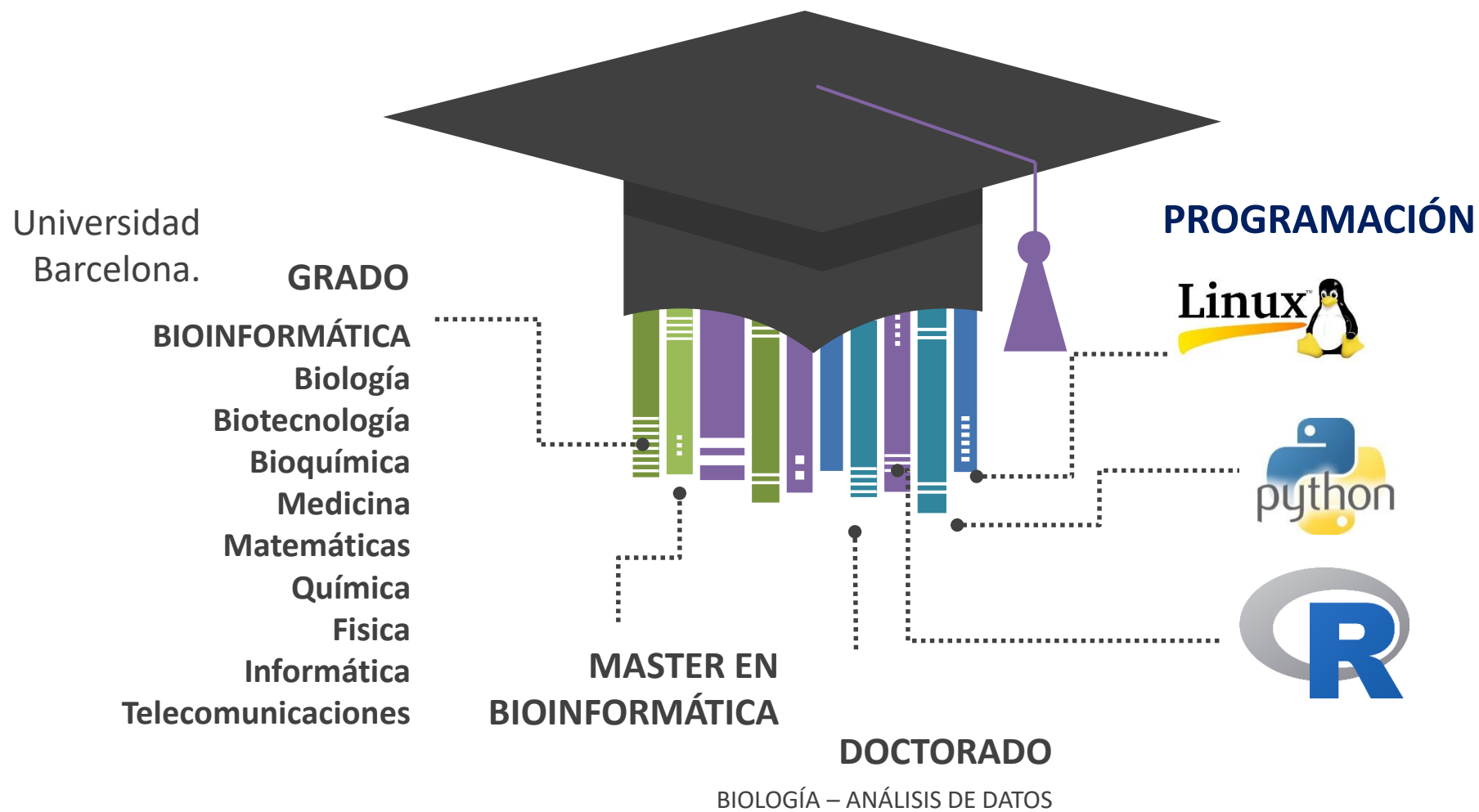


Hospitals Students

Roadmap: BU-ISCI Model



FORMACIÓN EN BIOINFORMÁTICA



¿Dónde trabaja un Bioinformático?



UNIVERSIDAD

Biociencias
Informática

CENTRO DE INVESTIGACIÓN

Biomedicina
Agricultura
Alimentación



EMPRESA

Bioinformática
Genética
Genómica

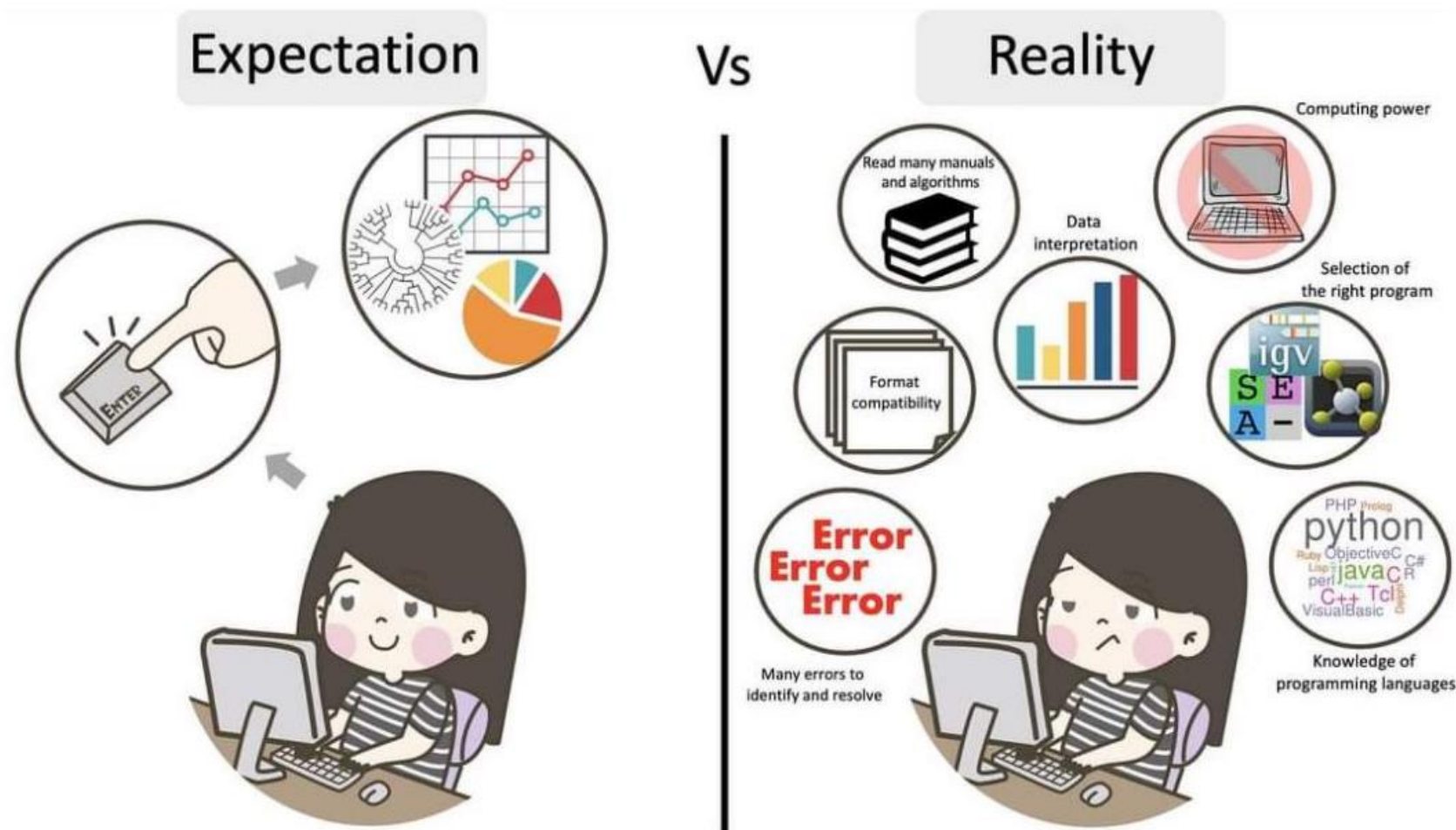
HOSPITAL

BIOINFORMÁTICO CLÍNICO

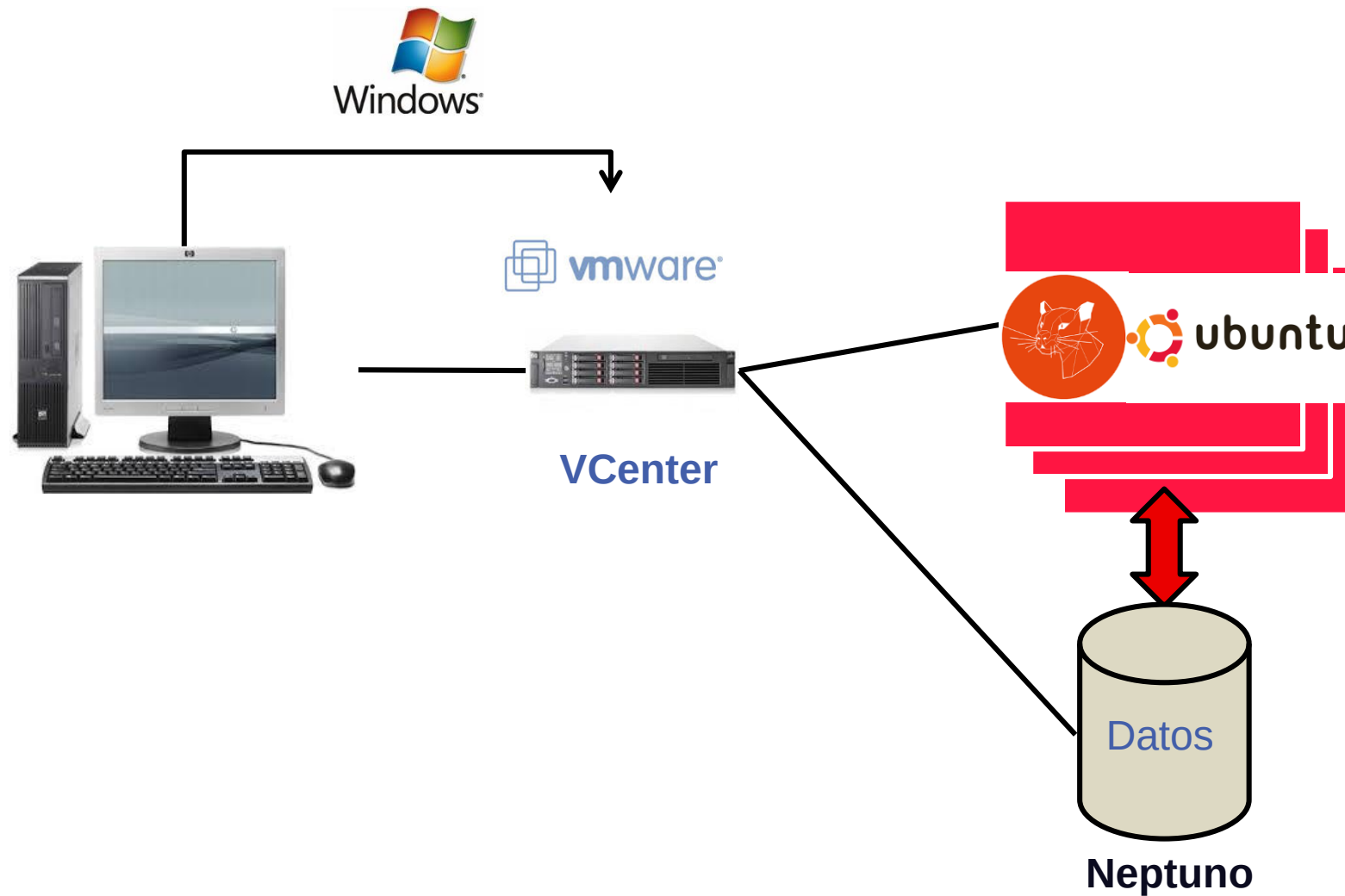
Genética
Oncología
Cardiología

The truth about bioinformatics

.image-100[



Recursos Informáticos para el curso



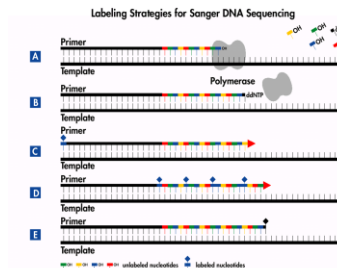
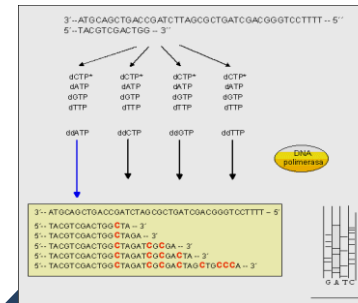
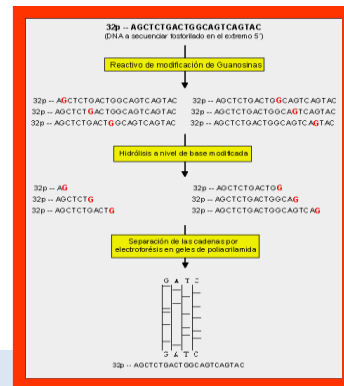
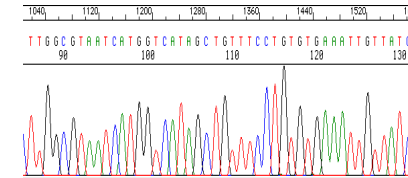
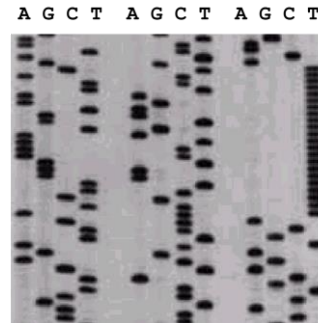
INDICE

❖ Unidad de Bioinformática
Servicios ofertados

❖ Evolución de la secuenciación

❖ Plataformas de secuenciación masiva
(NGS)

Métodos de secuenciación de DNA



SANGER

SANGER

MAXAM Y GILBERT

1ª GENERACIÓN

Evolution of DNA Revolution

A walk through the biological history: from Sanger to NGS

1953



Watson & Crick: The discovery of the molecular structure of DNA: the double helix (*Nature*, 171, 1953).

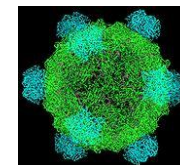
1972

Paul Berg: The first recombinant DNA molecule is built (*PNAS* 69, 1972).

Development of recombinant genetic engineering

1977

Gilbert & Maxam Sanger
Developed new techniques for rapid DNA sequencing.

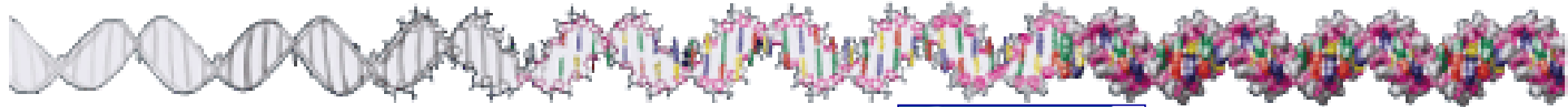


Bacteriophage Φ X174
5386nt
plus and minus method

Evolution of DNA Revolution

A walk through the biological history: from Sanger to NGS

1986

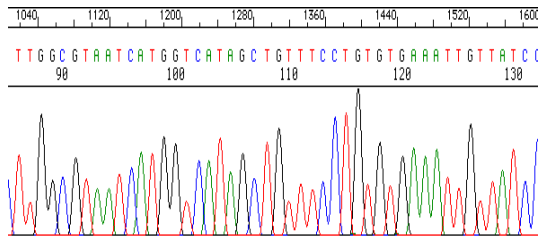


**Hood (Applied Biosystems Incorporated, ABI)
Developed the automatic sequencer
(Nature 321, 1986;
Science 254 1991)**

1990



**BLAST: Myers & Lipman
publish the first algorithm
to align DNA sequences
(JMB, 215 1990).**



Evolution of DNA Revolution

A walk through the biological history: from Sanger to NGS

1995

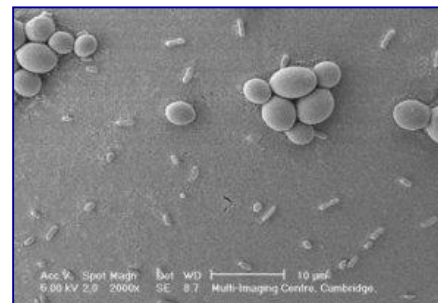
1996

2003



Venter (TIGR)
Haemophilus influenzae
Genome (1,8Mb),
Mycoplasma genitalium
(0,5Mb)
(Science 269, 1995)

**Consortium of 600
Scientist from Europe,
North America and Japan
Saccharomyces cerevisiae
genome (Science).**

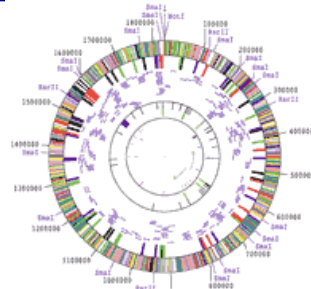


Bacterial Genomes

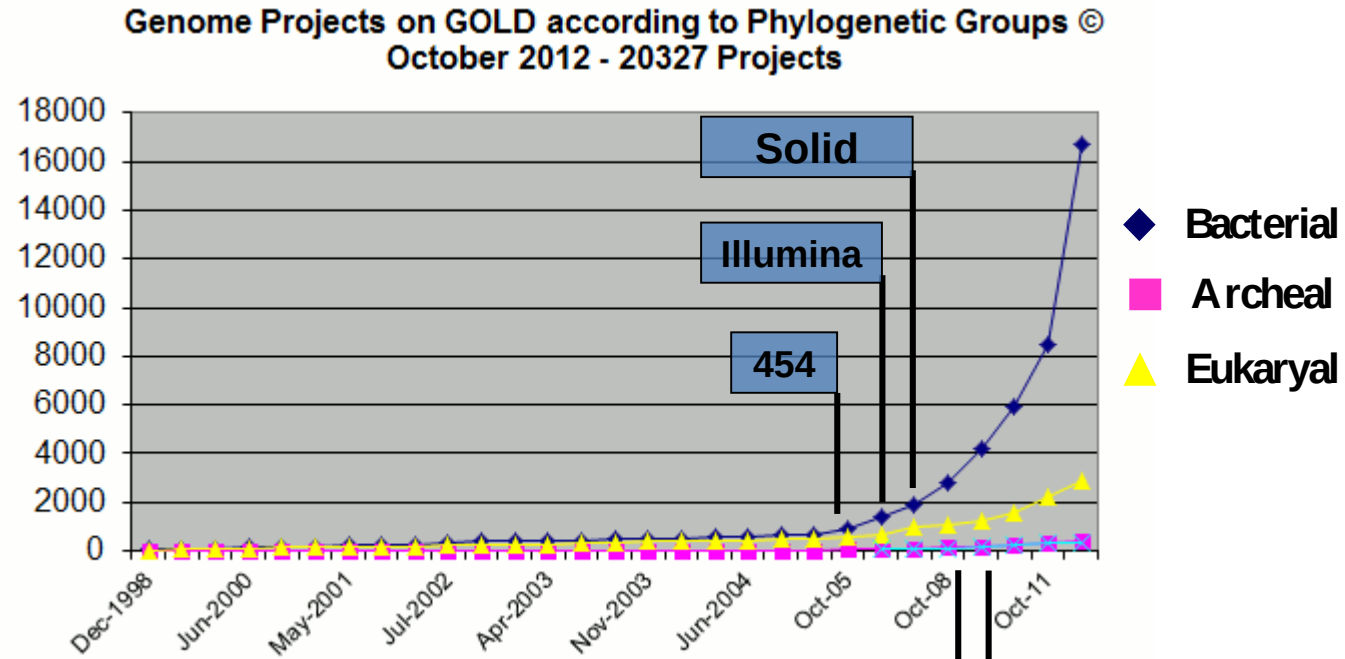
Bacillus subtilis
Escherichia coli

Eukaryotic Genomes

Caenorhabditis elegans
Arabidopsis thaliana
Drosophila melanogaster
Human genome (1986-2003)



Genomics Revolution Era



Source: <http://www.genomeonline.org>



1000 Genomes Project

NGS Platforms - Timeline

Popova and Carabetta, arxiv 03/2024

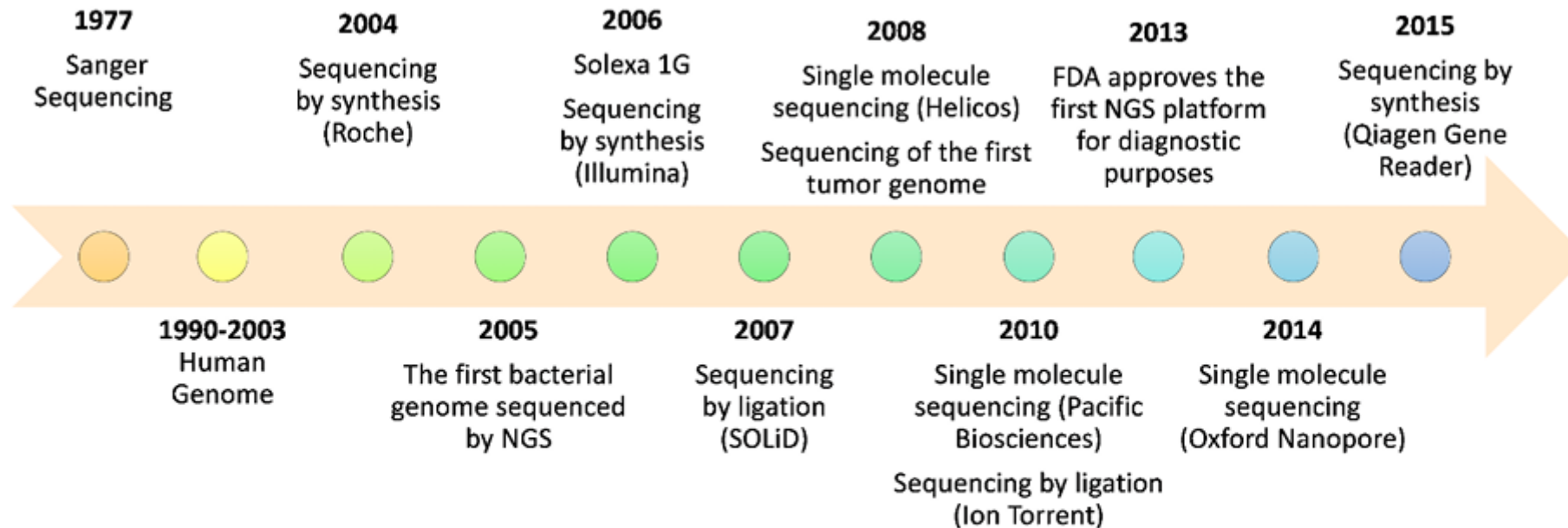
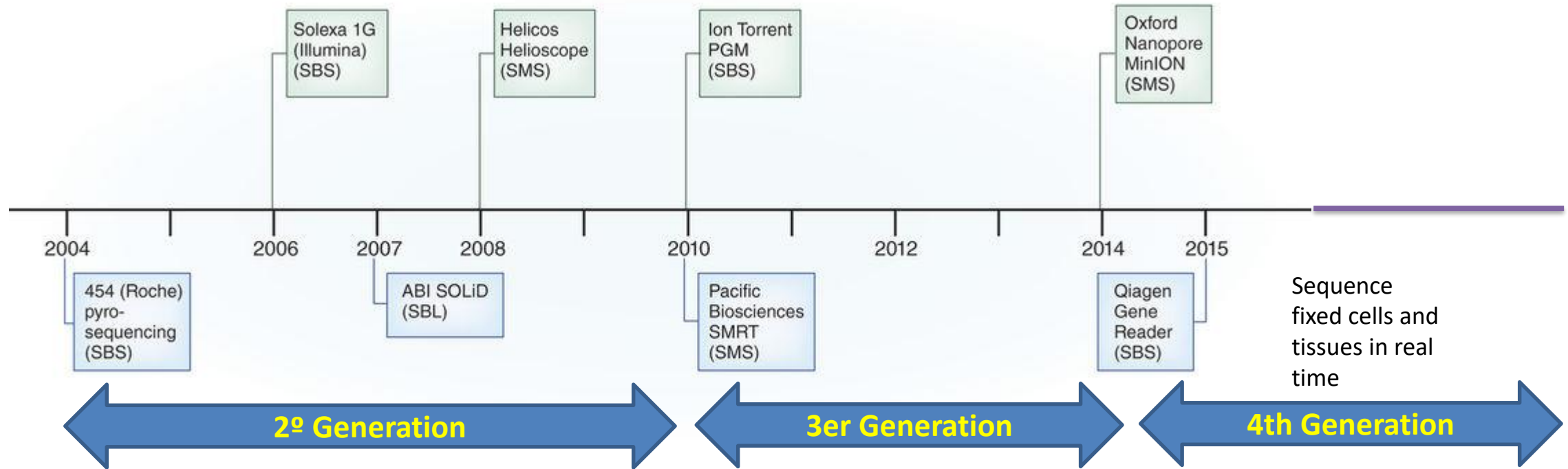


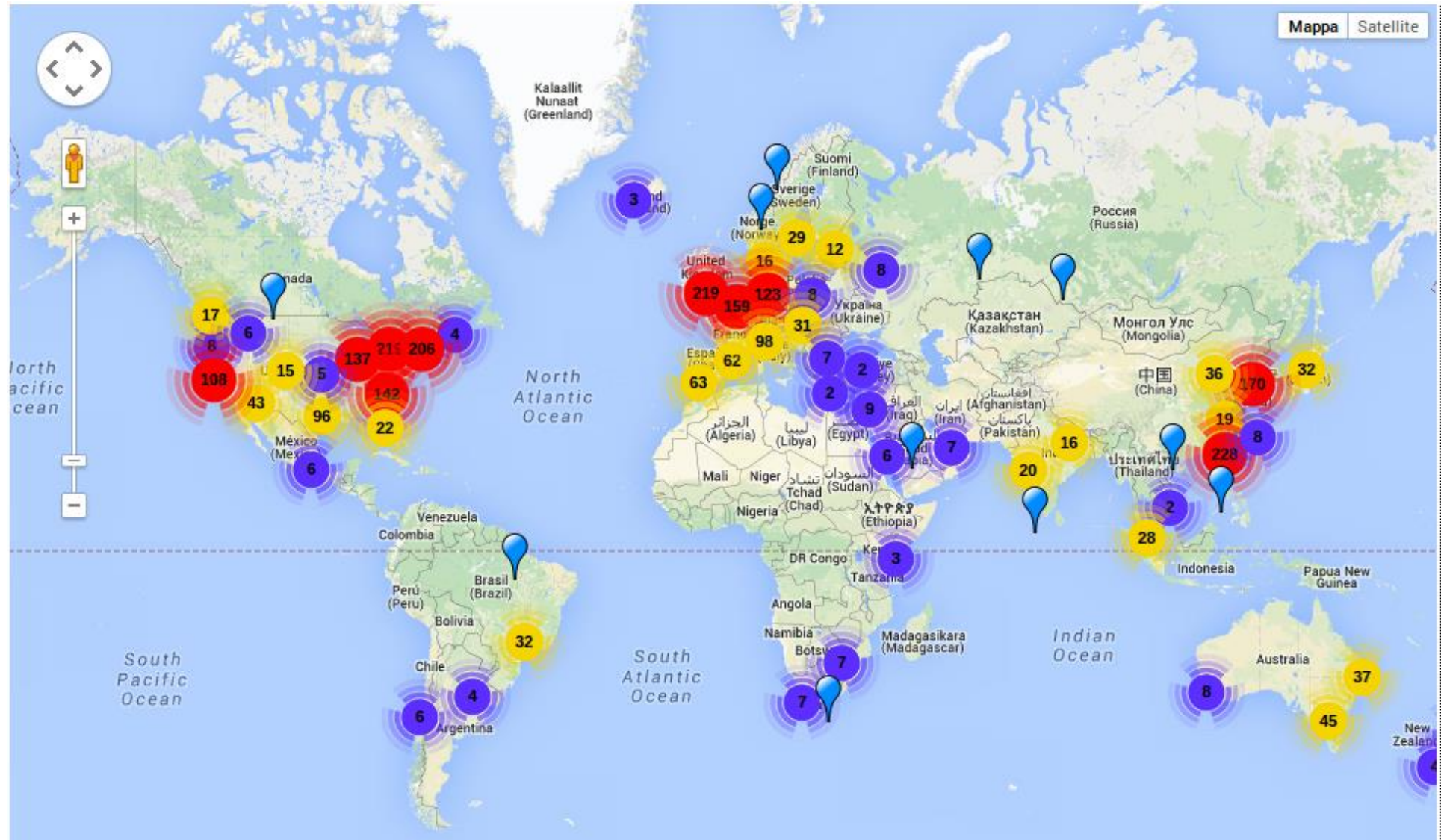
Figure 1. Timeline of commercial NGS platforms. A brief history of the instruments introduced to the market between 2004 and 2015 and developmental milestones. NGS; next-generation sequencing.

NGS Platforms - Timeline

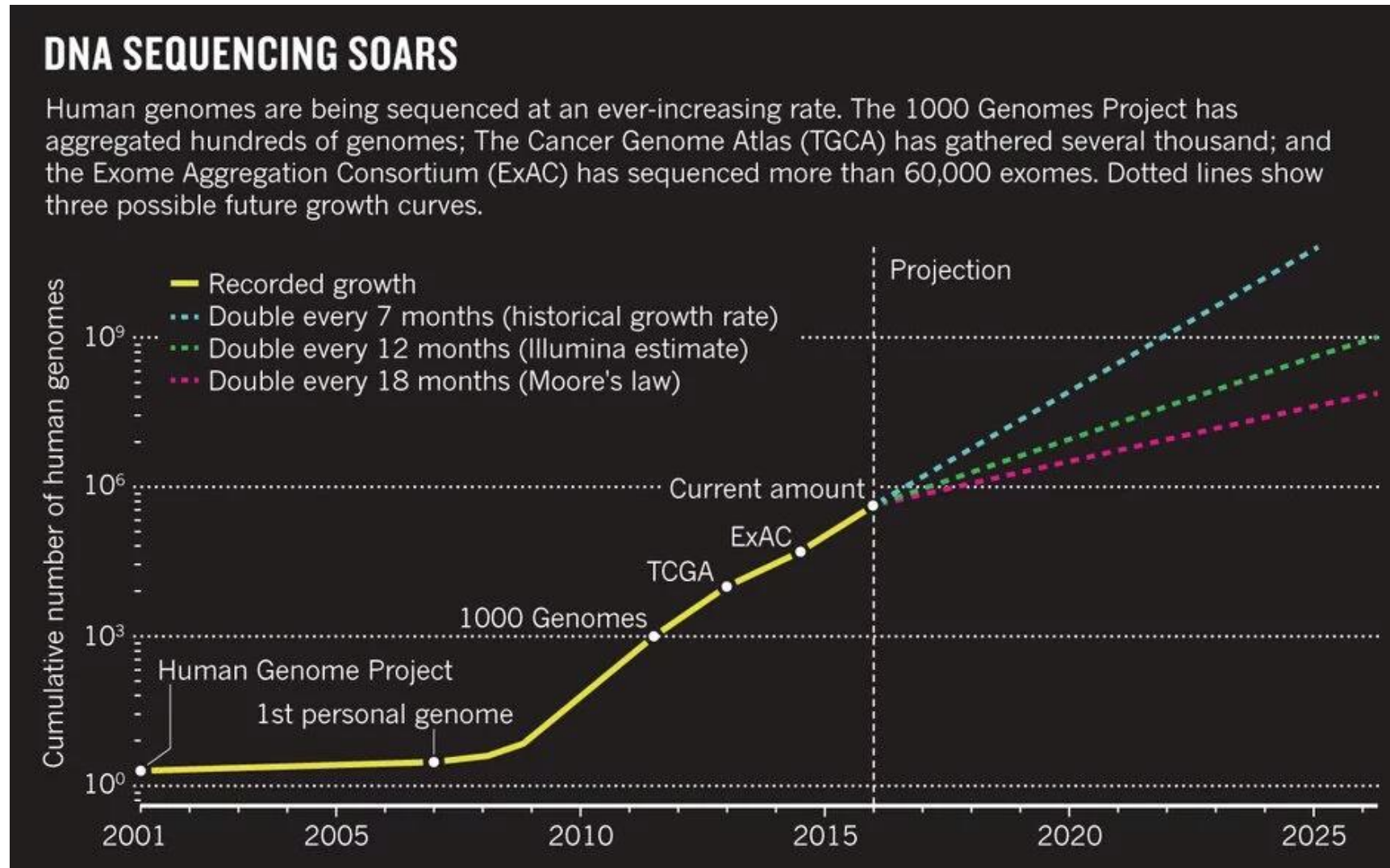


Democratización de la secuenciación.

Mapa de secuenciadores de alto rendimiento

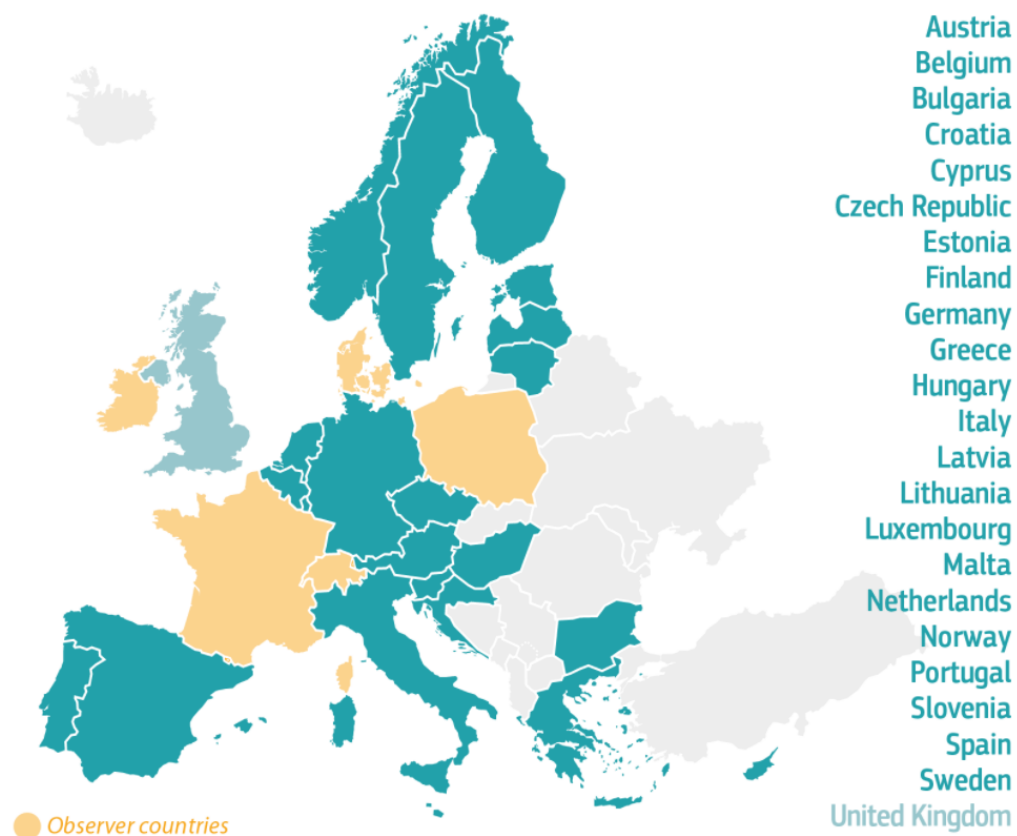


Sequencing projects



Sequencing projects

Countries that have signed the 1+MG Declaration since 2018



Declaration for delivering cross-border access to **genomic database**

- 1 million **genomes accessible** in the EU by 2022
- Linking access** to existing and future genomic database across the EU
- Providing a sufficient scale for **new clinically impactful** associations in research

The vision



Designing and testing



Scaling up and sustaining



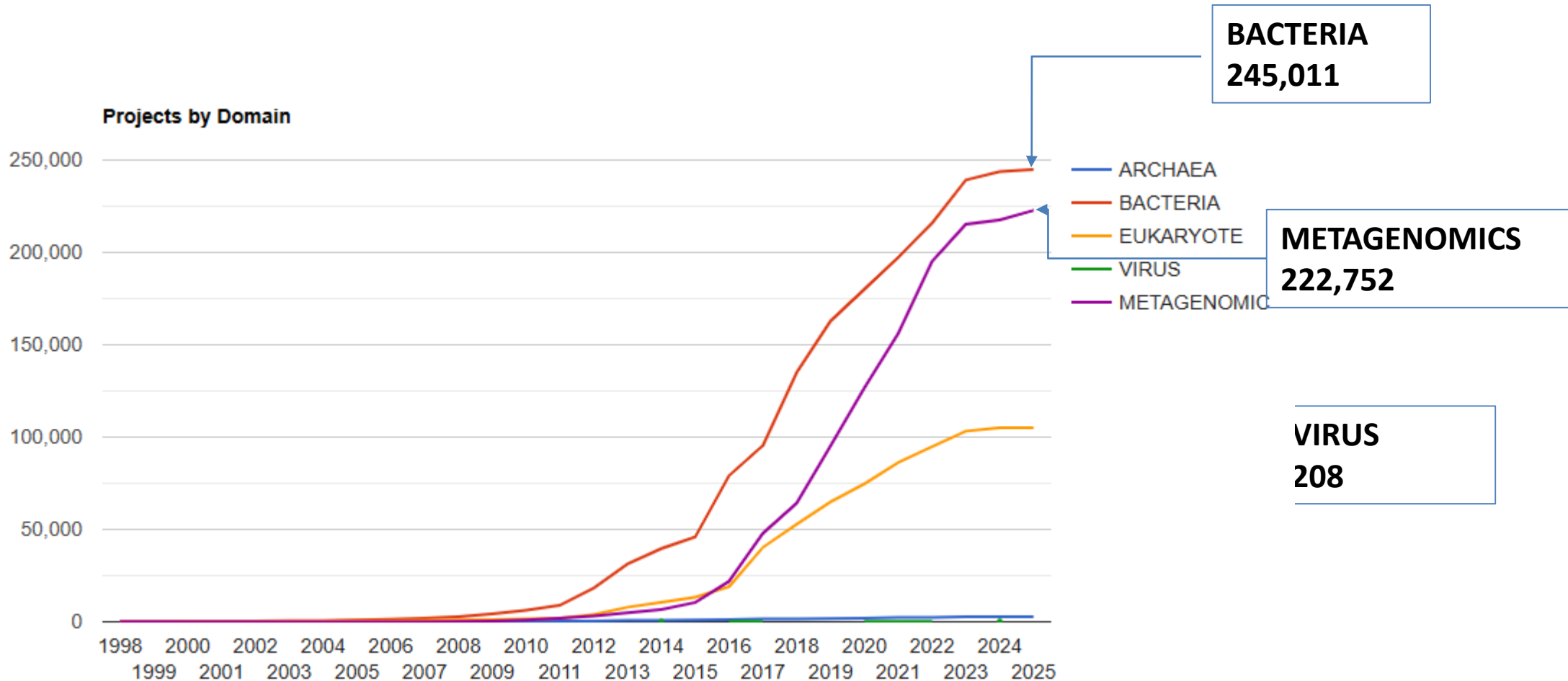
**European
Genomic Data
Infrastructure**

<https://gdi.onemilliongenomes.eu/>

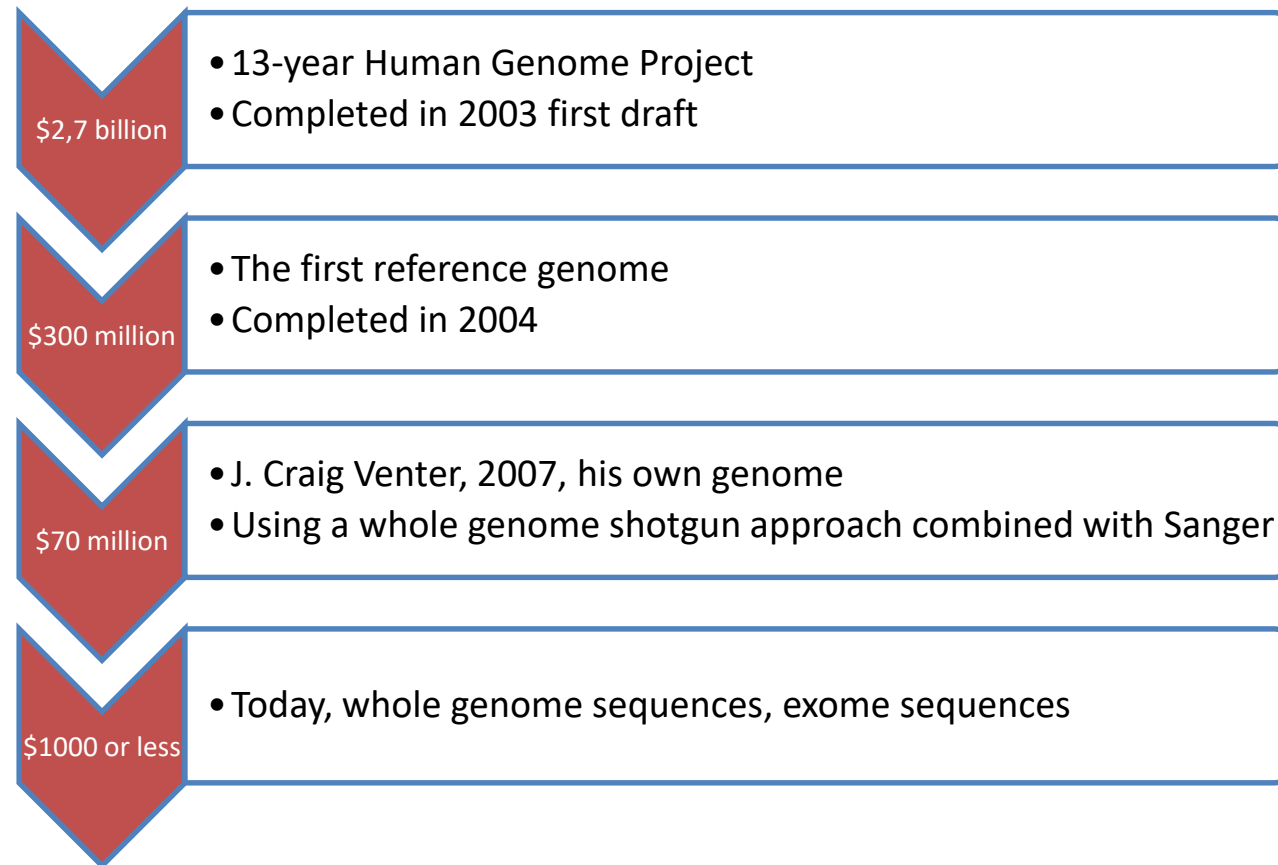
Sequencing projects

<https://gold.jgi.doe.gov/>

GOLD, Genome Online DataBase

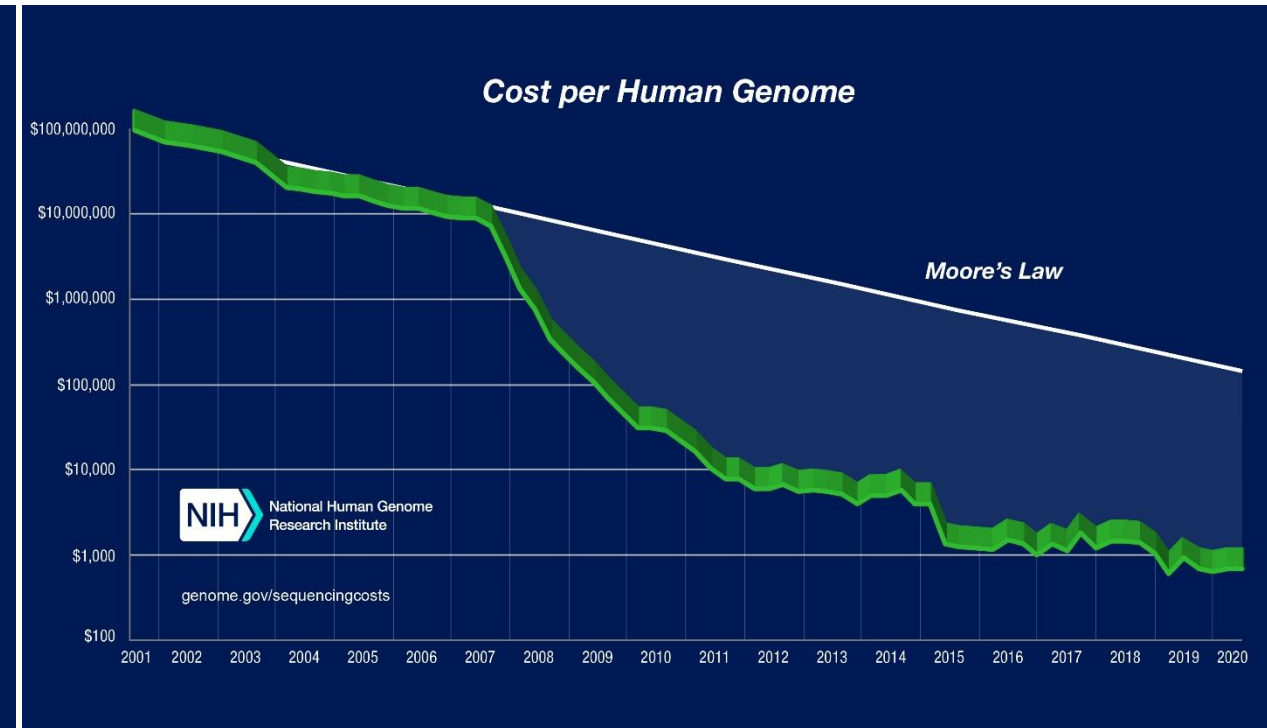
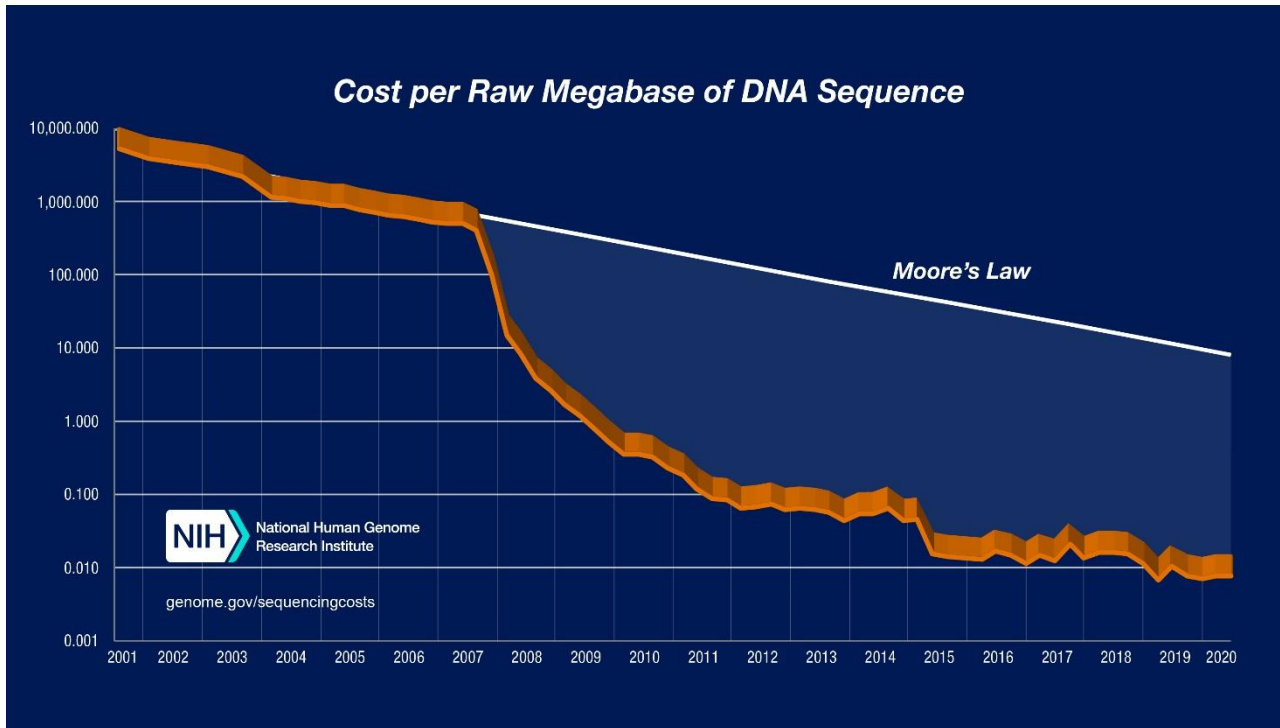


Evolución del coste de la secuenciación de un genoma humano



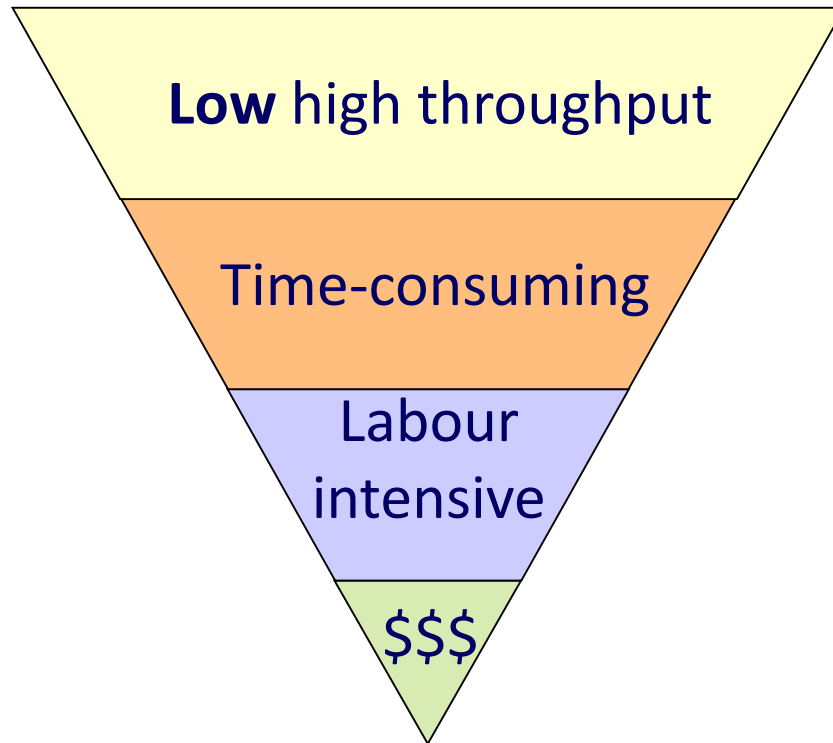
<https://www.aacc.org/publications/cln/articles/2012/april/sequencing>

Coste actual de la secuenciación

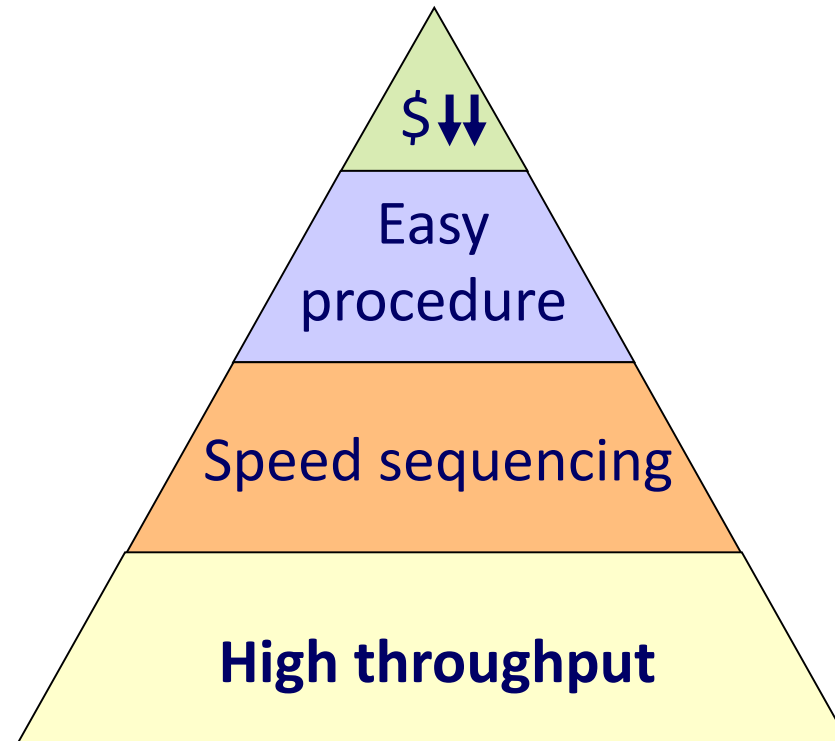


Sanger vs NGS

advantages of new technologies

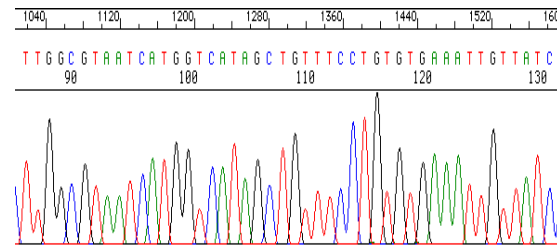


Semiautomatic **Sanger** capillary-based sequencing technology



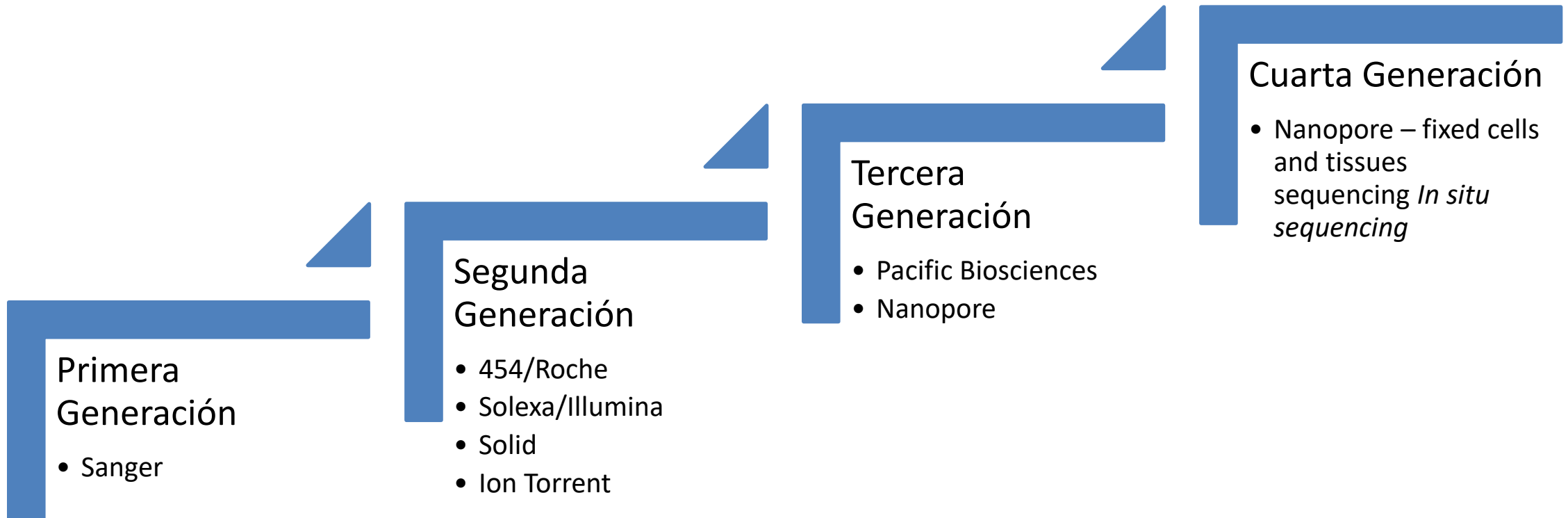
NGS
Next Generation Sequencing =
Now Generation Sequencing

BIG DATA

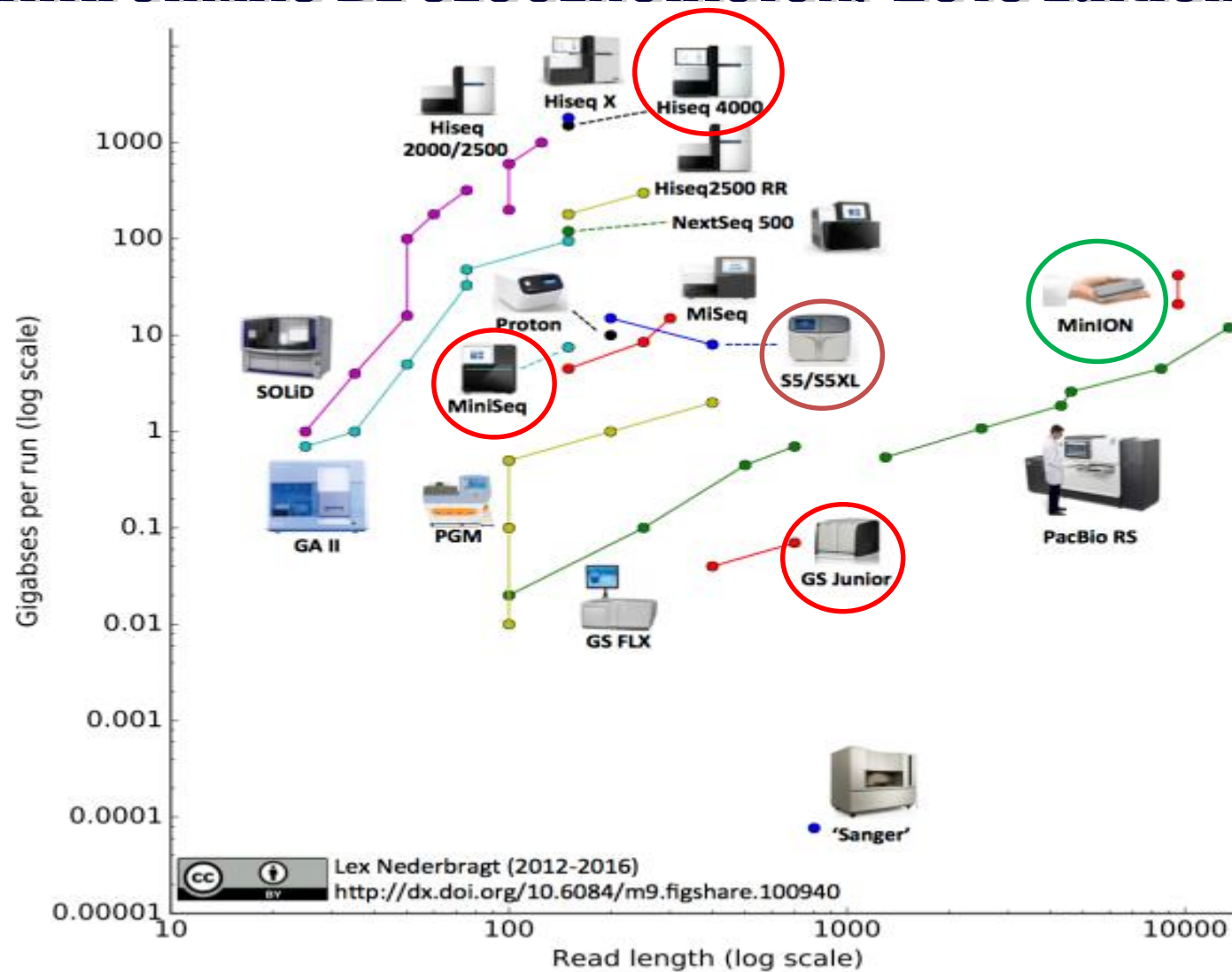


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Generaciones de Secuenciadores

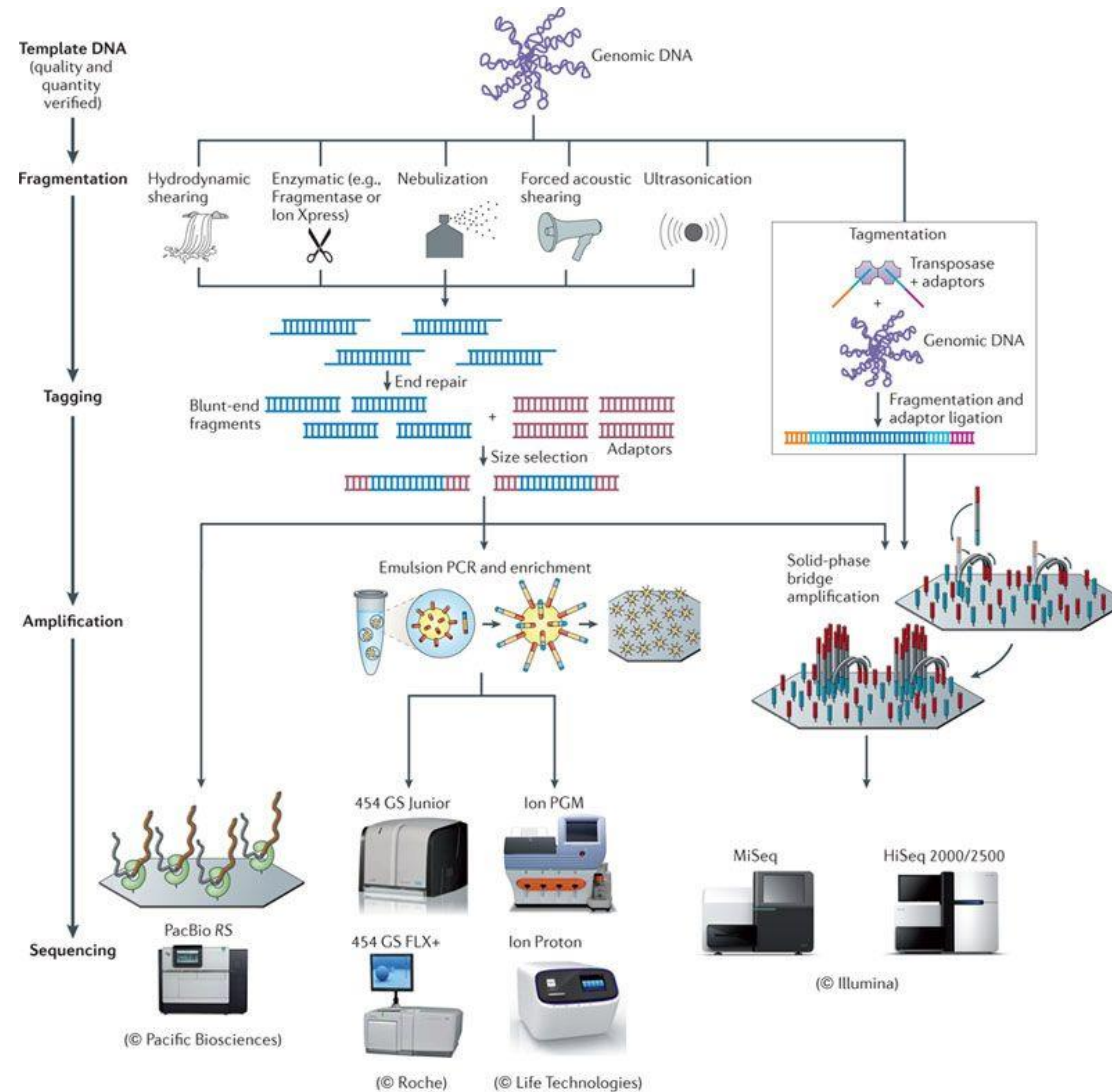


PLATAFORMAS DE SECUENCIACIÓN. 2016 Edition



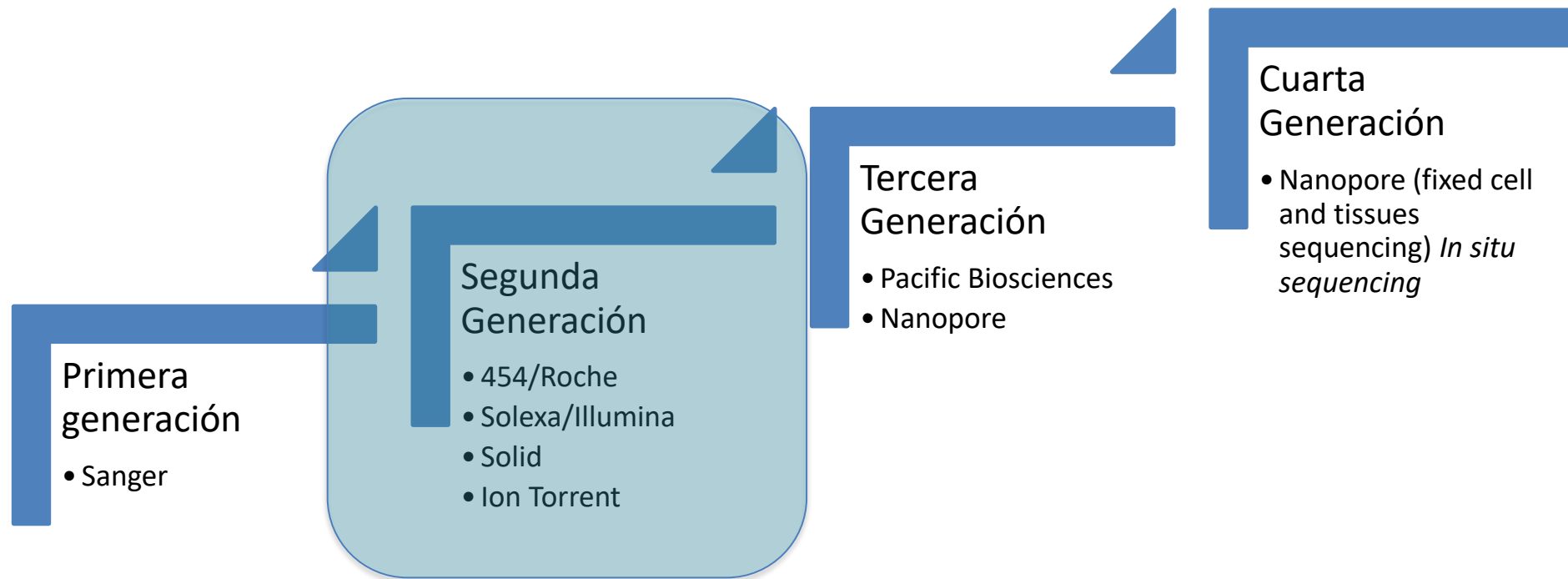
<https://flxlexblog.wordpress.com/>

High-throughput sequencing platforms



Nature Reviews | Microbiology Loman et al, 2012

Secuenciadores



High-throughput sequencing platforms 2nd GS



GS-FLX System



Genome Analyzer IIx

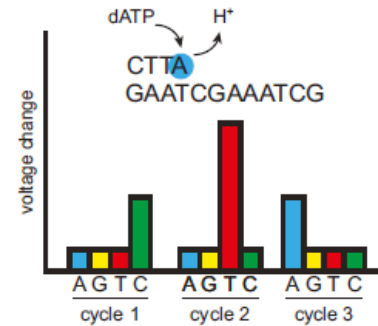
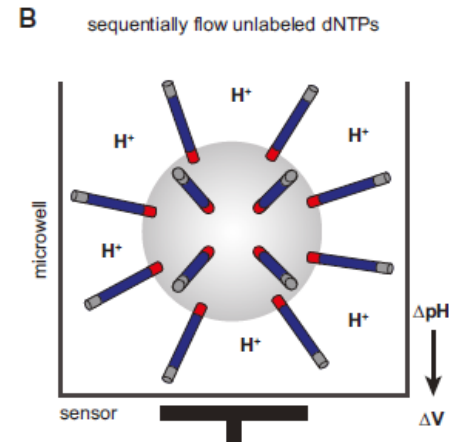
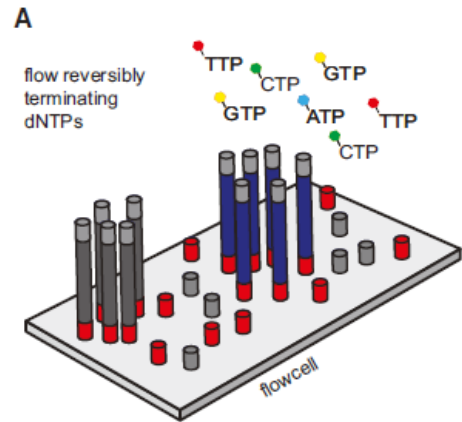


SOLID 3 Plus/4

Sequencing Chemistry	Sequencing by synthesis, pyrosequencing	Sequencing by synthesis with reversible terminators	Sequencing by ligation
Amplification approach	Emulsion PCR	Cluster amplification	Emulsion PCR
DNA support	25-35 µm bead	Flow cell surface	Bead (Solid 3 Plus/4) Flow cell surface (GA5500w)

Mardis et al., Trends in Genetics 2008, 24:3

The Second-generation Sequencing Technologies



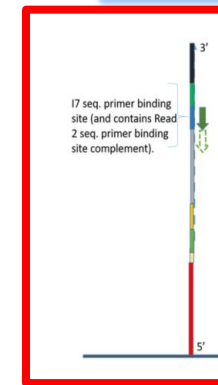
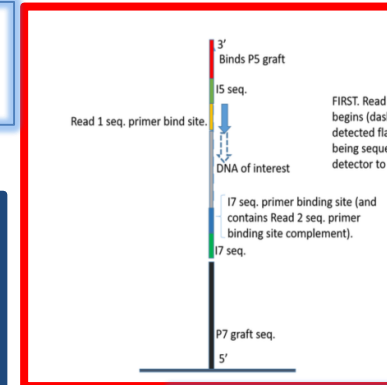
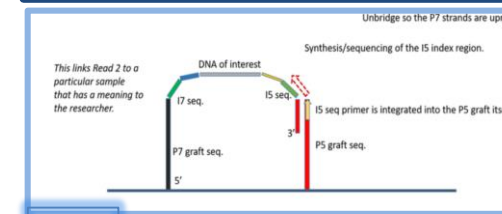
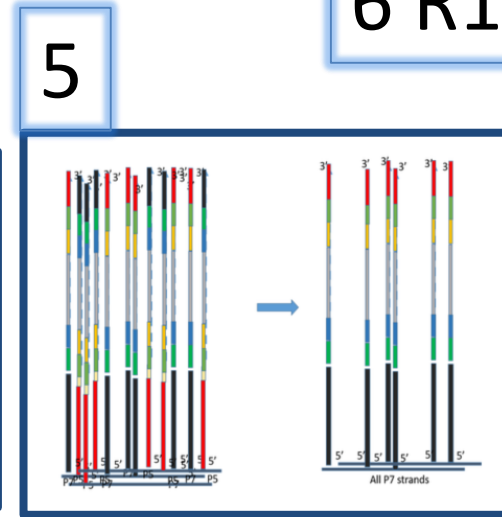
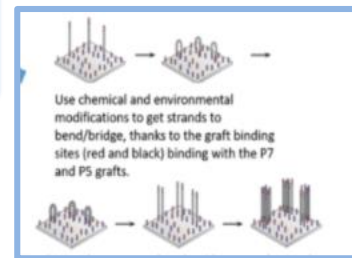
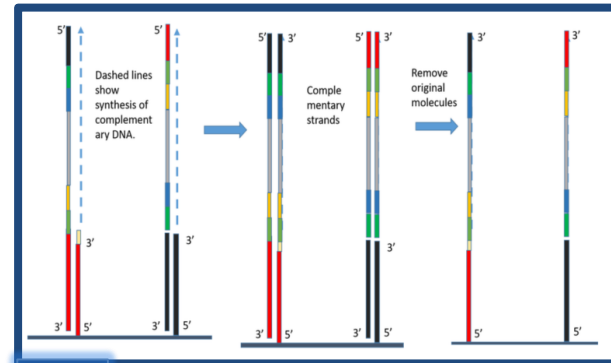
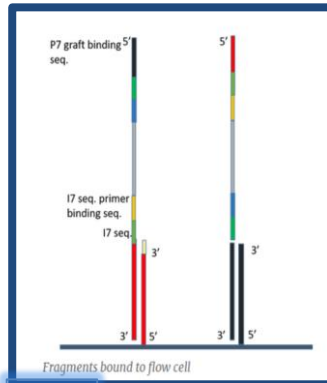
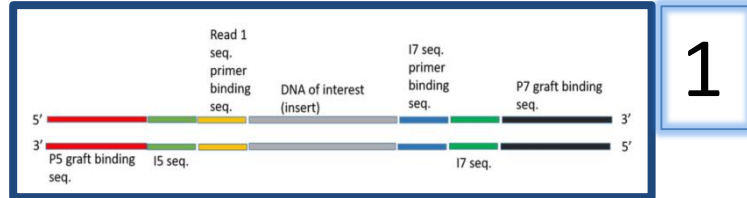
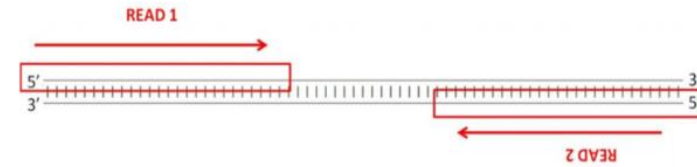
Clonal Amplification-Based Sequencing Platforms

(A) Illumina's four-color reversible termination sequencing method.

(B) Ion Torrent's semiconductor sequencing method.

Reuter et al., Mol Cell 2015

Illumina sequencing



<https://ksbioinformatics.wordpress.com/2017/02/13/illumina-sequencing-for-dummies-samples-are-sequenced/>

High-throughput sequencing platforms 2nd GS



GS-FLX System



**Genome
Analyzer IIx**



**HiSeq
2000/1000**



**HiSeq
2500/1500**



**NextSeq
500**



SOLID 3 Plus/4



**Genetic Analyzers
5500W Series**

Benchtop High-throughput sequencing platforms 2nd GS



Roche 454 GS Junior



MiSeq



iSeq 100



MiniSeq



Ion Proton™ System



Ion PGM™ System

Methods/applications	iSeq 100	MiniSeq	MiSeq series	NextSeq 550 series	Next Seq 1000 & 2000
Ideal for	Every size lab	TG sequencing	Long read applications	Exome and transcriptome sequencing	miRNA and sRNA analysis
Major applications	sWGS (microbes) and TGS	iSeq 100+TG EP and 16S MS	iSeq 100+16S MGS	iSeq 100+TCS	sWGS (microbes), ES, SC profiling, TS, miRNA, and sRNA analysis
Max. data quality	>85% > Q30	>85% > Q30	>90% > Q30	>80% > Q30	>90% > Q30
Run time	9.5–19 h	4–24 hours	4–55 hours	12–30 hours	11–48 hours
Maximum output	1.2 Gb	7.5 Gb	15 Gb	120 Gb	330 Gb*
Maximum reads per run	4 million	25 million	25 million	400 million	1.1 billion
Maximum read length	2 × 150 bp	2 × 150 bp	2 × 300 bp	2 × 150 bp	2 × 150 bp

Illumina Production scale Sequencers

Pervez et al., BioMed Research International 2022

Methods/applications	NextSeq 550	NextSeq 550Dx	NextSeq 1000 & 2000	NovaSeq 6000
Ideal for	Research	Research+in vitro diagnostic	Targeted sequencing	Long read applications
Major applications	sWGS (microbes), TGS, and TCS	NextSeq 550+clinical NGS applications	NextSeq 550 series+SCP	NextSeq 550 series+NextSeq 1000 & 2000+IWGS
Max. data quality	>80% > Q30	>75% > Q30	>90% > Q30	>90% > Q30
Run time	12-30 hours	35 hours	11-48 hours	13-44 hours
Maximum output	120 Gb	90 Gb	360 Gb	6000 Gb
Maximum reads per run	400 million	300 million	1.2 billion	20 billion
Maximum read length	2 × 150 bp	2 × 150 bp	2 × 150 bp	2 × 250 bp

Illumina Benchtop Sequencers



iSeq 100



MiniSeq



MiSeq Series



NextSeq 550 Series



NextSeq 1000 & 2000

Popular Applications & Methods	Key Application	Key Application	Key Application	Key Application	Key Application
Large Whole-Genome Sequencing (human, plant, animal)					
Small Whole-Genome Sequencing (microbe, virus)					
Exome & Large Panel Sequencing (enrichment-based)					
Targeted Gene Sequencing (amplicon-based, gene panel)					
Single-Cell Profiling (scRNA-Seq, scDNA-Seq, oligo tagging assays)					
Transcriptome Sequencing (total RNA-Seq, mRNA-Seq, gene expression profiling)					
Targeted Gene Expression Profiling					
miRNA & Small RNA Analysis					
DNA-Protein Interaction Analysis (ChIP-Seq)					
Methylation Sequencing					
16S Metagenomic Sequencing					
Metagenomic Profiling (shotgun metagenomics, metatranscriptomics)					
Cell-Free Sequencing & Liquid Biopsy Analysis					

Benchtop Sequencer Sheds Light on Ebola Outbreak

Local scientists use the iSeq 100 Sequencing System to analyze transmission patterns and trace the origin of an Ebola outbreak in the Democratic Republic of the Congo.

[Read Article >](#)

Run Time	9.5–19 hrs	4–24 hours	4–55 hours	12–30 hours	11–48 hours
Maximum Output	1.2 Gb	7.5 Gb	15 Gb	120 Gb	330 Gb*
Maximum Reads Per Run	4 million	25 million	25 million*	400 million	1.1 billion*
Maximum Read Length	2 × 150 bp	2 × 150 bp	2 × 300 bp	2 × 150 bp	2 × 150 bp

<https://emea.illumina.com/systems/sequencing-platforms.html>

Illumina Production-Scale Sequencers



NextSeq 550 Series



NextSeq 1000 & 2000



NovaSeq 6000

Popular Applications & Methods	Key Application	Key Application	Key Application
Large Whole-Genome Sequencing (human, plant, animal)			
Small Whole-Genome Sequencing (microbe, virus)			
Exome & Large Panel Sequencing (enrichment-based)			
Targeted Gene Sequencing (amplicon-based, gene panel)			
Single-Cell Profiling (scRNA-Seq, scDNA-Seq, oligo tagging assays)			
Transcriptome Sequencing (total RNA-Seq, mRNA-Seq, gene expression profiling)			
Chromatin Analysis (ATAC-Seq, ChIP-Seq)			
Methylation Sequencing			
Metagenomic Profiling (shotgun metagenomics, metatranscriptomics)			
Cell-Free Sequencing & Liquid Biopsy Analysis			

Optimized NGS Sample Tracking and Workflows

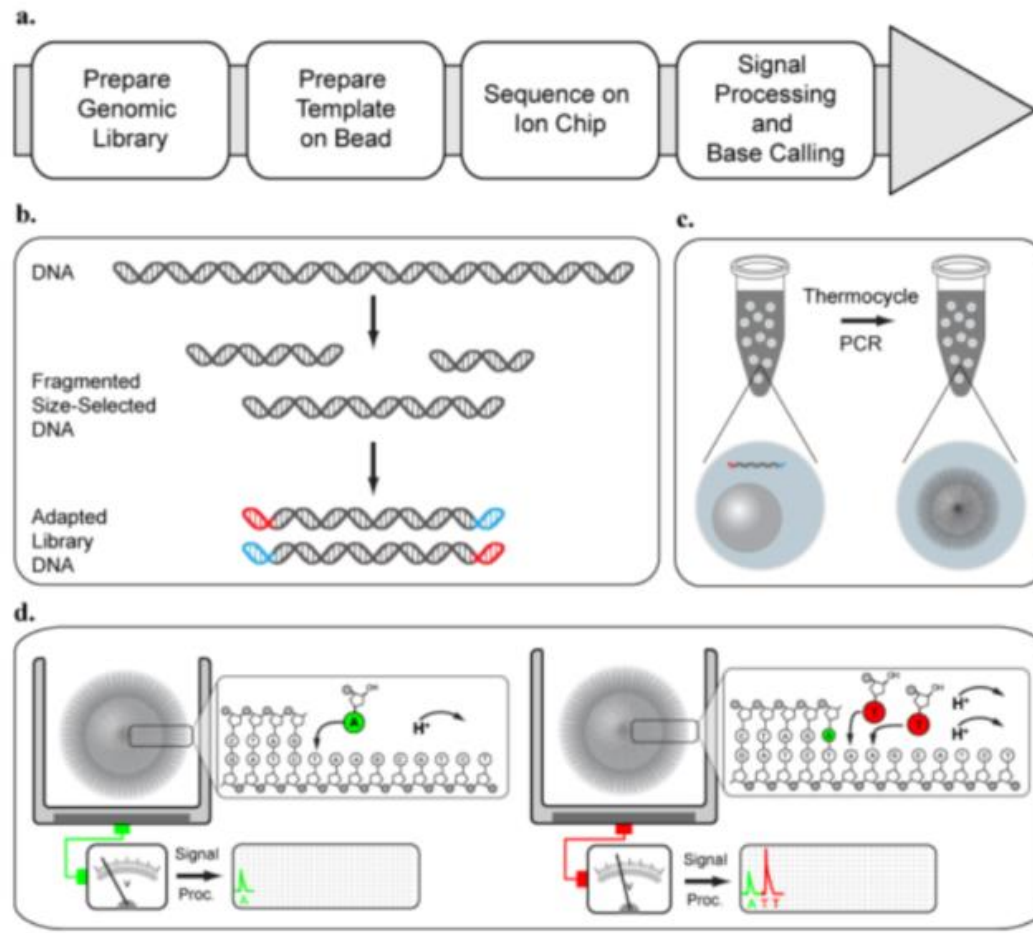
See how a Laboratory Information Management System (LIMS) enabled this large genomics lab to standardize lab procedures and cope with increasing sample volumes from diverse clients.

[Read Case Study](#)

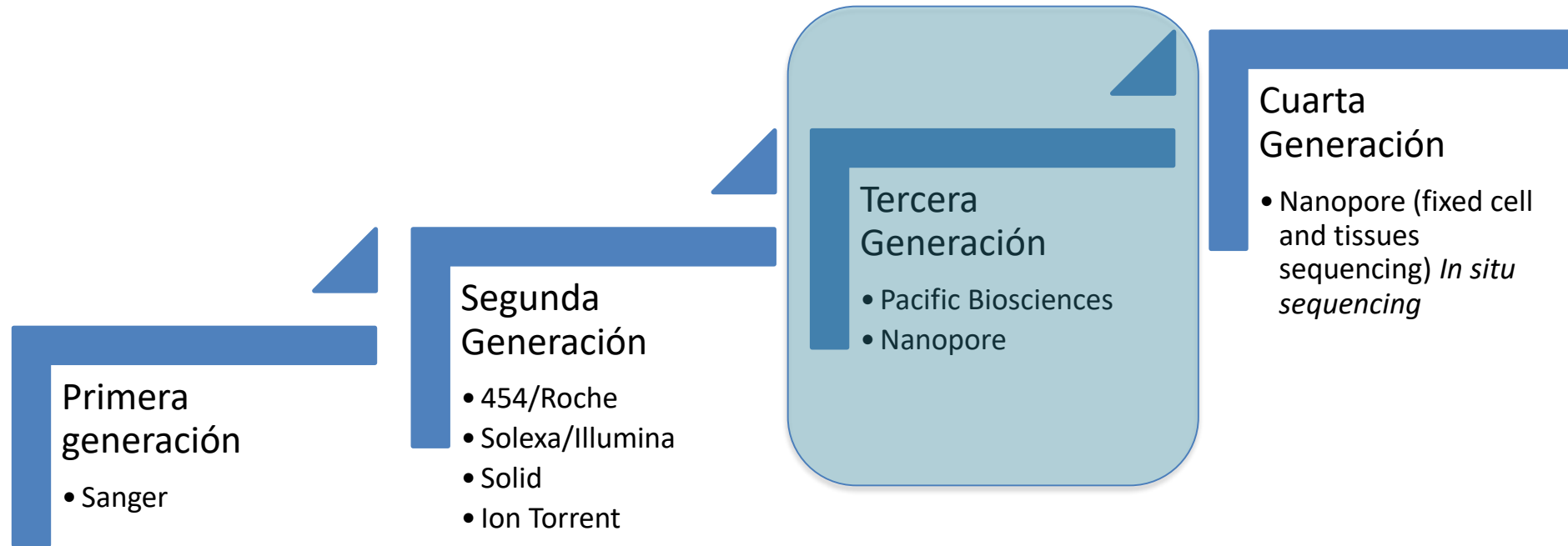
Run Time	12-30 hours	11-48 hours	-13 - 38 hours (dual SP flow cells) -13-25 hours (dual S1 flow cells) -16-36 hours (dual S2 flow cells) -44 hours (dual S4 flow cells)
Maximum Output	120 Gb	330 Gb*	6000 Gb
Maximum Reads Per Run	400 million	1.1 billion*	20 billion
Maximum Read Length	2 x 150 bp	2 x 150 bp	2 x 250**

Ion Torrent PGM

Personal Genome Machine



Secuenciadores



High-throughput single molecule sequencing platforms

3rd GS



The Helicos logo features a stylized 'H' made of orange and red ribbons. Below it, the text 'Helicos' is in a large, bold, sans-serif font, and 'BioSciences Corporation' is in a smaller font below that. Below the logo is a photograph of the Helicos Genetic Analysis System, a large, black, industrial-looking machine with a red stripe on the front.

Helicos Genetic Analysis System



The Pacific Biosciences logo consists of a red circle with a white 'pb' inside, followed by the text 'PACIFIC BIOSCIENCES®' in a bold, sans-serif font. Below the logo is a photograph of the PacBio RS II sequencing system, a large, white, industrial-looking machine with a screen on the front.

PacBio RS II is a Single Molecule, Real-Time (SMRT®) DNA Sequencing System



The Oxford Nanopore logo is a teal rectangle with the text 'Oxford Nanopore' in white. Below the logo is a photograph of the MinION device, a small, white, handheld sequencing device. Below the device is a photograph of a person using a pipette to add liquid to a yellow microfluidic chip, which is connected to the MinION device.

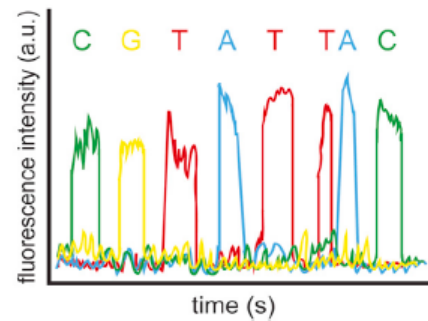
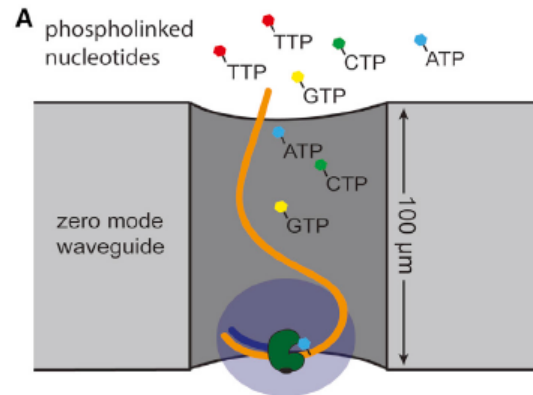
MinION™ Device

3ª GENERACIÓN: LECTURAS MAS LARGAS Y MOLECULA ÚNICA

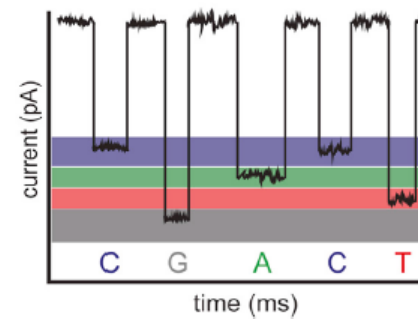
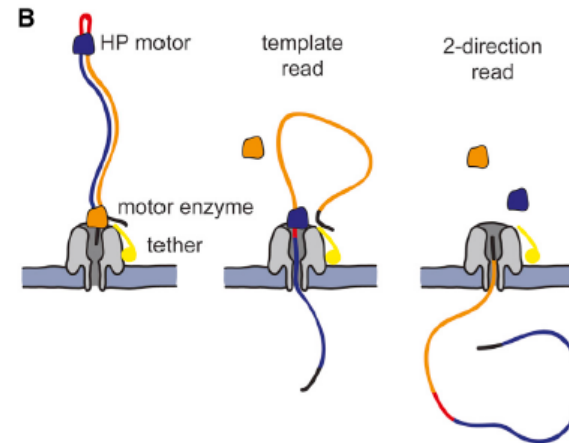
- PacBio, PACIFIC BIOSCIENCE
- MinION, GridION, OXFORD NANOPORE

The Third-generation Sequencing Technologies

Single Molecule Sequencing Platforms



Pacific Bioscience's SMRT sequencing



Oxford Nanopore's sequencing strategy

Reuter et al., Mol Cell 2015

PacBio sequencing and its applications

Rhoads & Au, Gen Prot Bioinf 2015



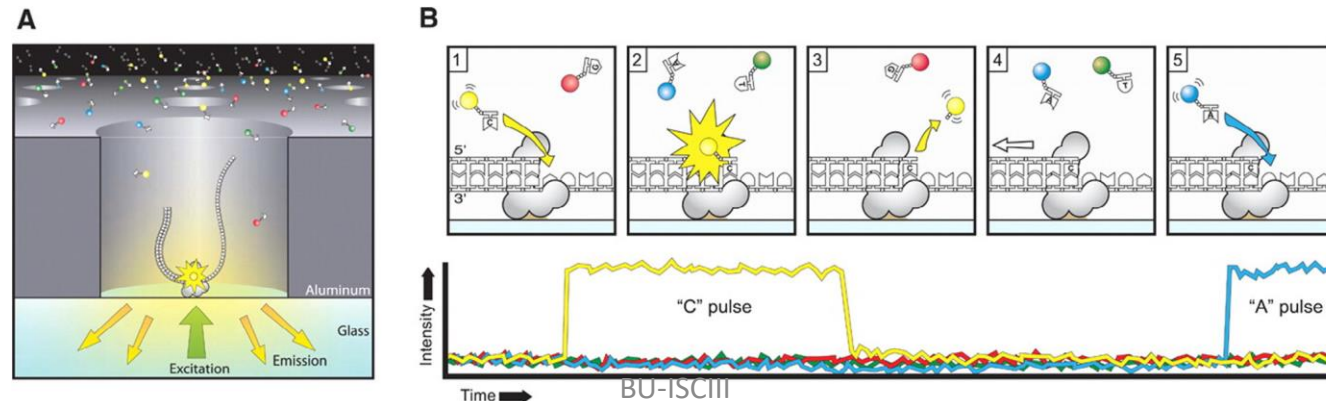
SMRTbell template: is a closed, single-stranded circular DNA that is created by ligating hairpin adaptors to both ends of a target dsDNA

Sequencing by light pulses: The replication processes in all ZMWs of a SMRTcell are recorder by a movie of light pulses, and the pulses corresponding to each ZMW can be interpreted to be a sequence of bases (**continuous long read, CLR**).

Both strands can be sequenced multiple times (passes) in a single CLR. CLR can be split to multiple reads (subreads) and CCS is the consensus sequence of multiple subreads



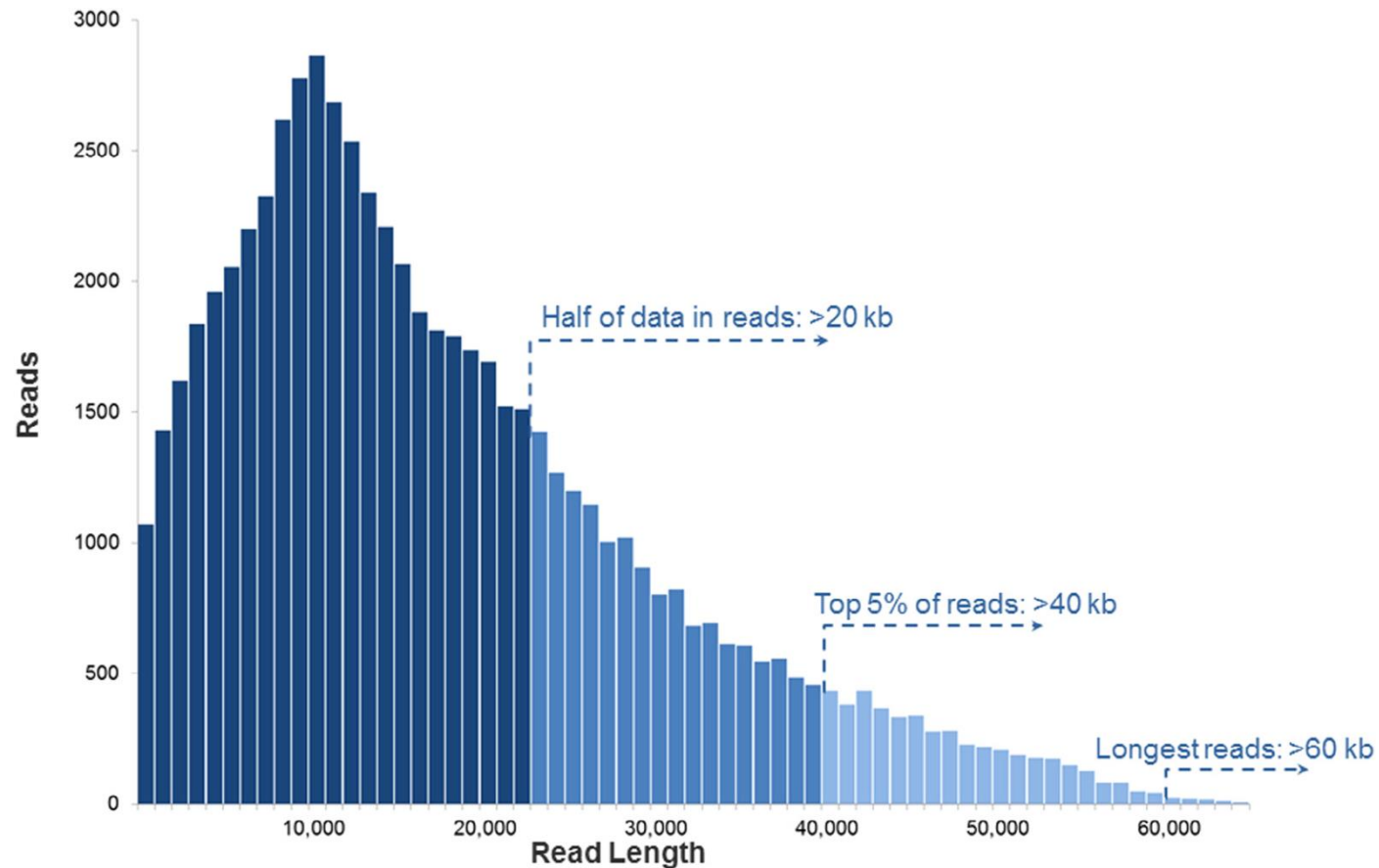
A single SMRT cell: this contains 150000 ZMWs (zero-mode waveguide). A SMRTbell diffuses into a ZMW. Approx 35000 -75000 ZMWs produce a read in a run lasting 0,5-4h resulting in 0,5-1Gb.



PacBio sequencing and its applications

Rhoads & Au, Gen Prot Bioinf 2015

PacBio RS II read length distribution using P6-C4 chemistry. Data are based on a 20kb size-selected E. coli library using a 4-h movie. A SMRTcell produces 0.5-1 billion bases.



PacBio sequencing and its applications

Rhoads & Au, Gen Prot Bioinf 2015

Table 2 *De novo* genome assemblies using hybrid sequencing or PacBio sequencing alone

Species	Method	Tools	SMRT cells	Coverage	Contigs	Achievements	Ref.
<i>Clostridium autoethanogenum</i>	PacBio	HGAP	2	179×	1	21 fewer contigs than using SGS; no collapsed repeat regions (≥ 4 using SGS)	[7]
<i>Potentilla micrantha</i> (chloroplast)	PacBio	HGAP, Celera, minimus2, SeqMan	26	320×	1	6 fewer contigs than with Illumina; 100% coverage (Illumina: 90.59%); resolved 187 ambiguous nucleotides in Illumina assembly; unambiguously assigned small differences in two > 25 kb inverted repeats	[33]
<i>Escherichia coli</i>	PacBio	PBcR, MHAP, Celera, Quiver	1	85×	1	4.6 CPU hours for genome assembly (10× improvement over BLASR)	[31]
<i>Saccharomyces cerevisiae</i>	PacBio	PBcR, MHAP, Celera	12	117×	21	27 CPU hours for genome assembly (8× improvement over BLASR); improved current reference of telomeres	[31]
<i>Arabidopsis thaliana</i>	PacBio	PBcR, MHAP, Celera	46	144×	38	1896 CPU hours for genome assembly	[31]
<i>Drosophila melanogaster</i>	PacBio	PBcR, MHAP, Celera, Quiver	42	121×	132	1060 CPU hours for genome assembly (593× improvement over BLASR); improved current reference of telomeres	[31]
<i>Homo sapiens</i> (CHM1htert)	PacBio	PBcR, MHAP, Celera	275	54×	3434	262,240 CPU hours for genome assembly; potentially closed 51 gaps in GRCh38; assembled MHC in 2 contigs (60 contigs with Illumina); reconstructed repetitive heterochromatic sequences in telomeres	[31]
<i>Homo sapiens</i> (CHM1tert)	PacBio	BLASR, Celera, Quiver	243	41×	N/A (local assembly)	Closed 50 gaps and extended into 40 additional gaps in GRCh37; added over 1 Mb of novel sequence to the genome; identified 26,079 indels at least 50 bp in length; cataloged 47,238 SV breakpoints	[32]
<i>Melopsittacus undulatus</i>	Hybrid	PBcR, Celera	3	5.5× PacBio + 15.4× 454 = 3.83× corrected	15,328	1st assembly of > 1 Gb parrot genome; N50 = 93,069	[34]
<i>Vibrio cholerae</i>	Hybrid	BLASR, Bambus, AHA	195	200× PacBio + 28× Illumina + 22× 454	2	No N's in contigs; 99.99% consensus accuracy; N50 = 3.01 Mb	[30]
<i>Helicobacter pylori</i>	PacBio	HGAP, Quiver, PGAP	8 per strain	446.5× average among strains	1 per strain	1 complete contig for each of 8 strains; methylation analysis associated motifs with genotypes of virulence factors	[35]

Note: N50, the contig length for which half of all bases are in contigs of this length or greater; MHC, major histocompatibility complex; SV, structural variation.

PacBio sequencing and its applications

Rhoads & Au, Gen Prot Bioinf 2015

Advantage

Closes gaps and completes genomes due to longer reads

Identifies non-SNP SVs

Achievements

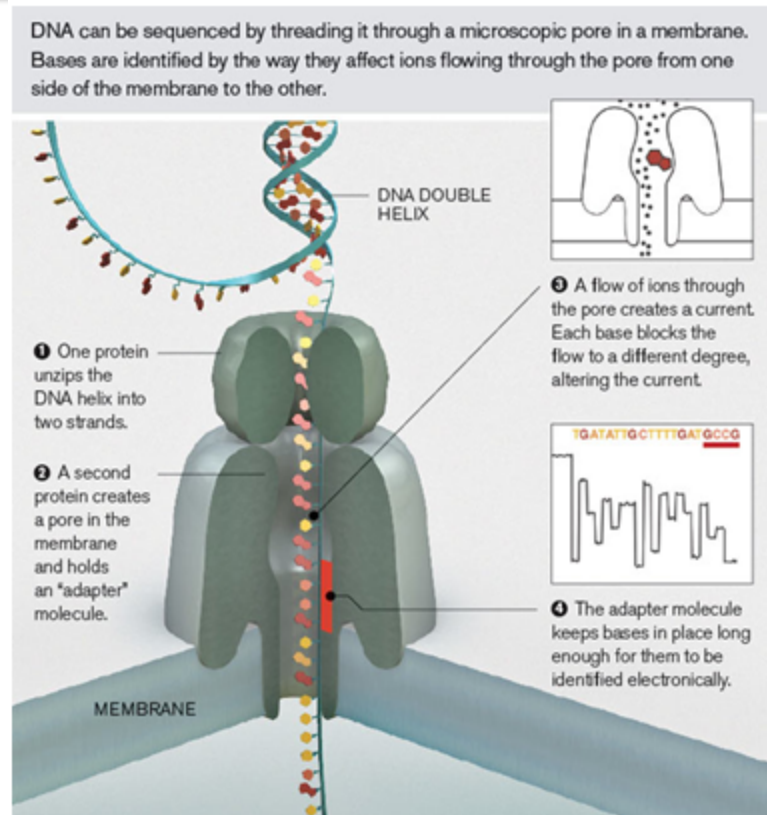
Produced highly-contiguous assemblies of bacterial and eukaryotic genomes

Discovered STRs (short tandem repeats)

Limitations

Both strands can be sequenced several times if the lifetime of the polymerase is long enough.

Nanopore-based fourth-generation DNA sequencing technology. ONT, Oxford Nanopore Technologies



Strand sequencing' is a technique that passes intact DNA polymers through a protein nanopore, sequencing in real time as the DNA translocates the pore.

Nanopore sequencing also offers, for the first time, direct RNA sequencing, as well as PCR or PCR-free cDNA sequencing.

<https://nanoporetech.com/applications/dna-nanopore-sequencing>

Feng et al , Gen Prot Bioinf 2015

Nanopore sequencing

The current state of Nanopore Sequencing. Arakawa. Methods in Molecular Biology 2023

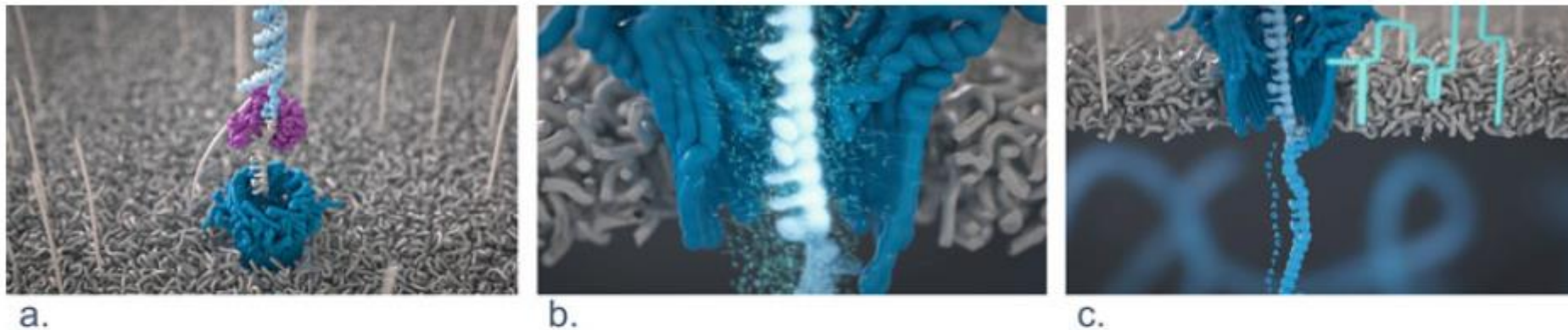


Fig. 1 The principle of nanopore sequencing. (a) A protein nanopore (blue) is imbedded into an electronically resistive lipid membrane (grey), before adapted DNA libraries containing a motor protein (purple) are introduced, and the motor feeds DNA progressively through the pore. (b) An ionic current (represented by light blue dots) is passed through the nanopore as the DNA translocates through the pore. (c) The bases within the nanopore block the current depending on their size and structure. As the strand moves progressively through the pore, a “squiggle” trace is produced, which is decoded into sequence data using artificial neural networks

MinIon, OXFORD NANOPORE



<https://nanoporetech.com/news/movies#movie-24-nanopore-dna-sequencing>

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03/06/2025

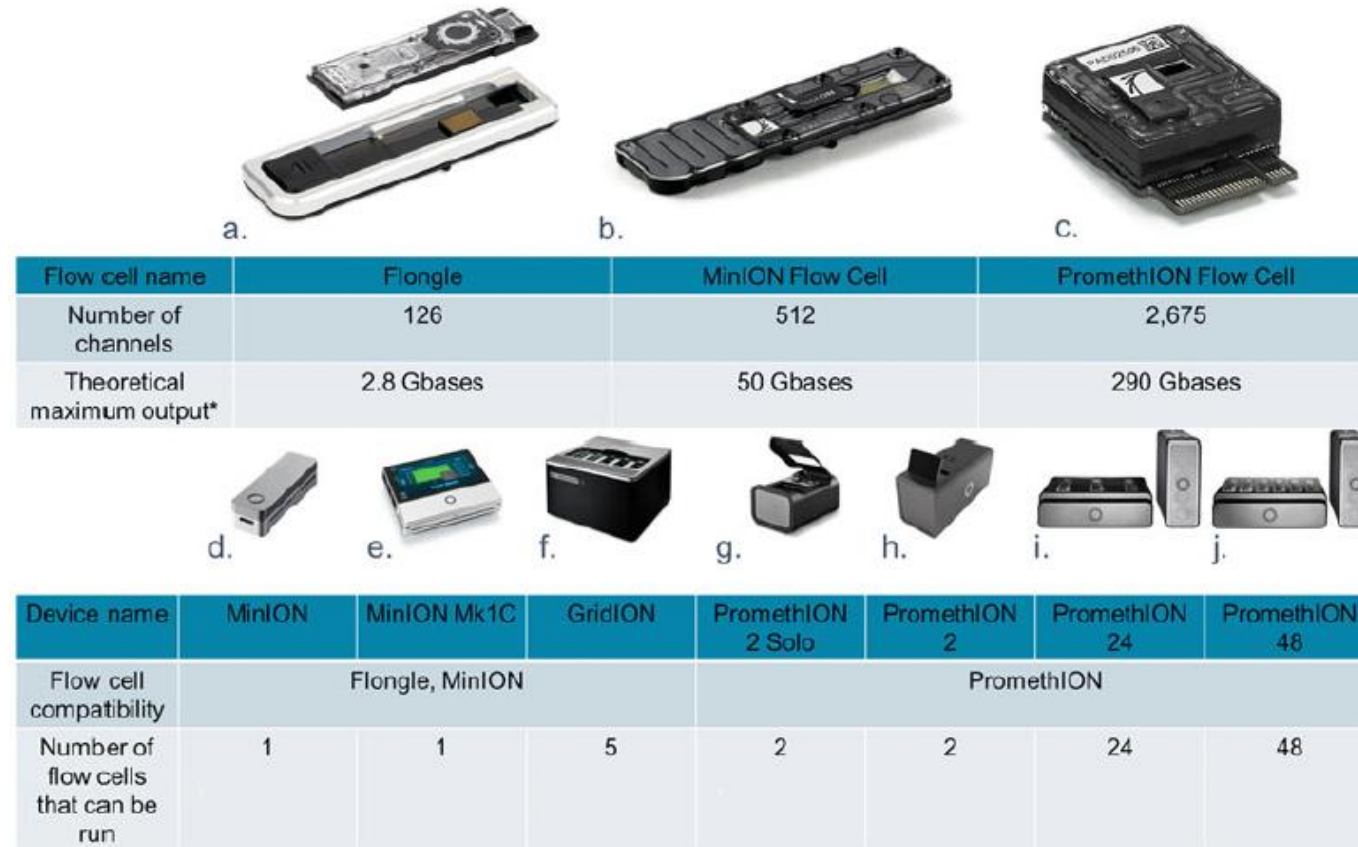


Fig. 2 The flow cells and devices for nanopore sequencing. The Flongle (a) consists of two parts, a reusable adapter, and a single-use flow cell. It has the same footprint as the MinION Flow Cell (b) meaning both can be run on the MinION (d), MinION Mk1C (e), or GridION (f) devices. Any combination of Flongle or MinION can be run on the GridION device. The PromethION Flow Cell (c) is compatible with all PromethION devices (g–j). With capacity for different numbers of flow cells, total device yields vary in line with the number of flow cells they can run. Where multiple flow cells can be run, all are individually controllable, meaning no requirement exists to run all flow cells at once and as a result samples can be run on demand. *Theoretical maximum output when flow cell or device is run 72 h (16 h for Flongle) at 420 bases/second. For devices, this is when all flow cells are run at once and the highest yielding flow cell option is chosen. Outputs may vary according to library type, run conditions, etc.

	Flongle	MinION Mk1B	MinION Mk1C	GridION Mk1	PromethION 24	PromethION 48
Number of channels per flow cell	126	512	512	512	3000	3000
Number of flow cells per device	1	1	1	5	24	48
Price per flow cell	\$90	\$900 - \$475	\$900 - \$475	\$900 - \$475	\$2000 - \$625	\$2000 - \$625
Run time	1 min - 16 hours	1 min - 72 hours	1 min - 72 hours	1 min - 72 hours	1 min - 72 hours	1 min - 72 hours
Yields in field are dependent on sample and preparation methods. Users can get outputs in the following ranges per flow cell when utilising the latest chemistries and protocols	1 - 2 Gb	10 - 30 - 50 Gb	10 - 30 - 50 Gb	10 - 30 - 50 Gb	100 - 200 - 300 Gb	100 - 200 - 300 Gb
Price per Gb for different flow cell yields (yields vary according to sample and preparation methods)	@1 - 2 Gb \$90 per flow cell: \$90 - 45	@10 - 30 - 50 Gb \$900 per flow cell: \$90 - 30 - 18 \$790 per flow cell: \$79 - 26 - 16 \$675 per flow cell: \$68 - 23 - 14 \$500 per flow cell: \$50 - 17 - 10 \$475 per flow cell: \$48 - 16 - 9.5	@10 - 30 - 50 Gb \$900 per flow cell: \$90 - 30 - 18 \$790 per flow cell: \$79 - 26 - 16 \$675 per flow cell: \$68 - 23 - 14 \$500 per flow cell: \$50 - 17 - 10 \$475 per flow cell: \$48 - 16 - 9.5	@10 - 30 - 50 Gb \$900 per flow cell: \$90 - 30 - 18 \$790 per flow cell: \$79 - 26 - 16 \$675 per flow cell: \$68 - 23 - 14 \$500 per flow cell: \$50 - 17 - 10 \$475 per flow cell: \$48 - 16 - 9.5	@100 - 200 - 300 Gb \$1,600 per flow cell: \$16 - 8 - 5 \$1,120 per flow cell: \$11 - 6 - 4 \$940 per flow cell: \$9 - 5 - 3.1 \$680 per flow cell: \$7 - 3.4 - 2.3 \$625 per flow cell: \$6 - 3 - 2	@100 - 200 - 300 Gb \$1,600 per flow cell: \$16 - 8 - 5 \$1,120 per flow cell: \$11 - 6 - 4 \$940 per flow cell: \$9 - 5 - 3.1 \$680 per flow cell: \$7 - 3.4 - 2.3 \$625 per flow cell: \$6 - 3 - 2

Library preparation



Oxford Nanopore has developed VolTRAX – a small device designed to perform library preparation automatically, so that a user can get a biological sample ready for analysis, hands-free. VolTRAX is designed as an alternative to a range of lab equipment, to allow consistent and varied, automated library prep options.

VolTRAX V2 Starter Pack

\$8,000.00

VolTRAX V2 is designed to automate all laboratory processes associated with Nanopore Sequencing from sample extraction to library preparation.

MinIT, Analysis



Eliminating the need for a dedicated laptop
for nanopore sequencing with MinION.
\$2400

MinIT Specifications:

Pre-installed software: Linux OS, MinKNOW, Guppy, EPI2ME

Bluetooth and Wi-Fi enabled; you can control your experiments using a laptop, tablet or smartphone

fastq or fast5 files are written to Onboard storage: 512 GB SSD

Processing: GPU accelerators (ARM processor 6 cores, 256 Core GPU), 8 GB RAM.

Small footprint, 290g

1 x USB 2.0 port, 1 x USB 3.0 port and 1 x Ethernet port (1 Gbit capacity)

MinIT has now been replaced by the MinION Mk1C, which combines the real-time, portable sequencing of MinION, with powerful integrated compute, a high-resolution touchscreen, and full connectivity.

SmidgION, Mobile analysis



Oxford Nanopore has now started developing an even smaller device, SmidgION.

potential applications may include remote monitoring of pathogens in a breakout or infectious disease; the on-site analysis of environmental samples such as water/metagenomics samples, real time species ID for analysis of food, timber, wildlife or even unknown samples; field-based analysis of agricultural environments, and much more.

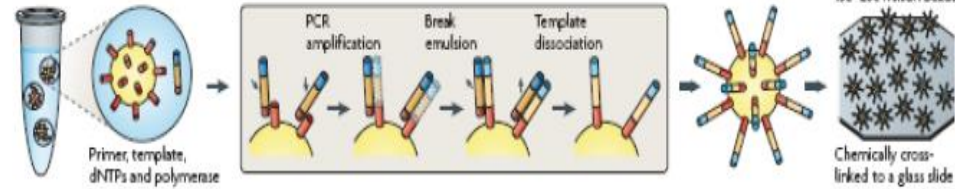
Long-read Sequencing Platforms characteristics

Company	Pacific Biosciences			Oxford Nanopore			
System Platform	Sequel	Sequel II	Sequel IIe	Flongle	MiniON	GridION	PromethION
Sequencing Principle	PacBio Single Molecule Sequencing			Nanopore Single molecule Sequencing			
Detection	Fluorescent			Electrical Conductivity			
Applications	Whole genome <i>de novo</i> assembly, variant detection, structural variation detection, full length transcript sequencing, targeted/amplicon sequencing, metagenomics sequencing			DNA, amplicons, cDNA, Direct RNA sequencing			
Maximum Read length (bases)	300 kb			Longest read so far: > 4 Mb			
Flow cells/device	12 SMRT Cells 1M can be used at a time, and 8 SMRT Cell 8M can be used serially			1 (126 channels per flow cell)	1 (512 channels per flow cell)	5 (512 channels per flow cell)	24 or 48 (3000 channels per flow cell)
Output (per flow cell)	75 Gb	600 Gb		1 - 2 Gb ^a	10 - 30 - 50 Gb ^a		100 - 200 - 300 Gb ^a
Sequencing Run time	Up to 20 hr	Up to 30 hr		1 min - 16 hr	1 min - 72 hr		
Accuracy/Quality Score	Number of HiFi Reads >99% Accuracy: Up to 5,000,000 reads	Number of HiFi Reads >99% Accuracy: Up to 4,000,000 reads		Single Molecule: R9 modal Accuracy >98.3%, R10 modal Accuracy >97.5%. New chemistry Accuracy >99% (coming soon) Consensus: R9.4.1: Current best Q45 (>99.99%) R10: Current Best Q50 (99.999%)			
Equipment Cost (USD)	approximately \$525,000			\$1,460 (12 flow cells included)	\$9,300	\$69,955	24 flow cells: \$335,455 48 flow cells: \$530,000
Dimensions	92.7 x 91.4 x 167.6 cm			105 x 23 x 8 mm	Mk1b: 105 x 23 x 33 mm; Mk1c: 140 x 30 x 116 mm	365 x 220 x 360 mm	Sequencer: 590 x 190 x 430 mm; Data Acquisition unit: 178 x 440 x 470 mm
Weight	362 kg			20 g	Mk1b: 87 g Mk1c: 450 g	11 kg	Sequencer: 28 kg; Data Acquisition unit: 25 kg
Advantages	Very long reads can help resolve ambiguities; no DNA amplification required, comparatively faster turnaround time			Fast-sequencing; Small instrument footprint; Portability; Real-time data analysis			
Disadvantages	Sequencing equipment is expensive, which could be cost-prohibitive for smaller clinical laboratories, large footprint of the equipment, historically higher error rate (continues to improve)			Historically higher error rate (continues to improve)			

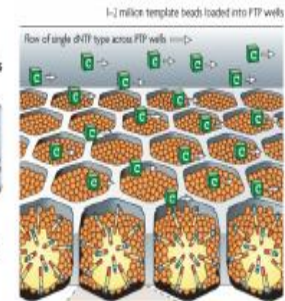
ESTRATEGIAS DE INMOVILIZACIÓN

a Roche/454, Life/APG, Polonator
Emulsion PCR

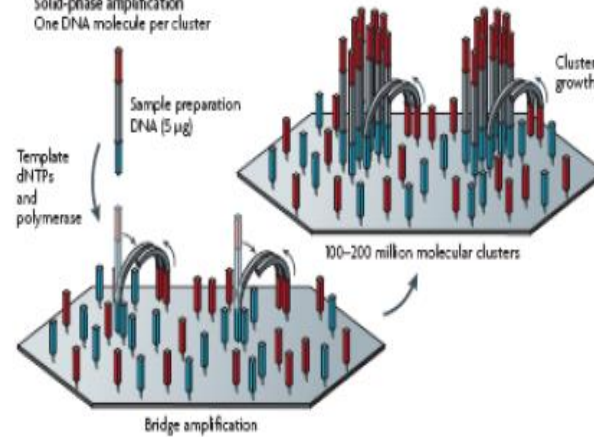
One DNA molecule per bead. Clonal amplification to thousands of copies occurs in microreactors in an emulsion



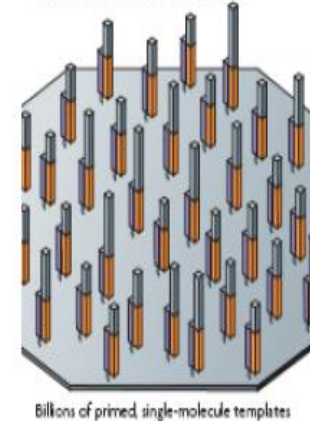
c Roche/454 — Pyrosequencing



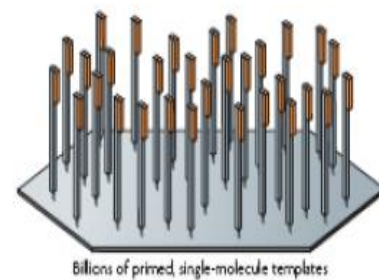
b Illumina/Solexa
Solid-phase amplification
One DNA molecule per cluster



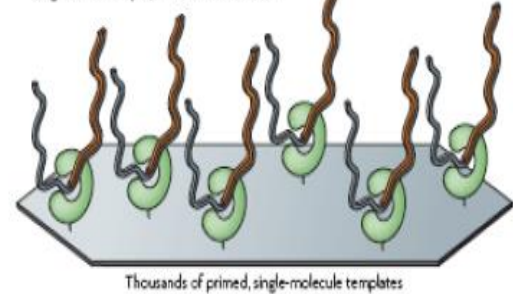
c Helicos BioSciences: one-pass sequencing
Single molecule: primer immobilized



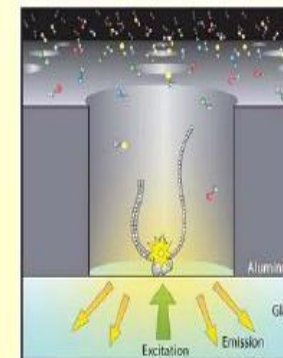
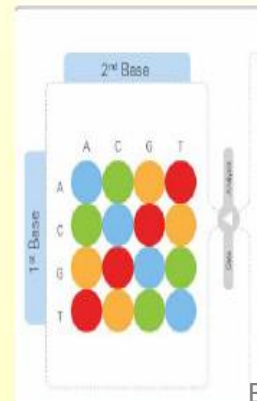
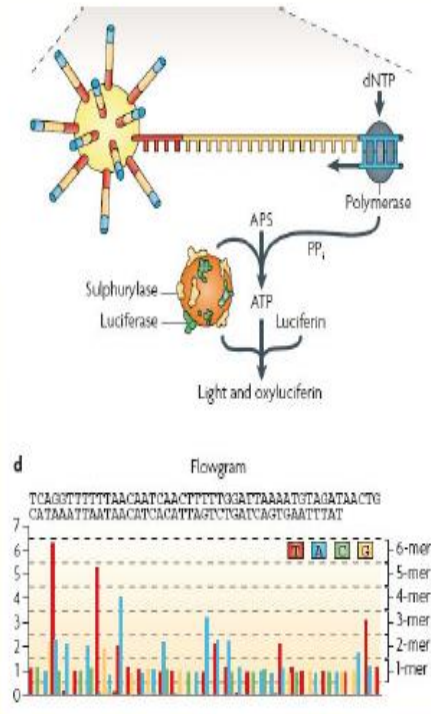
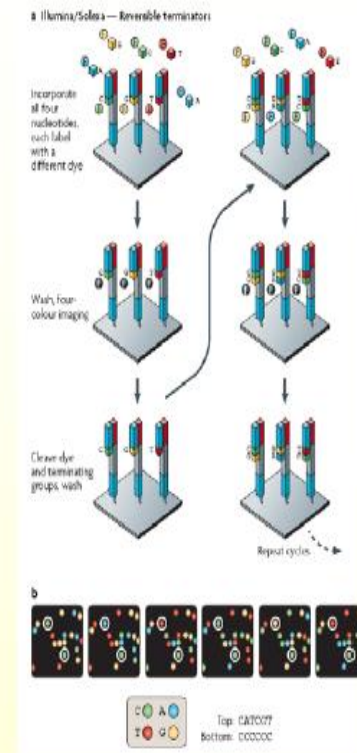
d Helicos BioSciences: two-pass sequencing
Single molecule: template immobilized



e Pacific Biosciences, Life/Visigen, LI-COR Biosciences
Single molecule: polymerase immobilized



ESTRATEGIAS DE LECTURA



BU-ISCIH

PacBio sequencing and its applications

Rhoads & Au, Gen Prot Bioinf 2015

Performance comparison of sequencing platforms of various generations

Method	Generation	Read length (bp)	Single pass error rate (%)	No. of reads per run	Time per run	Cost per million bases (USD)	Refs.
Sanger ABI 3730×1	1st	600–1000	0.001	96	0.5–3 h	500	[14,18–21]
Ion Torrent	2nd	200	1	8.2×10^7	2–4 h	0.1	[15,25]
454 (Roche) GS FLX +	2nd	700	1	1×10^6	23 h	8.57	[14,17,27]
Illumina HiSeq 2500 (High Output)	2nd	2×125	0.1	8×10^9 (paired)	7–60 h	0.03	[9,16,26]
Illumina HiSeq 2500 (Rapid Run)	2nd	2×250	0.1	1.2×10^9 (paired)	1–6 days	0.04	[9,16,26]
SOLiD 5500×1	2nd	2×60	5	8×10^8	6 days	0.11	[14,24]
PacBio RS II: P6-C4	3rd	$1.0\text{--}1.5 \times 10^4$ on average	13	$3.5\text{--}7.5 \times 10^4$	0.5–4 h	0.40–0.80	[5,12,15]
Oxford Nanopore MinION	3rd	$2\text{--}5 \times 10^3$ on average	38	$1.1\text{--}4.7 \times 10^4$	50 h	6.44–17.90	[22,23]

Characteristics, strengths and weaknesses of commonly used sequencing platforms

Table 2

Characteristics, strengths and weaknesses of commonly used sequencing platforms

Platform \ Instrument	Throughput range (Gb) ^a	Read length (bp)	Strength	Weakness
<i>Sanger sequencing</i>				
ABI 3500/3730	0.0003	Up to 1 kb	Read accuracy and length	Cost and throughput
<i>Illumina</i>				
MiniSeq	1.7–7.5	1×75 to ×150	Low initial investment	Run and read length
MiSeq	0.3–15	1×36 to 2×300	Read length, scalability	Run length
NextSeq	10–120	1×75 to 2×150	Throughput	Run and read length
HiSeq (2500)	10–1000	×50 to ×250	Read accuracy, throughput,	High initial investment, run
NovaSeq 5000/6000	2000–6000	2×50 to ×150	Read accuracy, throughput	High initial investment, run
<i>IonTorrent</i>				
PGM	0.08–2	Up to 400	Read length, speed	Throughput, homopolymers ^c
S5	0.6–15	Up to 400	Read length, speed,	Homopolymers ^c
Proton	10–15	Up to 200	Speed, throughput	Homopolymers ^c
<i>Pacific BioSciences</i>				
PacBio RSII	0.5–1 ^b	Up to 60 kb	Read length, speed (Average 10 kb, N50 20 kb)	High error rate and initial
Sequel	5–10 ^b	Up to 60 kb	Read length, speed (Average 10 kb, N50 20 kb)	High error rate
<i>Oxford Nanopore</i>				
MinION	0.1–1	Up to 100 kb	Read length, portability	High error rate, run length,

^a The throughput ranges are determined by available kits and run modes on a per run basis. As an example of a 15-GB throughput, thirty-five 5-MB genomes can be sequenced to a minimum coverage of 40× on the Illumina MiSeq using the v3 600 cycle chemistry.

^b Per one single-molecule real-time cell.

^c Results in increased error rate (increased proportion of reads containing errors among all reads) which in turn results in false-positive variant calling.

Besser et al., Clin Micr Infect, 2018

Comparison of various high-performing sequencing instruments

Pervez et al., BioMed Research International 2022

Manufacturer	Read length	Data output	Max. run time (hours)	Chemistry	Key applications**
Illumina (NovaSeq 6000)	300 PE	6 Tb (6000 Gb)	44	Sequencing by synthesis	SS-WGS and TGS, TGEF, 16sMGS, WES, SCP, LS-WGS, CA, MS, MGP, CFS, LBA
Thermo Fisher Scientific Ion Torrent (Ion GeneStudio S5 Prime)	600 SE	50 Gb	12	Sequencing by synthesis	WGS, WES, TGS
GenapSys (16 chips)	150 SE	2 Gb	24	Sequencing by synthesis	TS, SS-WGS, GEV, 16S rRNA sequencing, sRNA sequencing, TSCAS
QIAGEN (GeneReader)	100 SE	Not available	Not available	Sequencing by synthesis	Cancer research and identifying mutations
BGI/Complete Genomics	400 SE	6 Tb (6000 Gb)	40	DNA nanoball	Small and large WGS, WES and TGS
PacBio (HiFi Reads)	25 Kb	66.5 Gb	30	Real-time sequencing	DN sequencing, FT, identifying ASI, mutations, and EPM
Nanopore (PromethION)	4 Mb	14 Tb (14000 Gb)	72	Real-time sequencing	SV, GS, phasing, DNA and RNA base modifications, FT, and isoform detection

Advantages and disadvantages of sequencing generations

Pervez et al., BioMed Research International 2022

Sequencing generation	Advantages	Disadvantages
First generation	High accuracy Helps in validating findings of NGS	High cost Low throughput
Second generation	High throughput Low cost Have clinical applications Short run time	Short read length Difficult sample preparation PCR amplification Long run time
Third generation	No PCR amplification Require less starting material Longer read lengths Very low cost Low error rate during library preparation Advantages of 3 rd GS+	High sequencing error rate 10–15% in the PacBio and 5–20% in the ONT Fresh DNA requires for ensuring quality of ultralong reads Database systems and algorithms/tools are rare for analyzing 3 rd and 4 th GS data
Fourth generation	Ultrafast: scan of whole genome in 15 minutes Spatial distribution of the sequencing reads over the sample can be seen	

Advantages & disadvantages for short vs. Long read sequencing

	Advantages	Limitations
Short-read sequencing	· Higher sequence fidelity · Cheap · Can sequence fragmented DNA	· Not able to resolve structural variants, phasing alleles or distinguish highly homologous genomic regions · Unable to provide coverage of some repetitive regions
Long-read sequencing	· Able to sequence genetic regions that are difficult to characterize with short-read seq due to repeat sequences · Able to resolve structural rearrangements or homologous regions · Able to read through an entire RNA transcript to determine the specific isoform · Assists <i>de novo</i> genome assembly	· Lower per read accuracy · Bioinformatic challenges, caused by coverage biases, high error rates in base allocation, scalability and limited availability of appropriate pipelines

<https://www.technologynetworks.com/genomics/articles/an-overview-of-next-generation-sequencing-346532>

Advantages 3rd GS over 2nd GS

Pervez et al., BioMed Research International 2022

- Higher throughput
- Detecting haplotype directly
- Longer read lengths
- Better consensus accuracy to identify rare variants
- Whole chromosome phasing
- Small amount of sample are the salient features of the 3rd-generation sequencing which had it useful in clinical diagnostic

TABLE 5: An overview of human genome assembly quality metrics between PacBio system, Nanopore, and Illumina [49].

	Nanopore+Illumina	PacBio HiFi sequencing
Contiguity (N50)	32.3 Mb	98.7 Mb
Correctness (quality score)	Q34	Q51
Completeness (genome size)	2.8 Gb	3.1 Gb

TABLE 6: Overall costs for sequencing a human genome [49].

	Nanopore+Illumina	PacBio HiFi sequencing (US \$)
Consumables	4800	3800
Compute	5050	3850
Data storage	5200	3900

NGS PLATFORMS, main characteristics

- Numero de bases que secuencia
- Numero lecturas → aplicaciones
- Longitud de las lecturas -→ importante para las aplicaciones ensamblado genomas, de illumina a PacBio
- Error de la base → Corrección con profundidad de lectura y evolución de la tecnología
- Formato fichero salida
- Software dedicado, universal fastq

Mercado de secuenciadores desde 2023 (finaliza la patente de Illumina)

Polonis, Clinical Chemistry 71:1 97-108, 2025

Table 1. Short-read sequencing platforms by application range, performance characteristics, cost, and unique features (if available or applicable).								
Manufacturer		Illumina	Element Biosciences	MGI	PacBio	Singular Genomics	ThermoFisher Scientific	Ultima Genomics
Sequencing Platform		NovaSeq X Series	Aviti	DNBSEQ-T20X2	Onso	G4	Ion Torrent Genexus	UG 100
Advertised and Published applications	Whole genome	✓	✓	✓		✓		✓
	Exome large gene panels	✓	✓	✓	✓	✓		✓
	Targeted gene sequencing	✓	✓	✓	✓	✓	✓	✓
	Transcriptome sequencing	✓	✓	✓		✓		✓
	Methylation sequencing	✓	✓	✓		✓		✓
	Metagenomic profiling/16S sequencing	✓	✓	✓		✓	✓	✓
	Low-level variant detection	✓	✓	✓	✓	✓	✓	✓
	Single cell sequencing	✓	✓	✓	✓	✓		✓
Chemistry	Sequencing by Synthesis (SBS)	Avidity Sequencing	Nanoball Sequencing	SBB	4-color SBS	Semiconductor Sequencing	mostly natural Sequencing by Synthesis (mnSBS)	
Max read length	150 bp	300 bp	150 bp	200 bp	150 bp	approx. 400 bp	approx. 300 bp	
Max data output	16 Tb	300 Gb	72 Tb	150 Gb	200 Gb	24 Gb	28 Tb	
Max N of reads	52 B	1 B	40 B	1 B	1.6 B	100 M	8 B	
Max run time	48 h	48 h	80 h	48 h	24 h	30 h	14 h	
Data quality	≥85% Q30	>90% Q30	>85% Q30	≥90% Q40	≥85% Q30		>85% flows Q30	
Sequencer cost ^a		\$289 000		\$259 000	\$295 000	\$300 000		
\$ per Gb ^a	\$2	\$5	\$1	\$4	\$8	n/a	\$1	
	n/a	yes	yes	yes	yes	n/a	yes	
Continued								

Mercado de secuenciadores desde 2023 (finaliza la patente de Illumina)

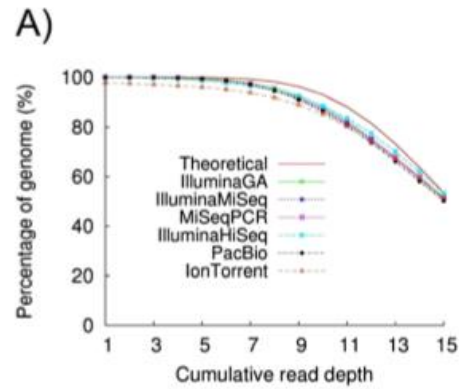
Polonis, Clinical Chemistry 71:1 97-108, 2025

Table 1. (continued)							
Chemistry	Sequencing by Synthesis (SBS)	Avidity Sequencing	Nanoball Sequencing	SBB	4-color SBS	Semiconductor Sequencing	mostly natural Sequencing by Synthesis (mnSBS)
Compatibility with third-party library solutions							
Website	https://www.illumina.com/systems/sequencing-platforms/novaseq-x-plus.html	https://www.elementbiosciences.com/	https://en.mgi-tech.com/ https://www.completegenomics.com/products/sequencing-platforms/	https://www.pacb.com/onso/	https://singulargenomics.com/	https://www.thermofisher.com/	https://www.ultimagenomics.com/ug-100-tm-sequencer
Unique features	Complete Long Reads technology generates linked sequencing reads	Independent dual flow cells per instrument; workflows for generating linked sequencing reads	Fully automated with integrated robotics		Can be converted to G4X allowing direct RNA sequencing, RNA transcript detection, multiplex protein analysis, digital hematoxylin and eosin	end-to-end NGS platform	24/7 automation

^aAs per information on respective websites (accessed June 2024).
 Abbreviations: approx., approximately; SBB, sequencing by binding; SBS, sequencing by synthesis; Tb, terabase.

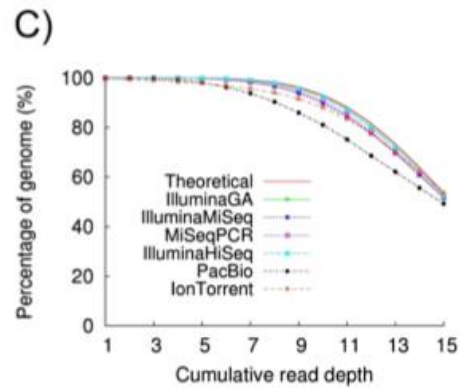
Uniformidad de cobertura a lo largo del genoma

%GC



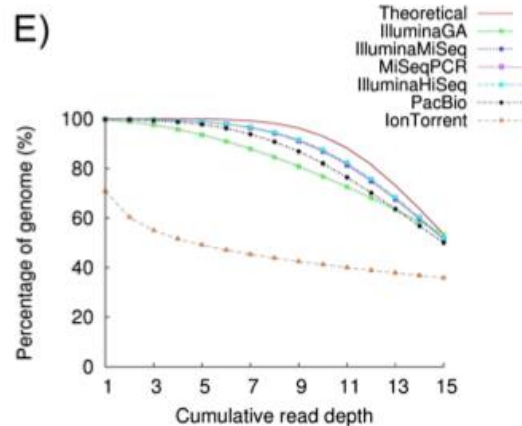
Bordetella pertussis

%GC



Staphylococcus aureus

%AT



Plasmodium falciparum

Quail et al., BMC Genomics 2012, 13:341



Performance assessment of DNA sequencing platforms in the ABRF Next-Generation Sequencing Study

Foxx et al., Nature Biotechnology 2021

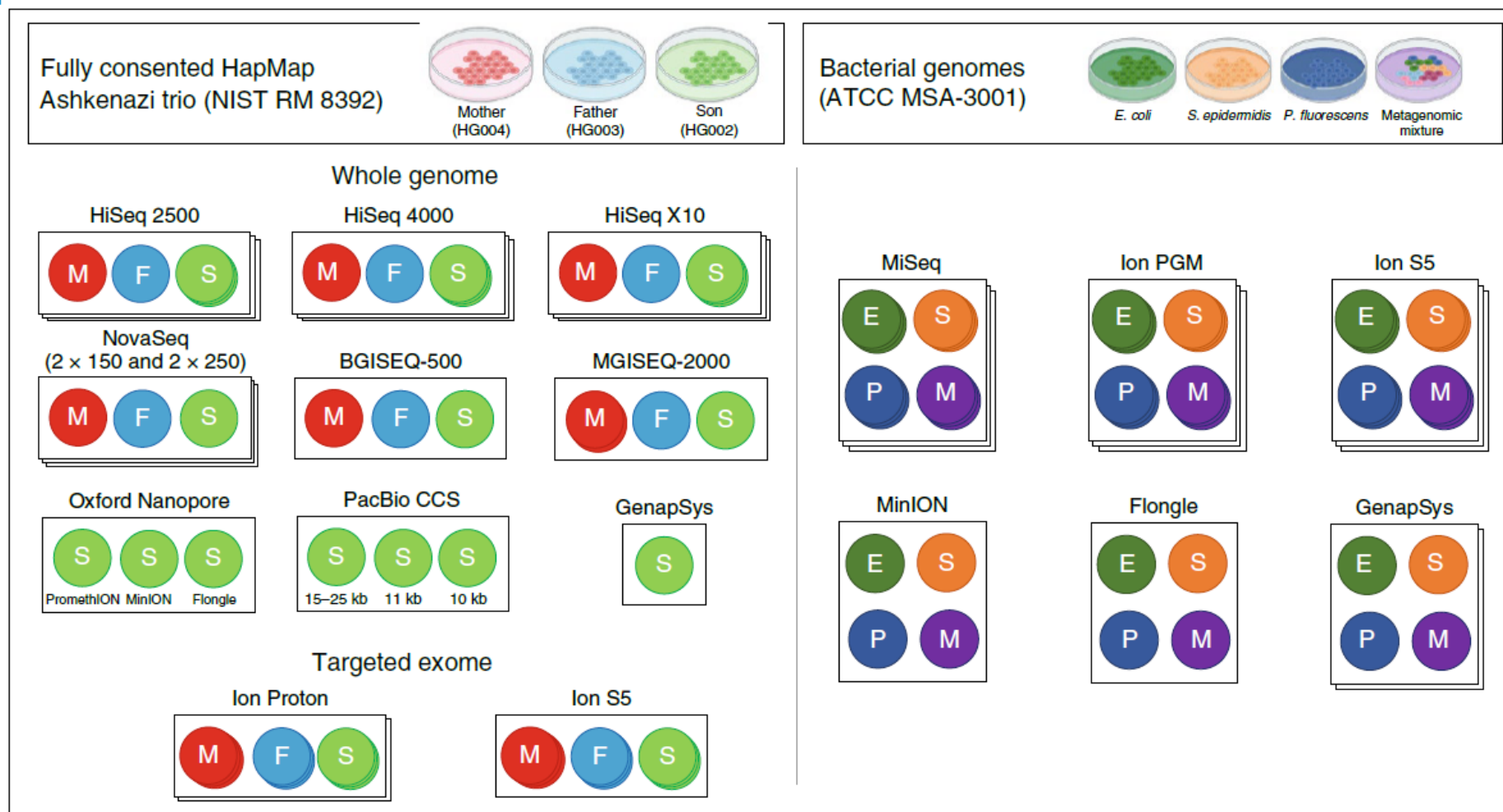
Interlaboratory and intralaboratory DNA-seq replicates of

- The Ashkenazi trio (NIST reference material)
- Three individual bacterial strains
- Metagenomic mixture of ten bacterial species

TO STUDY THE EFFECTS OF GC CONTENT AND LIBRARY COMPLEXITY.

These replicates were generated across

- five Illumina platforms
- Three ThermoFisher Ion Torrent platforms, the BGISEQ-500 and MGISEQ-2000 platforms, the GenapSys GS111 Platform
- Oxford Nanopore Technologies (ONT) Flongle, MinION and PromethION flow cells
- Publicly available PacBio CCS data for HG002.



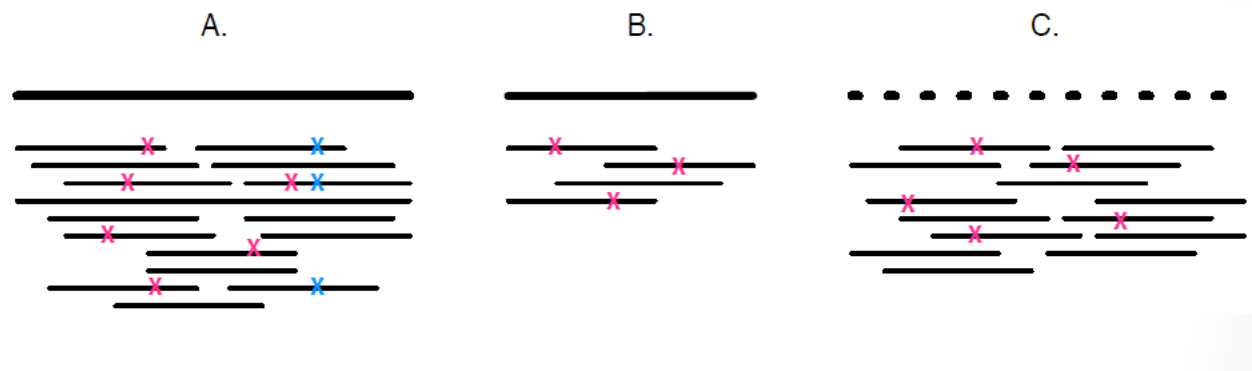
CONCLUSIONS Foox et al., Nature Biotechnology 2021

- Among short-read instruments, **HiSeq 4000 and X10** provided the **most consistent, highest genome coverage**, while **BGI/MGISEQ** provided the **lowest sequencing error rates**
- The long-read instrument **PacBio CCS** had the **highest reference-based mapping rate and lowest non-mapping rate**.
- The two long-read platforms **PacBio CCS and PromethION/MinION** showed the **best sequence mapping in repeat-rich areas and across homopolymers**.
- **NovaSeq 6000 using 2 × 250-bp** read chemistry was the most robust instrument for **capturing known insertion/deletion events**.

Algunos conceptos en secuenciación

Básicamente tres problemas

Resecuenciación, Conteo y ensamblado



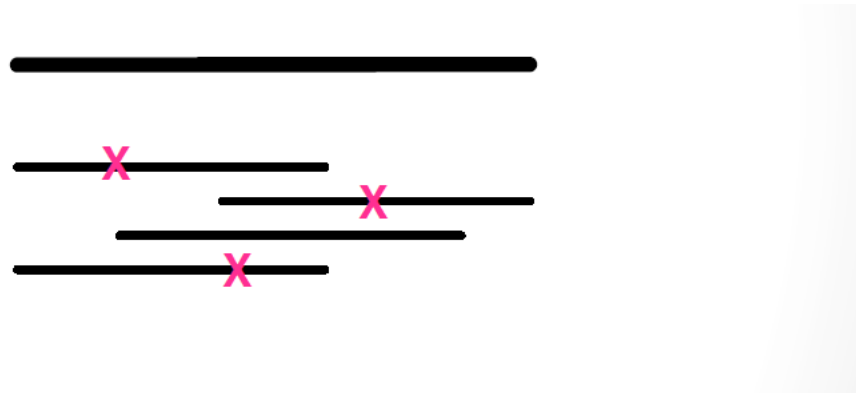
Resecuenciación

Conocemos el genoma, genoma de referencia, y queremos identificar variaciones (azul), en un background de errores (rosa)



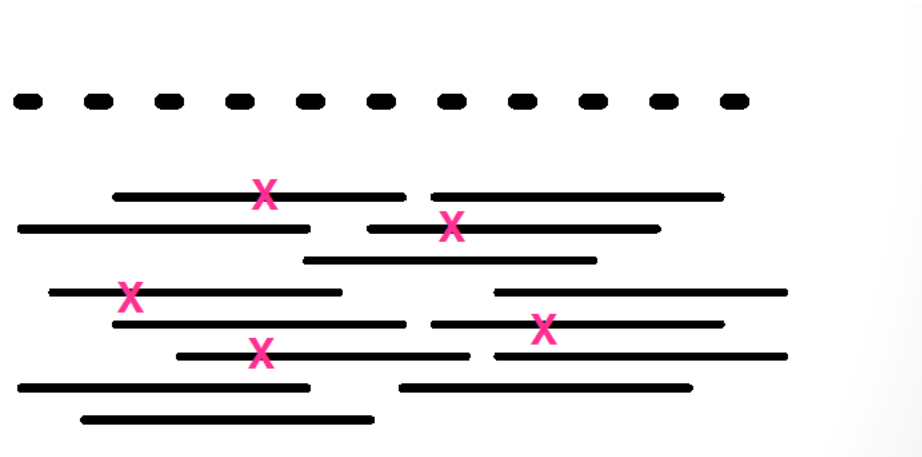
Conteo

Número de lecturas de un gen (amplicón) o mRNA (RNAseq). Equivalente a expresión en Microarrays.



Ensamblado

No hay genoma de referencia y lo construimos de novo



Sequencing terms

Breadth of coverage

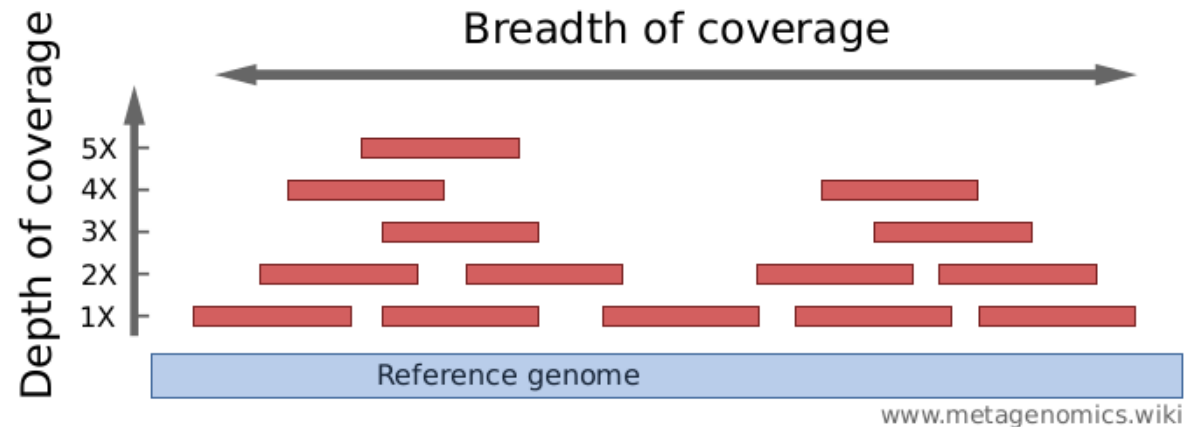
How much of a genome is "covered" by short reads? Are there regions that are not covered, even not by a single read?

Breadth of coverage is the percentage of bases of a reference genome that are covered with a certain depth. For example: 90% of a genome is covered at 1X depth; and still 70% is covered at 5X depth.

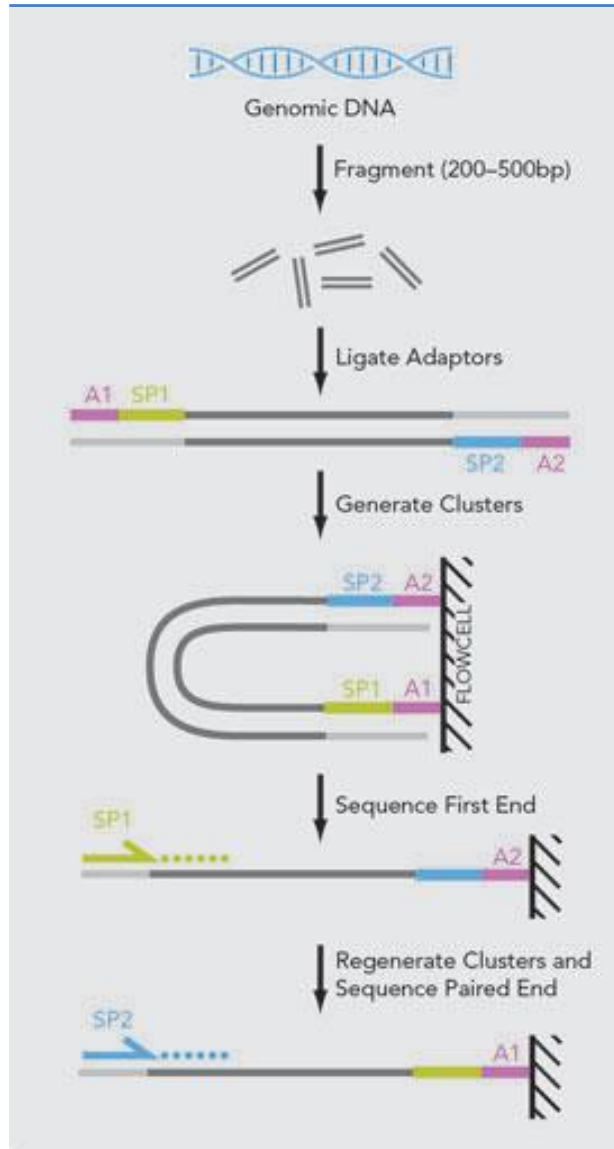
Depth of coverage

How strong is a genome "covered" by sequenced fragments (short reads)?

Per-base coverage is the average number of times a base of a genome is sequenced. The coverage depth of a genome is calculated as the number of bases of all short reads that match a genome divided by the length of this genome. It is often expressed as 1X, 2X, 3X,... (1, 2, or 3 times coverage).

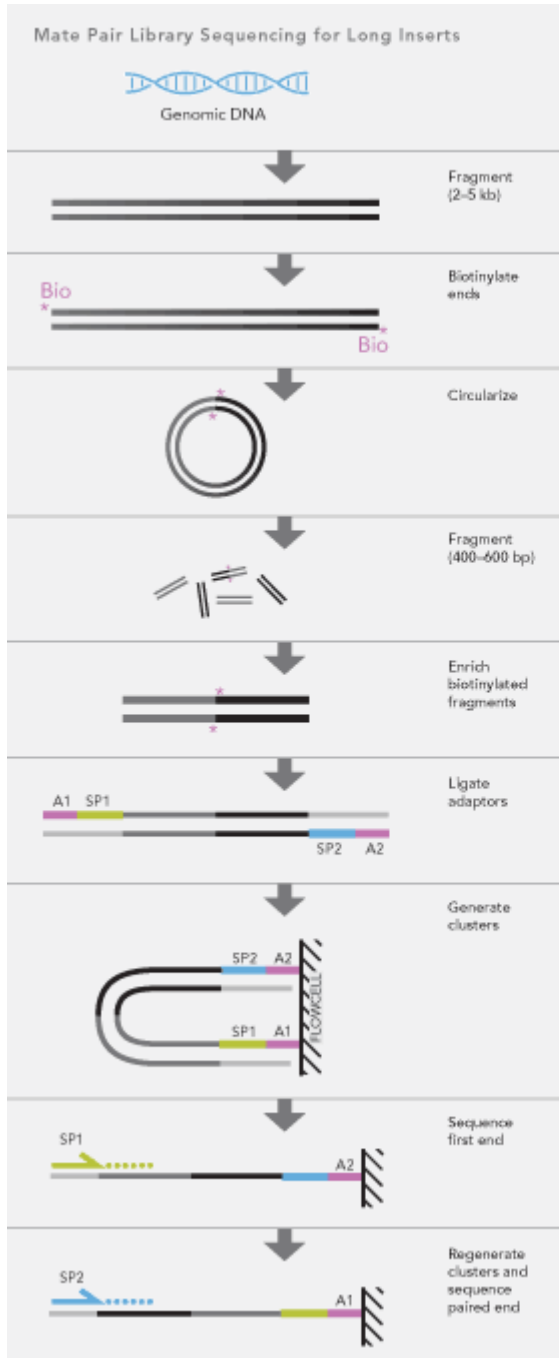


Que es Paired-end?



Secuenciación de un fragmento (bp)

**Modificación de single-read DNA,
Leyendo por ambos extremos, forward y reverse**



Mate Pair library preparation is designed to generate short fragments that consist of two segments that originally had a separation of several kilobases in the genome. Fragments of sample genomic DNA are end-biotinylated to tag the eventual mate pair segments. Self-circularization and refragmentation of these large fragments generates a population of small fragments, some of which contain both mate pair segments with no intervening sequence. These Mate Pair fragments are enriched using their biotin tag. Mate Pairs are sequenced using a similar two-adaptor strategy as described for paired-end sequencing.

Que es Mate-pair?

Secuenciación de dos fragmentos separados kb.

Util:

**Secuenciación de un Genoma de novo
Finalizar un genoma**

Detección de variantes estructurales

Resumen

Número de lecturas por muestra (cobertura y profundidad de cobertura)

Número de muestras que secuencio simultáneamente

Elegir secuenciador: Número de bases que lee un secuenciador y longitud de lectura, single o paired-end (kit de secuenciación)

Cobertura es importante para la llamada a variantes, RNAseq. **Plataforma con mayor rendimiento Illumina**

La longitud de las lecturas es importante para el ensamblado

PacBio y Minlon mayor longitud de lecturas (corrección de errores con Illumina)

Coverage and Read Depth Recommendations by Sequencing Application

Table 1: Coverage and Read Recommendations by Application

Category	Detection or Application	Recommended Coverage (x) or Reads (millions)	References
Whole genome sequencing	Homozygous SNVs	15x	Bentley et al., 2008
	Heterozygous SNVs	33x	Bentley et al., 2008
	INDELs	60x	Feng et al., 2014
	Genotype calls	35x	Ajay et al., 2011
	CNV	1-8x	Xie et al., 2009; Medvedev et al., 2010
Whole exome sequencing	Homozygous SNVs	100x (3x local depth)	Clark et al., 2011; Meynert et al., 2013
	Heterozygous SNVs	100x (13x local depth)	Clark et al., 2011; Meynert et al., 2013
	INDELs	not recommended	Feng et al., 2014
Transcriptome Sequencing	Differential expression profiling	10-25M	Liu Y. et al., 2014; ENCODE 2011 RNA-Seq
	Alternative splicing	50-100M	Liu Y. et al., 2013; ENCODE 2011 RNA-Seq
	Allele specific expression	50-100M	Liu Y. et al., 2013; ENCODE 2011 RNA-Seq
	De novo assembly	>100M	Liu Y. et al., 2013; ENCODE 2011 RNA-Seq

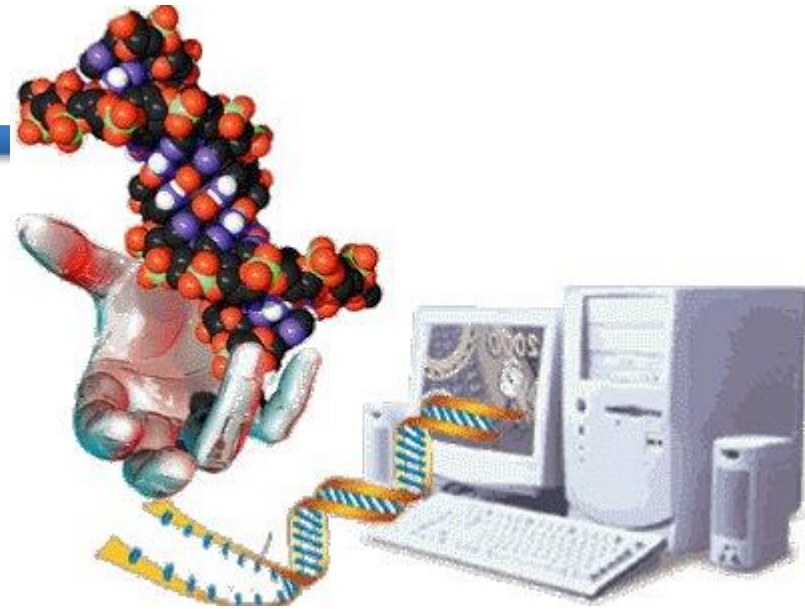
<https://genohub.com/recommended-sequencing-coverage-by-application/>

Coverage and Read Depth Recommendations by Sequencing Application

DNA Methylation Sequencing	CAP-Seq	>20M	Long, H.K. et al., 2013
	MeDIP-Seq	60M	Taiwo, O. et al., 2012
	RRBS (Reduced Representation Bisulfite Sequencing)	10X	ENCODE 2011 Genome
	Bisulfite-Seq	5-15X; 30X	Ziller, M.J et al., 2015; Epigenomics Road Map
RNA-Target-Based Sequencing	CLIP-Seq	10-40M	Cho J. et al., 2012; Eom T. et al., 2013; Sugimoto Y. et al., 2012
	iCLIP	5-15M	Sugimoto Y. et al., 2012; Rogelj B. et al., 2012
	PAR-CLIP	5-15M	Rogelj B. et al., 2012
	RIP-Seq	5-20M	Lu Z. et al., 2014
Small RNA (microRNA) Sequencing	Differential Expression	~1-2M	Metpally RPR et al., 2013; Campbell et al., 2015
	Discovery	~5-8M	Metpally RPR et al., 2013; Campbell et al., 2015

<https://genohub.com/recommended-sequencing-coverage-by-application/>

**Gracias por la
atención
Preguntas ???**



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